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Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced

Paper 1: Pure Mathematics 1

Wednesday 6 June 2018 – Morning

Time: 2 hours

Paper Reference

9MA0/01

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. Given that θ is small and is measured in radians, use the small angle approximations to find an approximate value of

$$\frac{1 - \cos 4\theta}{2\theta \sin 3\theta}$$

(3)

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Question 1 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 1 is 3 marks)



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(c) Determine the nature of this stationary point, giving a reason for your answer.

(2)

Question 2 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 2 is 7 marks)



3.

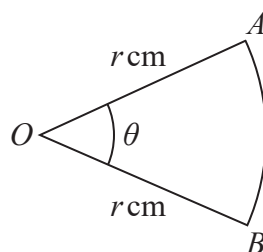


Figure 1

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is θ radians.

The area of the sector AOB is 11 cm^2

Given that the perimeter of the sector is 4 times the length of the arc AB , find the exact value of r .

(4)

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Question 3 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 3 is 4 marks)



4. The curve with equation $y = 2\ln(8 - x)$ meets the line $y = x$ at a single point, $x = \alpha$.

(a) Show that $3 < \alpha < 4$

(2)

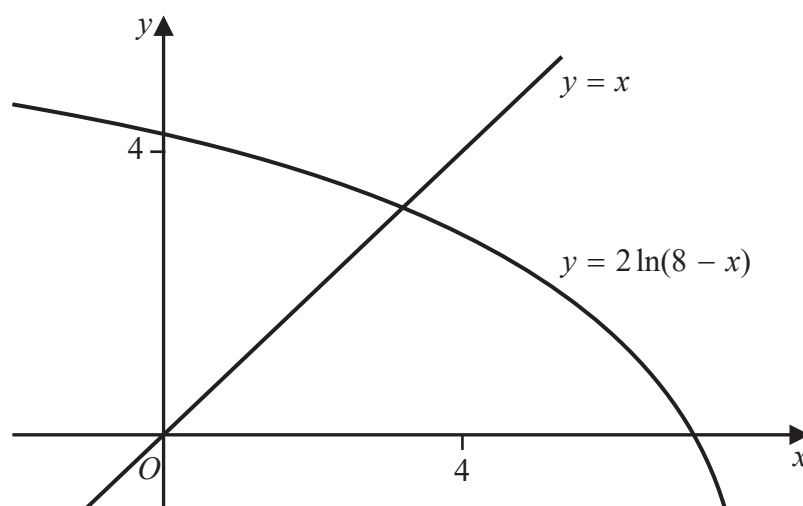


Figure 2

Figure 2 shows the graph of $y = 2 \ln(8 - x)$ and the graph of $y = x$.

A student uses the iteration formula

$$x_{n+1} = 2 \ln(8 - x_n), \quad n \in \mathbb{N}$$

in an attempt to find an approximation for α .

Using the graph and starting with $x_1 = 4$

(b) determine whether or not this iteration formula can be used to find an approximation for α , justifying your answer.

(2)



Question 4 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 4 is 4 marks)



5. Given that

$$y = \frac{3 \sin \theta}{2 \sin \theta + 2 \cos \theta} \quad -\frac{\pi}{4} < \theta < \frac{3\pi}{4}$$

show that

$$\frac{dy}{d\theta} = \frac{A}{1 + \sin 2\theta} \quad -\frac{\pi}{4} < \theta < \frac{3\pi}{4}$$

where A is a rational constant to be found.

(5)

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 6 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 6 is 10 marks)



7. Given that $k \in \mathbb{Z}^+$

(a) show that $\int_k^{3k} \frac{2}{(3x-k)} dx$ is independent of k ,

(b) show that $\int_k^{2k} \frac{2}{(2x-k)^2} dx$ is inversely proportional to k . (3)

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 7 marks)



8. The depth of water, D metres, in a harbour on a particular day is modelled by the formula

$$D = 5 + 2 \sin(30t)^\circ \quad 0 \leq t < 24$$

where t is the number of hours after midnight.

A boat enters the harbour at 6:30 am and it takes 2 hours to load its cargo.

The boat requires the depth of water to be at least 3.8 metres before it can leave the harbour.

- (a) Find the depth of the water in the harbour when the boat enters the harbour.

(1)

- (b) Find, to the nearest minute, the earliest time the boat can leave the harbour.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)



Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 5 marks)



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Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 10 marks)



10. The height above ground, H metres, of a passenger on a roller coaster can be modelled by the differential equation

$$\frac{dH}{dt} = \frac{H \cos(0.25t)}{40}$$

where t is the time, in seconds, from the start of the ride.

Given that the passenger is 5 m above the ground at the start of the ride,

- (a) show that $H = 5e^{0.1 \sin(0.25t)}$

- (b) State the maximum height of the passenger above the ground. (1)

The passenger reaches the maximum height, for the second time, T seconds after the start of the ride.

- (c) Find the value of T . (2)



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Question 10 continued

Lined area for writing the answer to Question 10.

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Question 10 continued

Lined area for writing the answer to Question 10.

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



Question 11 continued

Handwriting practice area with 20 horizontal lines.

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Question 11 continued

Lined area for writing the answer to Question 11.

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Question 11 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 11 is 10 marks)



12. The value, £ V , of a vintage car t years after it was first valued on 1st January 2001, is modelled by the equation

$$V = Ap^t \quad \text{where } A \text{ and } p \text{ are constants}$$

Given that the value of the car was £32 000 on 1st January 2005 and £50 000 on 1st January 2012

- (a) (i) find p to 4 decimal places,
- (ii) show that A is approximately 24 800
- (b) With reference to the model, interpret
- (i) the value of the constant A ,
- (ii) the value of the constant p .

Using the model,

- (c) find the year during which the value of the car first exceeds £100 000



Question 12 continued

Lined area for writing the answer to Question 12.

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Question 12 continued

Lined area for writing the answer to Question 12.

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 10 marks)



13. Show that

$$\int_0^2 2x\sqrt{x+2} \, dx = \frac{32}{15}(2 + \sqrt{2}) \quad (7)$$

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Question 13 continued

Handwriting practice area with 20 horizontal lines.

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Question 13 continued

Lined area for writing the answer to Question 13.

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



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Question 14 continued

Handwriting practice area with 20 horizontal lines.

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 10 marks)

TOTAL FOR PAPER IS 100 MARKS

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Centre Number

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Candidate Number

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Mathematics

Advanced

Paper 2: Pure Mathematics 2

Wednesday 13 June 2018 – Morning

Time: 2 hours

Paper Reference

9MA0/02

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Mathematical Formulae and Statistical Tables, calculator

Total Marks

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. Relative to a fixed origin O ,

the point A has position vector $(2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k})$,

the point B has position vector $(4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})$,

and the point C has position vector $(a\mathbf{i} + 5\mathbf{j} - 2\mathbf{k})$, where a is a constant and $a < 0$

D is the point such that $\overrightarrow{AB} = \overrightarrow{BD}$.

(a) Find the position vector of D .

(2)

Given $|\overrightarrow{AC}| = 4$

(b) find the value of a .

(3)

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Question 2 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 2 is 5 marks)



3. (a) “If m and n are irrational numbers, where $m \neq n$, then mn is also irrational.”

Disprove this statement by means of a counter example.

(2)

- (b) (i) Sketch the graph of $y = |x| + 3$

- (ii) Explain why $|x| + 3 \geq |x + 3|$ for all real values of x .

(3)

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4. (i) Show that $\sum_{r=1}^{16} (3 + 5r + 2^r) = 131\,798$ (4)

(ii) A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = \frac{1}{u_n}, \quad u_1 = \frac{2}{3}$$

Find the exact value of $\sum_{r=1}^{100} u_r$ (3)

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



5. The equation $2x^3 + x^2 - 1 = 0$ has exactly one real root.

(a) Show that, for this equation, the Newton-Raphson formula can be written

$$x_{n+1} = \frac{4x_n^3 + x_n^2 + 1}{6x_n^2 + 2x_n} \quad (3)$$

Using the formula given in part (a) with $x_1 = 1$

(b) find the values of x_2 and x_3 (2)

(c) Explain why, for this question, the Newton-Raphson method cannot be used with $x_1 = 0$



Question 5 continued

Handwriting practice area with 24 horizontal lines.

(Total for Question 5 is 6 marks)



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Question 6 continued

Handwriting practice area with 20 horizontal lines.

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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7. (i) Solve, for $0 \leq x < \frac{\pi}{2}$, the equation

$$4 \sin x = \sec x$$

(4)

- (ii) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$5 \sin \theta - 5 \cos \theta = 2$$

giving your answers to one decimal place.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Handwriting practice area with 25 horizontal lines.

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 9 marks)



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Question 8 continued

Handwriting practice area with 20 horizontal lines.

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Question 8 continued

Lined area for writing the answer to Question 8.

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Question 8 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 8 is 7 marks)



9. Given that θ is measured in radians, prove, from first principles, that

$$\frac{d}{d\theta}(\cos \theta) = -\sin \theta$$

You may assume the formula for $\cos(A \pm B)$ and that as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$
(5)

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 5 marks)



10. A spherical mint of radius 5 mm is placed in the mouth and sucked. Four minutes later, the radius of the mint is 3 mm.

In a simple model, the rate of decrease of the radius of the mint is inversely proportional to the square of the radius.

Using this model and all the information given,

- (a) find an equation linking the radius of the mint and the time.
(You should define the variables that you use.)

(5)

- (b) Hence find the total time taken for the mint to completely dissolve. Give your answer in minutes and seconds to the nearest second.

(2)

- (c) Suggest a limitation of the model.

(1)

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Question 10 continued

Lined area for writing the answer to Question 10.

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Question 10 continued

Lined area for writing the answer to Question 10.

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



11.

$$\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv A + \frac{B}{(x-3)} + \frac{C}{(1-2x)}$$

- (a) Find the values of the constants A , B and C .

(4)

$$f(x) = \frac{1 + 11x - 6x^2}{(x - 3)(1 - 2x)} \quad x > 3$$

- (b) Prove that $f(x)$ is a decreasing function.

(3)

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Question 11 continued

Lined area for writing the answer to Question 11.

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Question 11 continued

Lined area for writing the answer to Question 11.

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 7 marks)



12. (a) Prove that

$$1 - \cos 2\theta \equiv \tan \theta \sin 2\theta, \quad \theta \neq \frac{(2n+1)\pi}{2}, \quad n \in \mathbb{Z} \quad (3)$$

(b) Hence solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, the equation

$$(\sec^2 x - 5)(1 - \cos 2x) = 3 \tan^2 x \sin 2x$$

Give any non-exact answer to 3 decimal places where appropriate.

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Question 12 continued

Lined area for writing the answer to Question 12.

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Question 12 continued

Lined area for writing the answer to Question 12.

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 9 marks)



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Question 13 continued

Lined area for writing the answer to Question 13.

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Question 13 continued

Lined area for writing the answer to Question 13.

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Question 13 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 13 is 10 marks)



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Question 14 continued

Lined area for writing the answer to Question 14.

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Question 14 continued

Handwriting practice area with 30 horizontal lines.

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(Total for Question 14 is 10 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Wednesday 5 June 2019

Morning (Time: 2 hours)

Paper Reference **9MA0/01**

Mathematics

Advanced

Paper 1: Pure Mathematics 1

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

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- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1.

$$f(x) = 3x^3 + 2ax^2 - 4x + 5a$$

Given that $(x + 3)$ is a factor of $f(x)$, find the value of the constant a .

(3)

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Question 1 continued

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(Total for Question 1 is 3 marks)



2.

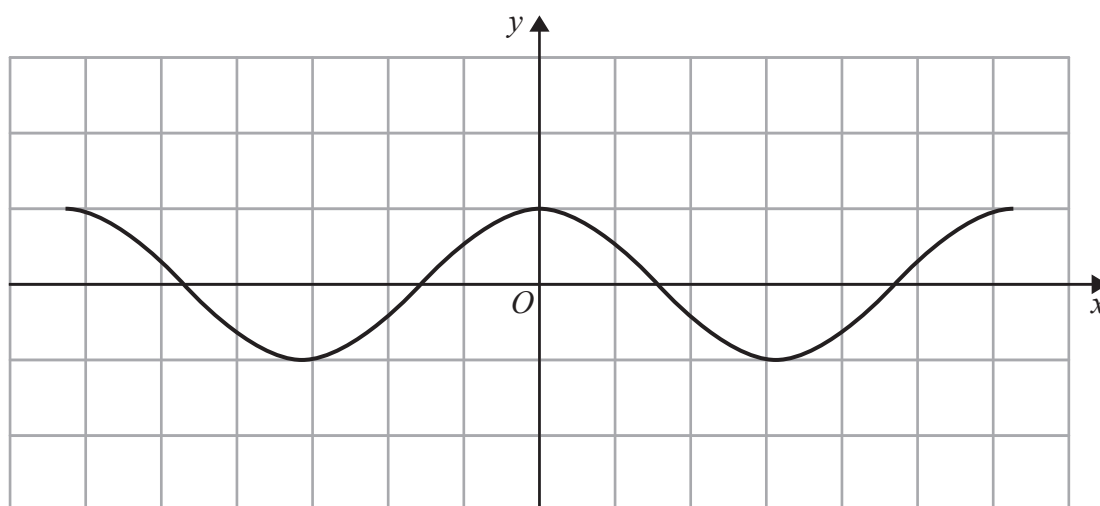


Figure 1

Figure 1 shows a plot of part of the curve with equation $y = \cos x$ where x is measured in radians. Diagram 1, on the opposite page, is a copy of Figure 1.

(a) Use Diagram 1 to show why the equation

$$\cos x - 2x - \frac{1}{2} = 0$$

has only one real root, giving a reason for your answer.

(2)

Given that the root of the equation is α , and that α is small,

(b) use the small angle approximation for $\cos x$ to estimate the value of α to 3 decimal places.

(3)

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Question 2 continued

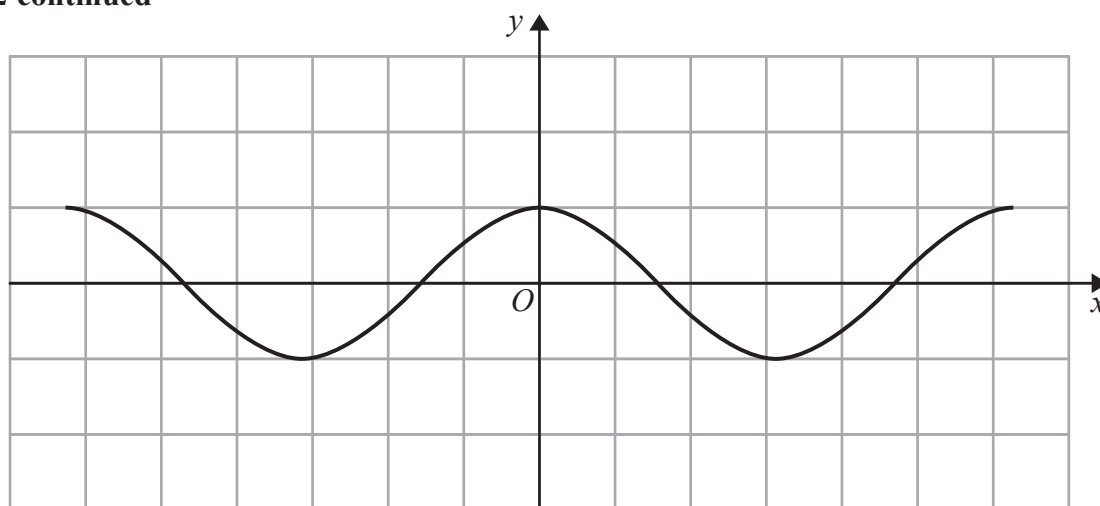


Diagram 1

(Total for Question 2 is 5 marks)



3.

$$y = \frac{5x^2 + 10x}{(x+1)^2} \quad x \neq -1$$

- (a) Show that $\frac{dy}{dx} = \frac{A}{(x+1)^n}$ where A and n are constants to be found. (4)

- (b) Hence deduce the range of values for x for which $\frac{dy}{dx} < 0$ (1)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4. (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\frac{1}{\sqrt{4-x}}$$

giving each coefficient in its simplest form.

(4)

The expansion can be used to find an approximation to $\sqrt{2}$
Possible values of x that could be substituted into this expansion are:

- $x = -14$ because $\frac{1}{\sqrt{4-x}} = \frac{1}{\sqrt{18}} = \frac{\sqrt{2}}{6}$
- $x = 2$ because $\frac{1}{\sqrt{4-x}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$
- $x = -\frac{1}{2}$ because $\frac{1}{\sqrt{4-x}} = \frac{1}{\sqrt{\frac{9}{2}}} = \frac{\sqrt{2}}{3}$

- (b) Without evaluating your expansion,

- (i) state, giving a reason, which of the three values of x should not be used

(1)

- (ii) state, giving a reason, which of the three values of x would lead to the most accurate approximation to $\sqrt{2}$

(1)



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5.

$$f(x) = 2x^2 + 4x + 9 \quad x \in \mathbb{R}$$

(a) Write $f(x)$ in the form $a(x + b)^2 + c$, where a , b and c are integers to be found. (3)

(b) Sketch the curve with equation $y = f(x)$ showing any points of intersection with the coordinate axes and the coordinates of any turning point. (3)

(c) (i) Describe fully the transformation that maps the curve with equation $y = f(x)$ onto the curve with equation $y = g(x)$ where

$$g(x) = 2(x - 2)^2 + 4x - 3 \quad x \in \mathbb{R}$$

(ii) Find the range of the function

$$h(x) = \frac{21}{2x^2 + 4x + 9} \quad x \in \mathbb{R} \quad (4)$$



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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 10 marks)



6. (a) Solve, for $-180^\circ \leq \theta \leq 180^\circ$, the equation

$$5 \sin 2\theta = 9 \tan \theta$$

giving your answers, where necessary, to one decimal place.

[Solutions based entirely on graphical or numerical methods are not acceptable.]

(6)

- (b) Deduce the smallest positive solution to the equation

$$5 \sin (2x - 50^\circ) = 9 \tan (x - 25^\circ)$$

(2)



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Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 8 marks)



7. In a simple model, the value, £ V , of a car depends on its age, t , in years.

The following information is available for car A

- its value when new is £20 000
- its value after one year is £16 000

(a) Use an exponential model to form, for car A , a possible equation linking V with t . (4)

The value of car A is monitored over a 10-year period.

Its value after 10 years is £2 000

(b) Evaluate the reliability of your model in light of this information. (2)

The following information is available for car B

- it has the same value, when new, as car A
- its value depreciates more slowly than that of car A

(c) Explain how you would adapt the equation found in (a) so that it could be used to model the value of car B . (1)



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Question 7 continued

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Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 7 marks)



8.

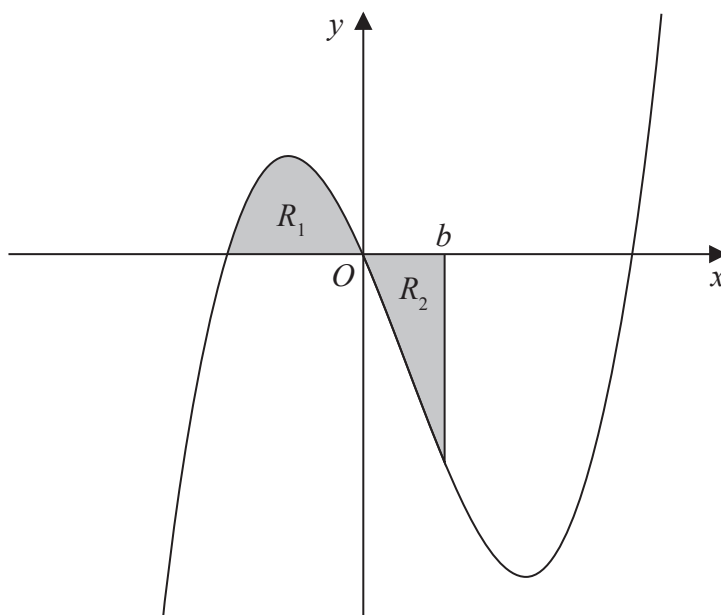


Figure 2

Figure 2 shows a sketch of part of the curve with equation $y = x(x + 2)(x - 4)$.

The region R_1 shown shaded in Figure 2 is bounded by the curve and the negative x -axis.

- (a) Show that the exact area of R_1 is $\frac{20}{3}$ (4)

The region R_2 also shown shaded in Figure 2 is bounded by the curve, the positive x -axis and the line with equation $x = b$, where b is a positive constant and $0 < b < 4$

Given that the area of R_1 is equal to the area of R_2

- (b) verify that b satisfies the equation

$$(b + 2)^2 (3b^2 - 20b + 20) = 0 \quad (4)$$

The roots of the equation $3b^2 - 20b + 20 = 0$ are 1.225 and 5.442 to 3 decimal places. The value of b is therefore 1.225 to 3 decimal places.

- (c) Explain, with the aid of a diagram, the significance of the root 5.442 (2)



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Question 8 continued

Handwriting practice area with horizontal lines.



Question 8 continued

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Question 8 continued

Handwriting practice lines for Question 8.

(Total for Question 8 is 10 marks)



9. Given that $a > b > 0$ and that a and b satisfy the equation

$$\log a - \log b = \log(a - b)$$

- (a) show that

$$a = \frac{b^2}{b - 1} \quad (3)$$

- (b) Write down the full restriction on the value of b , explaining the reason for this restriction. (2)



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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 5 marks)



10. (i) Prove that for all $n \in \mathbb{N}$, $n^2 + 2$ is not divisible by 4

(4)

(ii) "Given $x \in \mathbb{R}$, the value of $|3x - 28|$ is greater than or equal to the value of $(x - 9)$."

State, giving a reason, if the above statement is always true, sometimes true or never true.

(2)



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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 6 marks)



11. A competitor is running a 20 kilometre race.

She runs each of the first 4 kilometres at a steady pace of 6 minutes per kilometre.
After the first 4 kilometres, she begins to slow down.

In order to estimate her finishing time, the time that she will take to complete each subsequent kilometre is modelled to be 5% greater than the time that she took to complete the previous kilometre.

Using the model,

(a) show that her time to run the first 6 kilometres is estimated to be 36 minutes 55 seconds, (2)

(b) show that her estimated time, in minutes, to run the r th kilometre, for $5 \leq r \leq 20$, is

$$6 \times 1.05^{r-4} \quad (1)$$

(c) estimate the total time, in minutes and seconds, that she will take to complete the race. (4)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 7 marks)



12. $f(x) = 10e^{-0.25x} \sin x, \quad x \geq 0$

- (a) Show that the x coordinates of the turning points of the curve with equation $y = f(x)$ satisfy the equation $\tan x = 4$

(4)

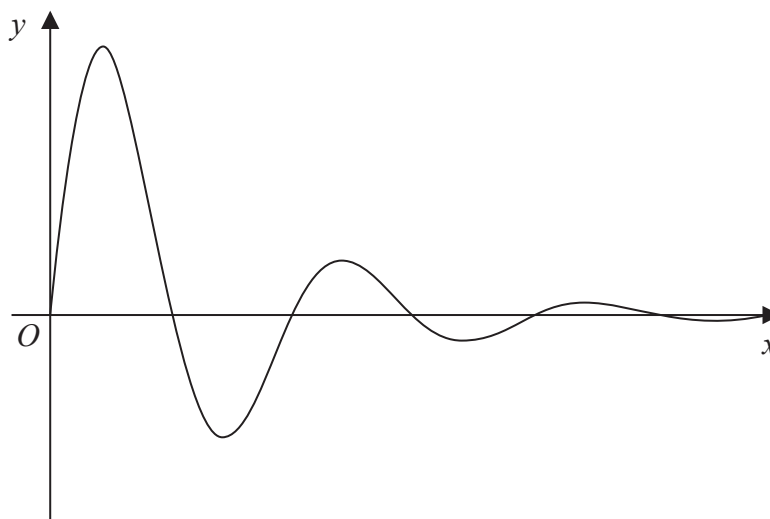


Figure 3

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$.

- (b) Sketch the graph of H against t where

$$H(t) = |10e^{-0.25t} \sin t| \quad t \geq 0$$

showing the long-term behaviour of this curve.

(2)

The function $H(t)$ is used to model the height, in metres, of a ball above the ground t seconds after it has been kicked.

Using this model, find

- (c) the maximum height of the ball above the ground between the first and second bounce.

(3)

- (d) Explain why this model should not be used to predict the time of each bounce.

(1)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 10 marks)



13. The curve C with equation

$$y = \frac{p - 3x}{(2x - q)(x + 3)} \quad x \in \mathbb{R}, x \neq -3, x \neq 2$$

where p and q are constants, passes through the point $\left(3, \frac{1}{2}\right)$ and has two vertical asymptotes with equations $x = 2$ and $x = -3$

(a) (i) Explain why you can deduce that $q = 4$

(ii) Show that $p = 15$

(3)

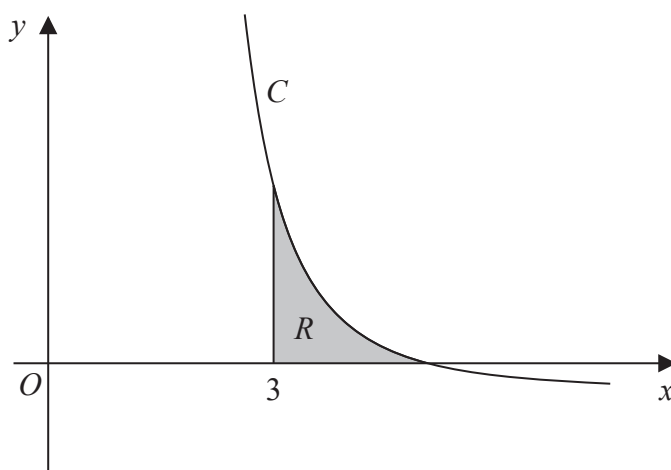


Figure 4

Figure 4 shows a sketch of part of the curve C . The region R , shown shaded in Figure 4, is bounded by the curve C , the x -axis and the line with equation $x = 3$

(b) Show that the exact value of the area of R is $a \ln 2 + b \ln 3$, where a and b are rational constants to be found.

(8)



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Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 11 marks)



14. The curve C , in the standard Cartesian plane, is defined by the equation

$$x = 4 \sin 2y \quad -\frac{\pi}{4} < y < \frac{\pi}{4}$$

The curve C passes through the origin O

(a) Find the value of $\frac{dy}{dx}$ at the origin. (2)

(b) (i) Use the small angle approximation for $\sin 2y$ to find an equation linking x and y for points close to the origin.

(ii) Explain the relationship between the answers to (a) and (b)(i). (2)

(c) Show that, for all points (x, y) lying on C ,

$$\frac{dy}{dx} = \frac{1}{a\sqrt{b-x^2}}$$

where a and b are constants to be found. (3)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 7 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

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Level 3 GCE**

Centre Number

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Candidate Number

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Wednesday 12 June 2019

Morning (Time: 2 hours)

Paper Reference **9MA0/02**

Mathematics

Advanced

Paper 2: Pure Mathematics 2

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Mathematical Formulae and Statistical Tables, calculator

Total Marks

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Answer ALL questions. Write your answers in the spaces provided.

1. Given

$$2^x \times 4^y = \frac{1}{2\sqrt{2}}$$

express y as a function of x .

(3)

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 3 marks)



2. The speed of a small jet aircraft was measured every 5 seconds, starting from the time it turned onto a runway, until the time when it left the ground.

The results are given in the table below with the time in seconds and the speed in ms^{-1} .

Time (s)	0	5	10	15	20	25
Speed (ms^{-1})	2	5	10	18	28	42

Using all of this information,

- (a) estimate the length of runway used by the jet to take off.

(3)

Given that the jet accelerated smoothly in these 25 seconds,

- (b) explain whether your answer to part (a) is an underestimate or an overestimate of the length of runway used by the jet to take off.

(1)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 4 marks)



3.

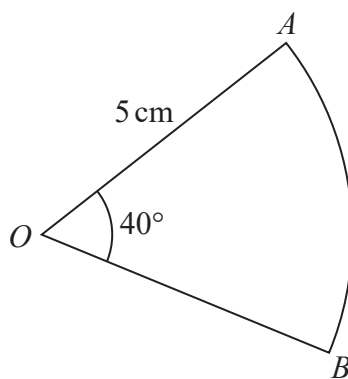


Figure 1

Figure 1 shows a sector AOB of a circle with centre O , radius 5 cm and angle $AOB = 40^\circ$

The attempt of a student to find the area of the sector is shown below.

$$\begin{aligned}\text{Area of sector} &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} \times 5^2 \times 40 \\ &= 500 \text{ cm}^2\end{aligned}$$

- (a) Explain the error made by this student.

(1)

- (b) Write out a correct solution.

(2)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 3 marks)



4.

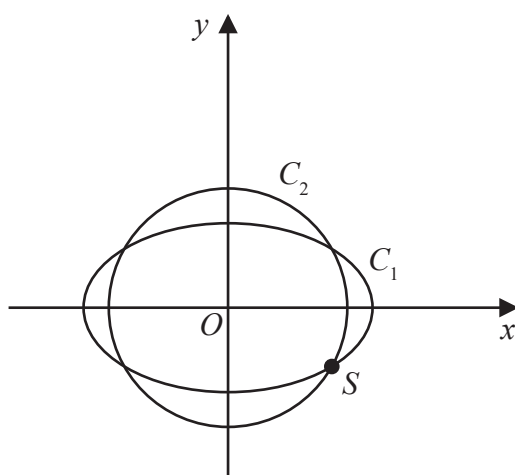


Figure 2

The curve C_1 with parametric equations

$$x = 10\cos t, \quad y = 4\sqrt{2}\sin t, \quad 0 \leq t < 2\pi$$

meets the circle C_2 with equation

$$x^2 + y^2 = 66$$

at four distinct points as shown in Figure 2.

Given that one of these points, S , lies in the 4th quadrant, find the Cartesian coordinates of S .
(6)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



5.

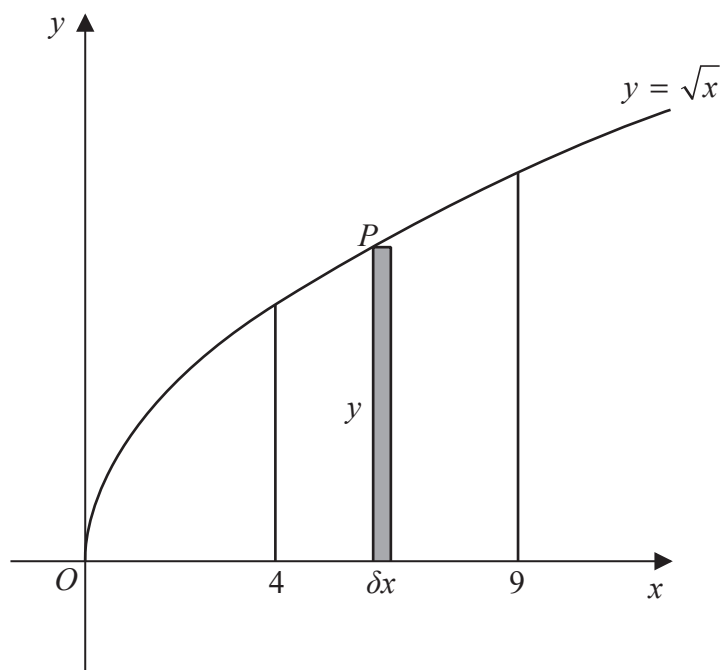
**Figure 3**

Figure 3 shows a sketch of the curve with equation $y = \sqrt{x}$

The point $P(x, y)$ lies on the curve.

The rectangle, shown shaded on Figure 3, has height y and width δx .

Calculate

$$\lim_{\delta x \rightarrow 0} \sum_{x=4}^9 \sqrt{x} \delta x$$

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 3 marks)



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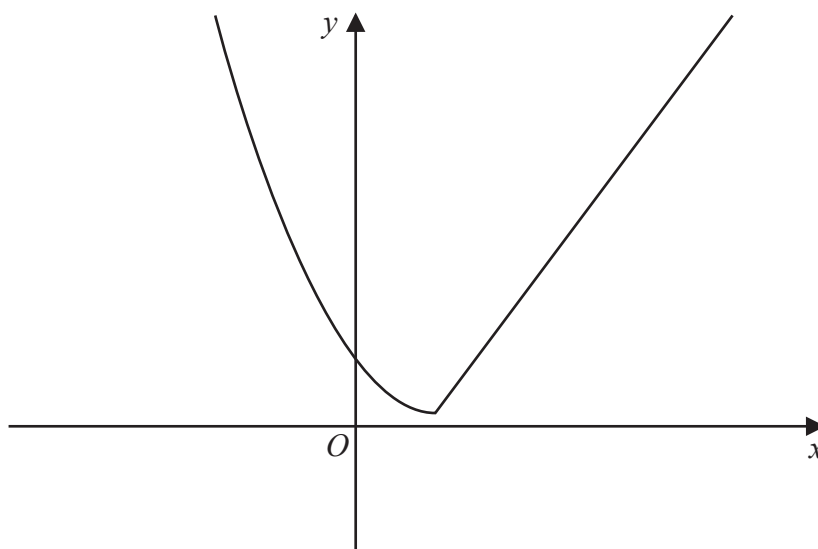


Figure 4

Figure 4 shows a sketch of the graph of $y = g(x)$, where

$$g(x) = \begin{cases} (x-2)^2 + 1 & x \leq 2 \\ 4x - 7 & x > 2 \end{cases}$$

(a) Find the value of $g(0)$.

(2)

(b) Find all values of x for which

$$g(x) > 28$$

(4)

The function h is defined by

$$h(x) = (x-2)^2 + 1 \quad x \leq 2$$

(c) Explain why h has an inverse but g does not.

(1)

(d) Solve the equation

$$h^{-1}(x) = -\frac{1}{2}$$

(3)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 10 marks)



7. A small factory makes bars of soap.

On any day, the total cost to the factory, £ y , of making x bars of soap is modelled to be the sum of two separate elements:

- a fixed cost
- a cost that is proportional to the number of bars of soap that are made that day

(a) Write down a general equation linking y with x , for this model.

(1)

The bars of soap are sold for £2 each.

On a day when 800 bars of soap are made and sold, the factory makes a profit of £500

On a day when 300 bars of soap are made and sold, the factory makes a loss of £80

Using the above information,

(b) show that $y = 0.84x + 428$

(3)

(c) With reference to the model, interpret the significance of the value 0.84 in the equation.

(1)

Assuming that each bar of soap is sold on the day it is made,

(d) find the least number of bars of soap that must be made on any given day for the factory to make a profit that day.

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 7 marks)



8. (i) Find the value of

$$\sum_{r=4}^{\infty} 20 \times \left(\frac{1}{2}\right)^r$$

(3)

(ii) Show that

$$\sum_{n=1}^{48} \log_5 \left(\frac{n+2}{n+1} \right) = 2$$

(3)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 6 marks)



9. A research engineer is testing the effectiveness of the braking system of a car when it is driven in wet conditions.

The engineer measures and records the braking distance, d metres, when the brakes are applied from a speed of $V \text{ km h}^{-1}$.

Graphs of d against V and $\log_{10} d$ against $\log_{10} V$ were plotted.

The results are shown below together with a data point from each graph.

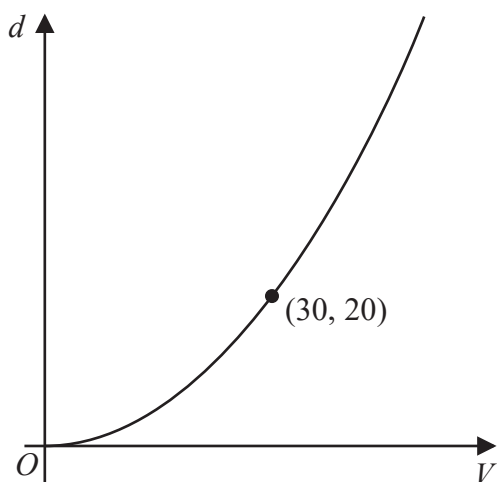


Figure 5

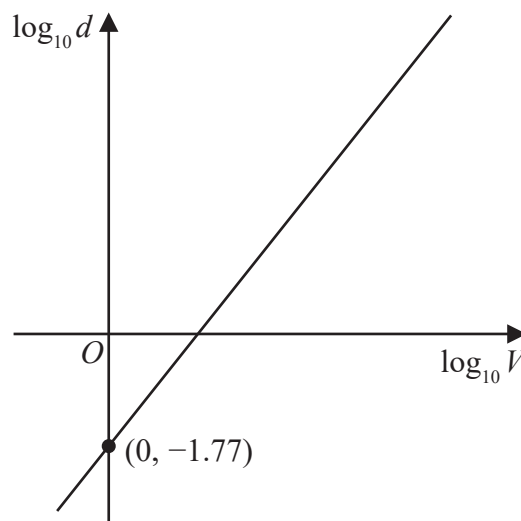


Figure 6

- (a) Explain how Figure 6 would lead the engineer to believe that the braking distance should be modelled by the formula

$$d = kV^n \quad \text{where } k \text{ and } n \text{ are constants}$$

with $k \approx 0.017$

(3)

Using the information given in Figure 5, with $k = 0.017$

- (b) find a complete equation for the model giving the value of n to 3 significant figures.

(3)

Sean is driving this car at 60 km h^{-1} in wet conditions when he notices a large puddle in the road 100 m ahead. It takes him 0.8 seconds to react before applying the brakes.

- (c) Use your formula to find out if Sean will be able to stop before reaching the puddle.

(3)

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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 9 marks)



10.

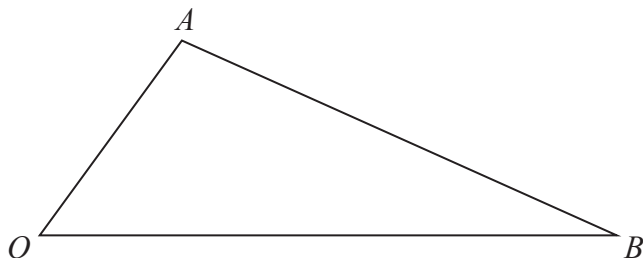


Figure 7

Figure 7 shows a sketch of triangle OAB .

The point C is such that $\vec{OC} = 2\vec{OA}$.

The point M is the midpoint of AB .

The straight line through C and M cuts OB at the point N .

Given $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$

(a) Find $\vec{CM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ (2)

(b) Show that $\vec{ON} = \left(2 - \frac{3}{2}\lambda\right)\mathbf{a} + \frac{1}{2}\lambda\mathbf{b}$, where λ is a scalar constant. (2)

(c) Hence prove that $ON:NB = 2:1$ (2)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 6 marks)



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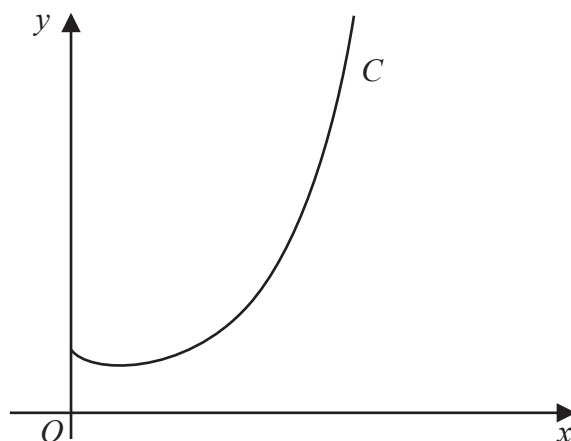
**Figure 8**

Figure 8 shows a sketch of the curve C with equation $y = x^x$, $x > 0$

(a) Find, by firstly taking logarithms, the x coordinate of the turning point of C .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(5)

The point $P(\alpha, 2)$ lies on C .

(b) Show that $1.5 < \alpha < 1.6$

(2)

A possible iteration formula that could be used in an attempt to find α is

$$x_{n+1} = 2x_n^{1-x_n}$$

Using this formula with $x_1 = 1.5$

(c) find x_4 to 3 decimal places,

(2)

(d) describe the long-term behaviour of x_n

(2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 11 marks)



12. (a) Prove

$$\frac{\cos 3\theta}{\sin \theta} + \frac{\sin 3\theta}{\cos \theta} \equiv 2 \cot 2\theta \quad \theta \neq (90n)^\circ, n \in \mathbb{Z} \quad (4)$$

(b) Hence solve, for $90^\circ < \theta < 180^\circ$, the equation

$$\frac{\cos 3\theta}{\sin \theta} + \frac{\sin 3\theta}{\cos \theta} = 4$$

giving any solutions to one decimal place.

(3)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 7 marks)



13.

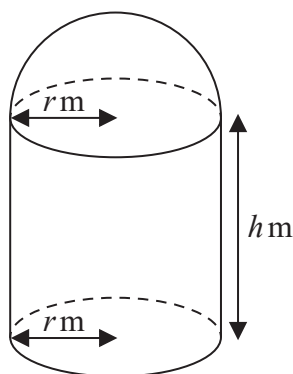


Figure 9

[A sphere of radius r has volume $\frac{4}{3}\pi r^3$ and surface area $4\pi r^2$]

A manufacturer produces a storage tank.

The tank is modelled in the shape of a hollow circular cylinder closed at one end with a hemispherical shell at the other end as shown in Figure 9.

The walls of the tank are assumed to have negligible thickness.

The cylinder has radius r metres and height h metres and the hemisphere has radius r metres.

The volume of the tank is 6 m^3 .

(a) Show that, according to the model, the surface area of the tank, in m^2 , is given by

$$\frac{12}{r} + \frac{5}{3}\pi r^2 \quad (4)$$

The manufacturer needs to minimise the surface area of the tank.

(b) Use calculus to find the radius of the tank for which the surface area is a minimum. (4)

(c) Calculate the minimum surface area of the tank, giving your answer to the nearest integer. (2)



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Question 13 continued

Handwriting practice area with horizontal lines.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 10 marks)



14. (a) Use the substitution $u = 4 - \sqrt{h}$ to show that

$$\int \frac{dh}{4 - \sqrt{h}} = -8 \ln|4 - \sqrt{h}| - 2\sqrt{h} + k$$

where k is a constant

(6)

A team of scientists is studying a species of slow growing tree.

The rate of change in height of a tree in this species is modelled by the differential equation

$$\frac{dh}{dt} = \frac{t^{0.25}(4 - \sqrt{h})}{20}$$

where h is the height in metres and t is the time, measured in years, after the tree is planted.

- (b) Find, according to the model, the range in heights of trees in this species.

(2)

One of these trees is one metre high when it is first planted.

According to the model,

- (c) calculate the time this tree would take to reach a height of 12 metres, giving your answer to 3 significant figures.

(7)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 15 marks)

TOTAL FOR PAPER IS 100 MARKS



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Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Wednesday 7 October 2020

Morning (Time: 2 hours)

Paper Reference **9MA0/01**

Mathematics

Advanced

Paper 1: Pure Mathematics 1

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

**Candidates may use any calculator allowed by Pearson regulations.
Calculators must not have the facility for symbolic algebra manipulation,
differentiation and integration, or have retrievable mathematical formulae
stored in them.**

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. (a) Find the first four terms, in ascending powers of x , of the binomial expansion of

$$(1 + 8x)^{\frac{1}{2}}$$

giving each term in simplest form.

(3)

- (b) Explain how you could use $x = \frac{1}{32}$ in the expansion to find an approximation for $\sqrt{5}$

There is no need to carry out the calculation.

(2)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 5 marks)



2. By taking logarithms of both sides, solve the equation

$$4^{3p-1} = 5^{210}$$

giving the value of p to one decimal place.

(3)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 3 marks)



3. Relative to a fixed origin O

- point A has position vector $2\mathbf{i} + 5\mathbf{j} - 6\mathbf{k}$
- point B has position vector $3\mathbf{i} - 3\mathbf{j} - 4\mathbf{k}$
- point C has position vector $2\mathbf{i} - 16\mathbf{j} + 4\mathbf{k}$

- (a) Find $\overrightarrow{AB}$ (2)
- (b) Show that quadrilateral $OABC$ is a trapezium, giving reasons for your answer. (2)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 4 marks)



4. The function f is defined by

$$f(x) = \frac{3x - 7}{x - 2} \quad x \in \mathbb{R}, x \neq 2$$

(a) Find $f^{-1}(7)$

(2)

(b) Show that $ff(x) = \frac{ax + b}{x - 3}$ where a and b are integers to be found.

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



5. A car has six forward gears.

The fastest speed of the car

- in 1st gear is 28 km h^{-1}
- in 6th gear is 115 km h^{-1}

Given that the fastest speed of the car in successive gears is modelled by an **arithmetic sequence**,

(a) find the fastest speed of the car in 3rd gear.

(3)

Given that the fastest speed of the car in successive gears is modelled by a **geometric sequence**,

(b) find the fastest speed of the car in 5th gear.

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



6. (a) Express $\sin x + 2 \cos x$ in the form $R \sin(x + \alpha)$ where R and α are constants, $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R and give the value of α in radians to 3 decimal places.

(3)

The temperature, $\theta^{\circ}\text{C}$, inside a room on a given day is modelled by the equation

$$\theta = 5 + \sin\left(\frac{\pi t}{12} - 3\right) + 2\cos\left(\frac{\pi t}{12} - 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after midnight.

Using the equation of the model and your answer to part (a),

- (b) deduce the maximum temperature of the room during this day, (1)

- (c) find the time of day when the maximum temperature occurs, giving your answer to the nearest minute.

(3)

[illegible]

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 7 marks)



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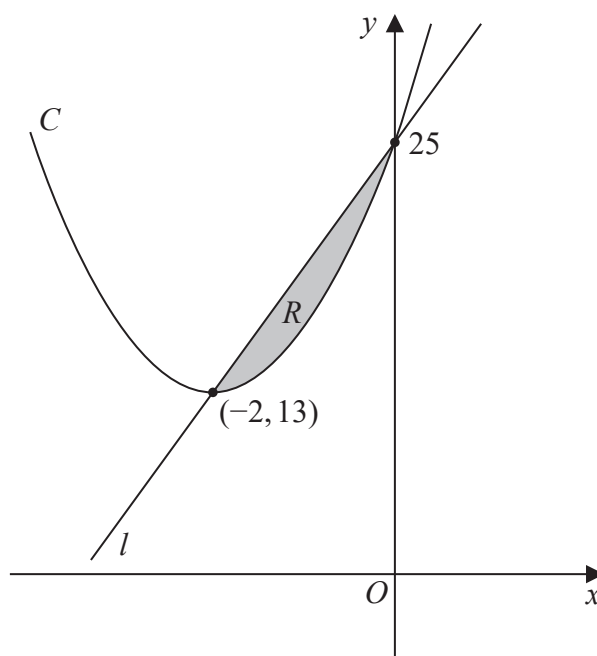
**Figure 1**

Figure 1 shows a sketch of a curve C with equation $y = f(x)$ and a straight line l .

The curve C meets l at the points $(-2, 13)$ and $(0, 25)$ as shown.

The shaded region R is bounded by C and l as shown in Figure 1.

Given that

- $f(x)$ is a quadratic function in x
- $(-2, 13)$ is the minimum turning point of $y = f(x)$

use inequalities to define R .

(5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 5 marks)



8. A new smartphone was released by a company.

The company monitored the total number of phones sold, n , at time t days after the phone was released.

The company observed that, during this time,

the rate of increase of n was proportional to n

Use this information to write down a suitable equation for n in terms of t .

(You do not need to evaluate any unknown constants in your equation.)

(2)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 2 marks)



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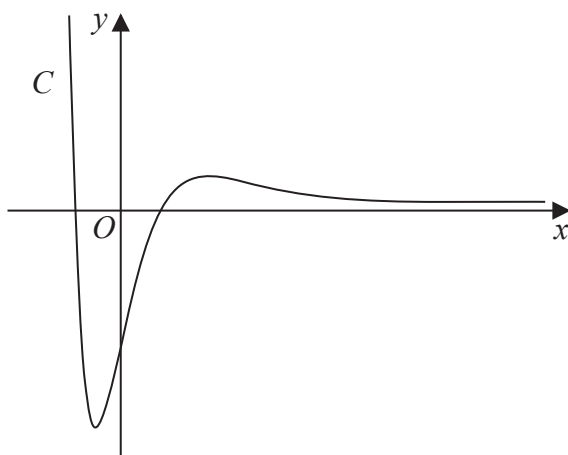


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = f(x)$ where

$$f(x) = 4(x^2 - 2)e^{-2x} \quad x \in \mathbb{R}$$

(a) Show that $f'(x) = 8(2 + x - x^2)e^{-2x}$ (3)

(b) Hence find, in simplest form, the exact coordinates of the stationary points of C . (3)

The function g and the function h are defined by

$$g(x) = 2f(x) \quad x \in \mathbb{R}$$

$$h(x) = 2f(x) - 3 \quad x \geq 0$$

(c) Find
 (i) the range of g
 (ii) the range of h (3)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 9 marks)



10. (a) Use the substitution $x = u^2 + 1$ to show that

$$\int_5^{10} \frac{3 \, dx}{(x-1)(3+2\sqrt{x-1})} = \int_p^q \frac{6 \, du}{u(3+2u)}$$

where p and q are positive constants to be found.

(4)

(b) Hence, using algebraic integration, show that

$$\int_5^{10} \frac{3 \, dx}{(x-1)(3+2\sqrt{x-1})} = \ln a$$

where a is a rational constant to be found.

(6)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 10 marks)



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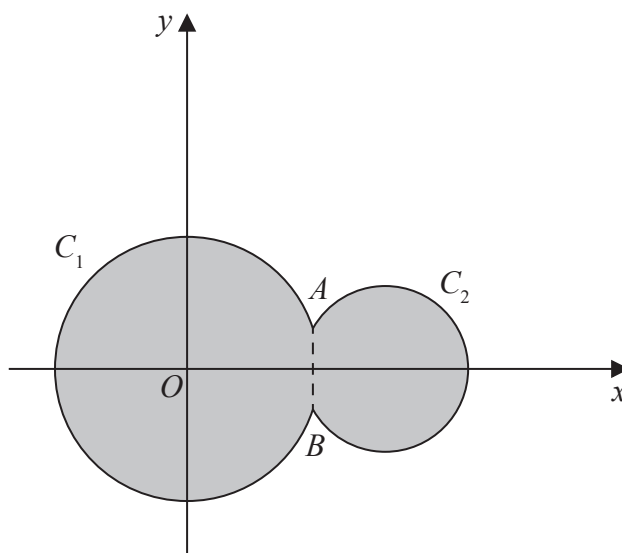


Figure 3

Circle C_1 has equation $x^2 + y^2 = 100$

Circle C_2 has equation $(x - 15)^2 + y^2 = 40$

The circles meet at points A and B as shown in Figure 3.

(a) Show that angle $AOB = 0.635$ radians to 3 significant figures, where O is the origin.

(4)

The region shown shaded in Figure 3 is bounded by C_1 and C_2

(b) Find the perimeter of the shaded region, giving your answer to one decimal place.

(4)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing answers.

(Total for Question 11 is 8 marks)



12.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that

$$\operatorname{cosec} \theta - \sin \theta \equiv \cos \theta \cot \theta \quad \theta \neq (180n)^\circ \quad n \in \mathbb{Z} \quad (3)$$

(b) Hence, or otherwise, solve for $0 < x < 180^\circ$

$$\operatorname{cosec} x - \sin x = \cos x \cot(3x - 50^\circ) \quad (5)$$



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 8 marks)



13. A sequence of numbers $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = \frac{k(a_n + 2)}{a_n} \quad n \in \mathbb{N}$$

where k is a constant.

Given that

- the sequence is a periodic sequence of order 3
- $a_1 = 2$

(a) show that

$$k^2 + k - 2 = 0 \quad (3)$$

(b) For this sequence explain why $k \neq 1$ (1)

(c) Find the value of

$$\sum_{r=1}^{80} a_r \quad (3)$$



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



14. A large spherical balloon is deflating.

At time t seconds the balloon has radius r cm and volume V cm³

The volume of the balloon is modelled as decreasing at a constant rate.

(a) Using this model, show that

$$\frac{dr}{dt} = -\frac{k}{r^2}$$

where k is a positive constant.

(3)

Given that

- the initial radius of the balloon is 40 cm
- after 5 seconds the radius of the balloon is 20 cm
- the volume of the balloon continues to decrease at a constant rate until the balloon is empty

(b) solve the differential equation to find a complete equation linking r and t .

(5)

(c) Find the limitation on the values of t for which the equation in part (b) is valid.

(2)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 10 marks)



15. The curve C has equation

$$x^2 \tan y = 9 \qquad 0 < y < \frac{\pi}{2}$$

(a) Show that

$$\frac{dy}{dx} = \frac{-18x}{x^4 + 81} \quad (4)$$

(b) Prove that C has a point of inflection at $x = \sqrt[4]{27}$ (3)

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.

(Total for Question 15 is 7 marks)



16. Prove by contradiction that there are no positive integers p and q such that

$$4p^2 - q^2 = 25$$

(4)

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Question 16 continued

Lined area for writing the answer to Question 16.



Question 16 continued

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(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Wednesday 14 October 2020			
Afternoon (Time: 2 hours)		Paper Reference 9MA0/02	
Mathematics Advanced Paper 2: Pure Mathematics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

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- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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- The values of y are given to 4 significant figures.

x	0.5	1	1.5	2	2.5
y	0.5774	0.7071	0.7746	0.8165	0.8452

- (a) Use the trapezium rule, with all the values of y in the table, to find an estimate for

$$\int_{0.5}^{2.5} \sqrt{\frac{x}{1+x}} \, dx$$

giving your answer to 3 significant figures.

(3)

- (b) Using your answer to part (a), deduce an estimate for $\int_{0.5}^{2.5} \sqrt{\frac{9x}{1+x}} \, dx$

(1)

Given that

$$\int_{0.5}^{2.5} \sqrt{\frac{9x}{1+x}} \, dx = 4.535 \text{ to 4 significant figures}$$

- (c) comment on the accuracy of your answer to part (b).

(1)

Question 1 continued

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(Total for Question 1 is 5 marks)



2. Relative to a fixed origin, points P , Q and R have position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively.

Given that

- P , Q and R lie on a straight line
- Q lies one third of the way from P to R

show that

$$\mathbf{q} = \frac{1}{3}(\mathbf{r} + 2\mathbf{p}) \quad (3)$$



Question 2 continued

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(Total for Question 2 is 3 marks)



3. (a) Given that

$$2\log(4-x) = \log(x+8)$$

show that

$$x^2 - 9x + 8 = 0$$

(3)

(b) (i) Write down the roots of the equation

$$x^2 - 9x + 8 = 0$$

(ii) State which of the roots in (b)(i) is **not** a solution of

$$2\log(4-x) = \log(x+8)$$

giving a reason for your answer.

(2)



Question 3 continued

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(Total for Question 3 is 5 marks)



4. In the binomial expansion of

$$(a + 2x)^7 \quad \text{where } a \text{ is a constant}$$

the coefficient of x^4 is 15 120

Find the value of a .

(3)



Question 4 continued

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(Total for Question 4 is 3 marks)



5. The curve with equation $y = 3 \times 2^x$ meets the curve with equation $y = 15 - 2^{x+1}$ at the point P .
Find, using algebra, the exact x coordinate of P .

(4)

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 4 marks)



6. (a) Given that

$$\frac{x^2 + 8x - 3}{x + 2} \equiv Ax + B + \frac{C}{x + 2} \quad x \in \mathbb{R} \quad x \neq -2$$

find the values of the constants A , B and C

(3)

(b) Hence, using algebraic integration, find the exact value of

$$\int_0^6 \frac{x^2 + 8x - 3}{x + 2} dx$$

giving your answer in the form $a + b \ln 2$ where a and b are integers to be found.

(4)



Question 6 continued

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Question 6 continued

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Question 6 continued

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(Total for Question 6 is 7 marks)



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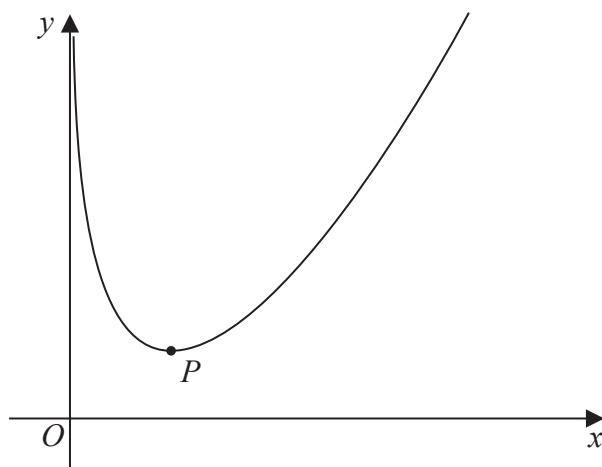


Figure 1

Figure 1 shows a sketch of the curve C with equation

$$y = \frac{4x^2 + x}{2\sqrt{x}} - 4\ln x \quad x > 0$$

(a) Show that

$$\frac{dy}{dx} = \frac{12x^2 + x - 16\sqrt{x}}{4x\sqrt{x}} \quad (4)$$

The point P , shown in Figure 1, is the minimum turning point on C .

(b) Show that the x coordinate of P is a solution of

$$x = \left(\frac{4}{3} - \frac{\sqrt{x}}{12} \right)^{\frac{2}{3}} \quad (3)$$

(c) Use the iteration formula

$$x_{n+1} = \left(\frac{4}{3} - \frac{\sqrt{x_n}}{12} \right)^{\frac{2}{3}} \quad \text{with } x_1 = 2$$

to find (i) the value of x_2 to 5 decimal places,

(ii) the x coordinate of P to 5 decimal places.

(3)



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Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8. A curve C has equation $y = f(x)$

Given that

- $f'(x) = 6x^2 + ax - 23$ where a is a constant
- the y intercept of C is -12
- $(x + 4)$ is a factor of $f(x)$

find, in simplest form, $f(x)$

(6)



Question 8 continued

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Question 8 continued

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Question 8 continued

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(Total for Question 8 is 6 marks)





Question 9 continued

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(Total for Question 9 is 6 marks)



10.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\cos 3A \equiv 4\cos^3 A - 3\cos A \quad (4)$$

(b) Hence solve, for $-90^\circ \leq x \leq 180^\circ$, the equation

$$1 - \cos 3x = \sin^2 x \quad (4)$$



Question 10 continued

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Question 10 continued

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(Total for Question 10 is 8 marks)



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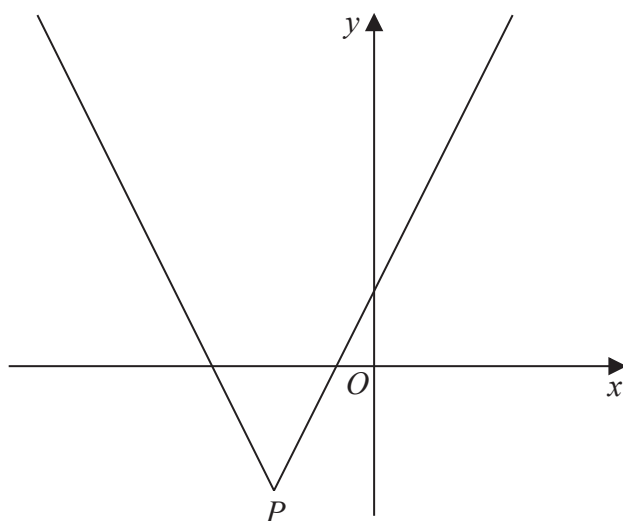


Figure 2

Figure 2 shows a sketch of the graph with equation

$$y = 2|x + 4| - 5$$

The vertex of the graph is at the point P , shown in Figure 2.

- (a) Find the coordinates of P .

(2)

- (b) Solve the equation

$$3x + 40 = 2|x + 4| - 5$$

(2)

A line l has equation $y = ax$, where a is a constant.

Given that l intersects $y = 2|x + 4| - 5$ at least once,

- (c) find the range of possible values of a , writing your answer in set notation.

(3)



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

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Question 11 continued

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(Total for Question 11 is 7 marks)



12.

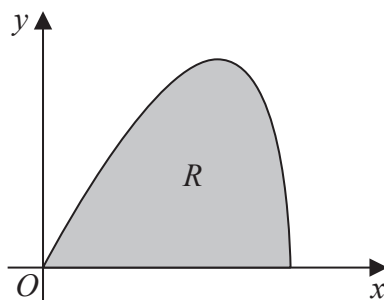


Figure 3

The curve shown in Figure 3 has parametric equations

$$x = 6 \sin t \quad y = 5 \sin 2t \quad 0 \leq t \leq \frac{\pi}{2}$$

The region R , shown shaded in Figure 3, is bounded by the curve and the x -axis.

(a) (i) Show that the area of R is given by $\int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt$

(3)

(ii) Hence show, by algebraic integration, that the area of R is exactly 20

(3)

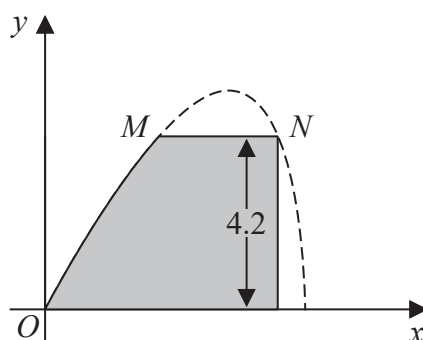


Figure 4

Part of the curve is used to model the profile of a small dam, shown shaded in Figure 4. Using the model and given that

- x and y are in metres
- the vertical wall of the dam is 4.2 metres high
- there is a horizontal walkway of width MN along the top of the dam

(b) calculate the width of the walkway.

(5)



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

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Question 12 continued

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(Total for Question 12 is 11 marks)



13. The function g is defined by

$$g(x) = \frac{3\ln(x) - 7}{\ln(x) - 2} \quad x > 0 \quad x \neq k$$

where k is a constant.

(a) Deduce the value of k .

(1)

(b) Prove that

$$g'(x) > 0$$

for all values of x in the domain of g .

(3)

(c) Find the range of values of a for which

$$g(a) > 0$$

(2)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

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(Total for Question 13 is 6 marks)



14. A circle C with radius r

- lies only in the 1st quadrant
- touches the x -axis and touches the y -axis

The line l has equation $2x + y = 12$

(a) Show that the x coordinates of the points of intersection of l with C satisfy

$$5x^2 + (2r - 48)x + (r^2 - 24r + 144) = 0 \quad (3)$$

Given also that l is a tangent to C ,

(b) find the two possible values of r , giving your answers as fully simplified surds. (4)



Question 14 continued

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Question 14 continued

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Question 14 continued

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(Total for Question 14 is 7 marks)



Solutions relying entirely on calculator technology are not acceptable.

(a) prove that

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad (4)$$

Given also that S_{10} is four times S_5

(b) find the exact value of r . (4)

Question 15 continued

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Question 15 continued

Handwriting practice area with 30 horizontal lines.

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Question 15 continued

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(Total for Question 15 is 8 marks)



16. Use algebra to prove that the square of any natural number is **either** a multiple of 3 **or** one more than a multiple of 3

(4)

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Question 16 continued

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Question 16 continued

Lined area for writing answers.

(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS

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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Wednesday 6 October 2021 – Afternoon

Time 2 hours

Paper
reference

9MA0/01

Mathematics

Advanced

PAPER 1: Pure Mathematics 1

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

$$f(x) = ax^3 + 10x^2 - 3ax - 4$$

Given that $(x - 1)$ is a factor of $f(x)$, find the value of the constant a .

You must make your method clear.

(3)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 3 marks)



2. Given that

$$f(x) = x^2 - 4x + 5 \quad x \in \mathbb{R}$$

(a) express $f(x)$ in the form $(x + a)^2 + b$ where a and b are integers to be found.

(2)

The curve with equation $y = f(x)$

- meets the y -axis at the point P
- has a minimum turning point at the point Q

(b) Write down

(i) the coordinates of P

(ii) the coordinates of Q

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3. The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = k - \frac{24}{u_n} \quad u_1 = 2$$

where k is an integer.

Given that $u_1 + 2u_2 + u_3 = 0$

- (a) show that

$$3k^2 - 58k + 240 = 0 \quad (3)$$

- (b) Find the value of k , giving a reason for your answer. (2)

- (c) Find the value of u_3 (1)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)

4. The curve with equation $y = f(x)$ where

$$f(x) = x^2 + \ln(2x^2 - 4x + 5)$$

has a single turning point at $x = \alpha$

- (a) Show that α is a solution of the equation

$$2x^3 - 4x^2 + 7x - 2 = 0 \quad (4)$$

The iterative formula

$$x_{n+1} = \frac{1}{7}(2 + 4x_n^2 - 2x_n^3)$$

is used to find an approximate value for α .

Starting with $x_1 = 0.3$

- (b) calculate, giving each answer to 4 decimal places,

(i) the value of x_2

(ii) the value of x_4

(3)

Using a suitable interval and a suitable function that should be stated,

- (c) show that α is 0.341 to 3 decimal places.

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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5.

In this question you should show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A company made a profit of £20 000 in its first year of trading, Year 1

A model for future trading predicts that the yearly profit will increase by 8% each year, so that the yearly profits will form a geometric sequence.

According to the model,

- (a) show that the profit for Year 3 will be £23 328 (1)
- (b) find the first year when the yearly profit will exceed £65 000 (3)
- (c) find the total profit for the first 20 years of trading, giving your answer to the nearest £1000 (2)



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(Total for Question 5 is 6 marks)



6.

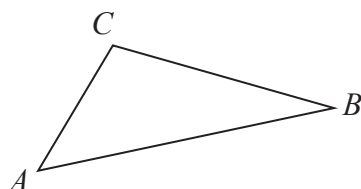


Figure 1

Figure 1 shows a sketch of triangle ABC .

Given that

- $\vec{AB} = -3\mathbf{i} - 4\mathbf{j} - 5\mathbf{k}$
- $\vec{BC} = \mathbf{i} + \mathbf{j} + 4\mathbf{k}$

(a) find $\vec{AC}$

(2)

(b) show that $\cos ABC = \frac{9}{10}$

(3)



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Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 5 marks)



7. The circle C has equation

$$x^2 + y^2 - 10x + 4y + 11 = 0$$

(a) Find

- the coordinates of the centre of C ,
- the exact radius of C , giving your answer as a simplified surd.

(4)

The line l has equation $y = 3x + k$ where k is a constant.

Given that l is a tangent to C ,

- (b) find the possible values of k , giving your answers as simplified surds.

(5)

[illegible]

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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8. A scientist is studying the growth of two different populations of bacteria.

The number of bacteria, N , in the **first** population is modelled by the equation

$$N = Ae^{kt} \quad t \geq 0$$

where A and k are positive constants and t is the time in hours from the start of the study.

Given that

- there were 1000 bacteria in this population at the start of the study
- it took exactly 5 hours from the start of the study for this population to double

(a) find a complete equation for the model.

(4)

(b) Hence find the rate of increase in the number of bacteria in this population exactly 8 hours from the start of the study. Give your answer to 2 significant figures.

(2)

The number of bacteria, M , in the **second** population is modelled by the equation

$$M = 500e^{1.4kt} \quad t \geq 0$$

where k has the value found in part (a) and t is the time in hours from the start of the study.

Given that T hours after the start of the study, the number of bacteria in the two different populations was the same,

(c) find the value of T .

(3)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 9 marks)



9.

$$f(x) = \frac{50x^2 + 38x + 9}{(5x + 2)^2(1 - 2x)} \quad x \neq -\frac{2}{5} \quad x \neq \frac{1}{2}$$

Given that $f(x)$ can be expressed in the form

$$\frac{A}{5x + 2} + \frac{B}{(5x + 2)^2} + \frac{C}{1 - 2x}$$

where A , B and C are constants

(a) (i) find the value of B and the value of C

(ii) show that $A = 0$

(4)

(b) (i) Use binomial expansions to show that, in ascending powers of x

$$f(x) = p + qx + rx^2 + \dots$$

where p , q and r are simplified fractions to be found.

(ii) Find the range of values of x for which this expansion is valid.

(7)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 11 marks)



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Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



11.

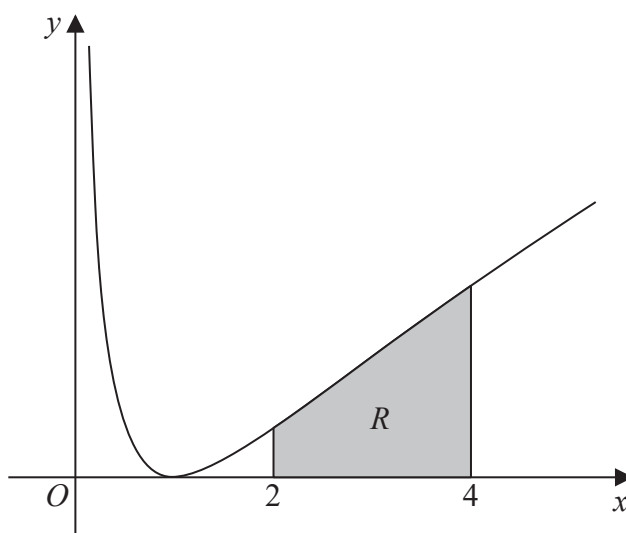


Figure 2

Figure 2 shows a sketch of part of the curve with equation

$$y = (\ln x)^2 \quad x > 0$$

The finite region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 4$

The table below shows corresponding values of x and y , with the values of y given to 4 decimal places.

x	2	2.5	3	3.5	4
y	0.4805	0.8396	1.2069	1.5694	1.9218

- (a) Use the trapezium rule, with all the values of y in the table, to obtain an estimate for the area of R , giving your answer to 3 significant figures.

(3)

- (b) Use algebraic integration to find the exact area of R , giving your answer in the form

$$y = a(\ln 2)^2 + b \ln 2 + c$$

where a , b and c are integers to be found.

(5)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

(Total for Question 11 is 8 marks)



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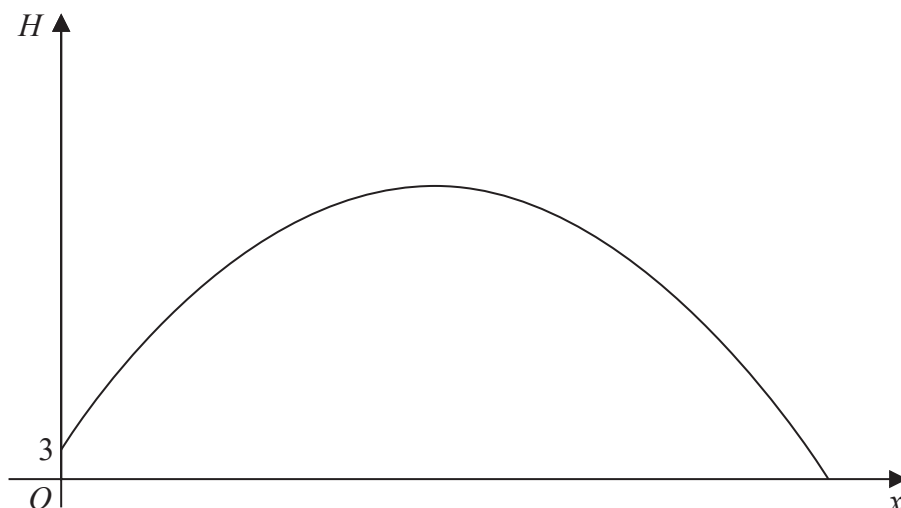


Figure 3

Figure 3 is a graph of the trajectory of a golf ball after the ball has been hit until it first hits the ground.

The vertical height, H metres, of the ball above the ground has been plotted against the horizontal distance travelled, x metres, measured from where the ball was hit.

The ball is modelled as a particle travelling in a vertical plane above horizontal ground.

Given that the ball

- is hit from a point on the top of a platform of vertical height 3 m above the ground
- reaches its maximum vertical height after travelling a horizontal distance of 90 m
- is at a vertical height of 27 m above the ground after travelling a horizontal distance of 120 m

Given also that H is modelled as a **quadratic** function in x

- (a) find H in terms of x (5)
- (b) Hence find, according to the model,
- the maximum vertical height of the ball above the ground,
 - the horizontal distance travelled by the ball, from when it was hit to when it first hits the ground, giving your answer to the nearest metre. (3)
- (c) The possible effects of wind or air resistance are two limitations of the model. Give one other limitation of this model. (1)

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Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 9 marks)



13. A curve C has parametric equations

$$x = \frac{t^2 + 5}{t^2 + 1} \quad y = \frac{4t}{t^2 + 1} \quad t \in \mathbb{R}$$

Show that all points on C satisfy

$$(x - 3)^2 + y^2 = 4 \quad (3)$$



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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 3 marks)



14. Given that

$$y = \frac{x-4}{2+\sqrt{x}} \quad x > 0$$

show that

$$\frac{dy}{dx} = \frac{1}{A\sqrt{x}} \quad x > 0$$

where A is a constant to be found.

(4)



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Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 4 marks)



15. (i) Use proof by exhaustion to show that for $n \in \mathbb{N}$, $n \leq 4$

$$(n+1)^3 > 3^n \quad (2)$$

(ii) Given that $m^3 + 5$ is odd, use proof by contradiction to show, using algebra, that m is even.

(4)



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Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 6 marks)

TOTAL FOR PAPER IS 100 MARKS



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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Wednesday 13 October 2021 – Afternoon

Time 2 hours

Paper
reference

9MA0/02

Mathematics

Advanced

PAPER 2: Pure Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

1. In an arithmetic series

- the first term is 16
- the 21st term is 24

(a) Find the common difference of the series.

(2)

(b) Hence find the sum of the first 500 terms of the series.

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. The functions f and g are defined by

$$f(x) = 7 - 2x^2 \quad x \in \mathbb{R}$$

$$g(x) = \frac{3x}{5x-1} \quad x \in \mathbb{R} \quad x \neq \frac{1}{5}$$

(a) State the range of f

(1)

(b) Find $gf(1.8)$

(2)

(c) Find $g^{-1}(x)$

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)

3. Using the laws of logarithms, solve the equation

$$\log_3 (12y + 5) - \log_3 (1 - 3y) = 2$$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 3 marks)

4. Given that θ is small and measured in radians, use the small angle approximations to show that

$$4 \sin \frac{\theta}{2} + 3 \cos^2 \theta \approx a + b\theta + c\theta^2$$

where a , b and c are integers to be found.

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 3 marks)



5. The curve C has equation

$$y = 5x^4 - 24x^3 + 42x^2 - 32x + 11 \quad x \in \mathbb{R}$$

(a) Find

(i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$

(3)

(b) (i) Verify that C has a stationary point at $x = 1$

(ii) Show that this stationary point is a point of inflection, giving reasons for your answer.

(4)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



6.

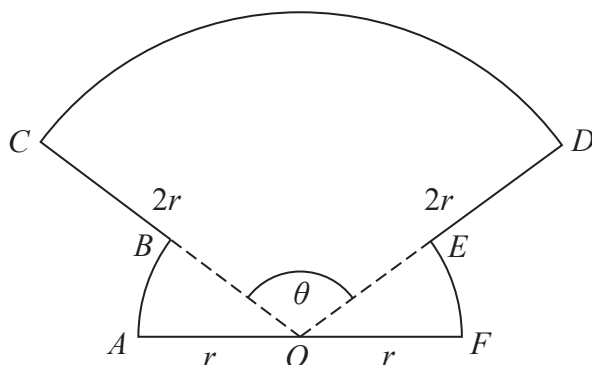


Figure 1

The shape $OABCDEFO$ shown in Figure 1 is a design for a logo.

In the design

- OAB is a sector of a circle centre O and radius r
- sector OFE is congruent to sector OAB
- ODC is a sector of a circle centre O and radius $2r$
- AOF is a straight line

Given that the size of angle COD is θ radians,

(a) write down, in terms of θ , the size of angle AOB

(1)

(b) Show that the area of the logo is

$$\frac{1}{2} r^2 (3\theta + \pi)$$

(2)

(c) Find the perimeter of the logo, giving your answer in simplest form in terms of r , θ and π .

(2)



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Question 6 continued

Handwriting practice area with horizontal lines.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 5 marks)



7.

In this question you should show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

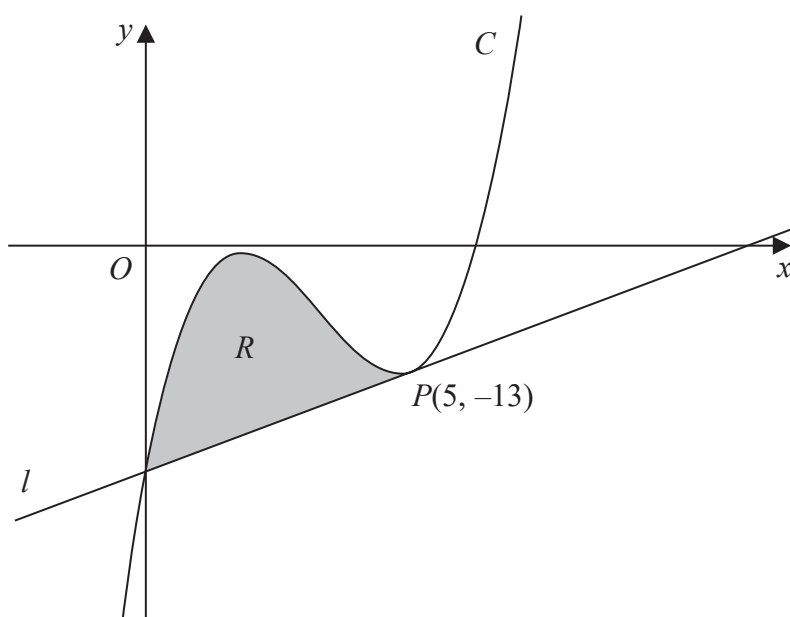


Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = x^3 - 10x^2 + 27x - 23$$

The point $P(5, -13)$ lies on C The line l is the tangent to C at P

- (a) Use differentiation to find the equation of l , giving your answer in the form $y = mx + c$ where m and c are integers to be found. (4)

- (b) Hence verify that l meets C again on the y -axis. (1)

The finite region R , shown shaded in Figure 2, is bounded by the curve C and the line l .

- (c) Use algebraic integration to find the exact area of R . (4)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 9 marks)



8. The curve C has equation

$$px^3 + qxy + 3y^2 = 26$$

where p and q are constants.

(a) Show that

$$\frac{dy}{dx} = \frac{apx^2 + bqy}{qx + cy}$$

where a , b and c are integers to be found.

(4)

Given that

- the point $P(-1, -4)$ lies on C
- the normal to C at P has equation $19x + 26y + 123 = 0$

(b) find the value of p and the value of q .

(5)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 9 marks)



9. Show that

$$\sum_{n=2}^{\infty} \left(\frac{3}{4}\right)^n \cos(180n)^{\circ} = \frac{9}{28}$$

(3)

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 3 marks)



10. The time, T seconds, that a pendulum takes to complete one swing is modelled by the formula

$$T = al^b$$

where l metres is the length of the pendulum and a and b are constants.

(a) Show that this relationship can be written in the form

$$\log_{10} T = b \log_{10} l + \log_{10} a \quad (2)$$

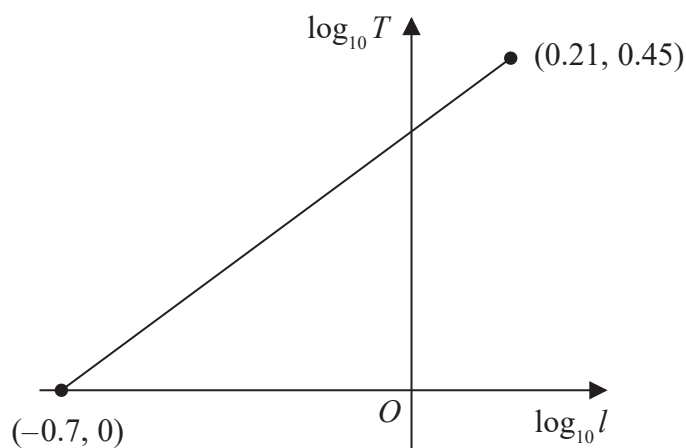


Figure 3

A student carried out an experiment to find the values of the constants a and b .

The student recorded the value of T for different values of l .

Figure 3 shows the linear relationship between $\log_{10} l$ and $\log_{10} T$ for the student's data.

The straight line passes through the points $(-0.7, 0)$ and $(0.21, 0.45)$

Using this information,

(b) find a complete equation for the model in the form

$$T = al^b$$

giving the value of a and the value of b , each to 3 significant figures.

(3)

(c) With reference to the model, interpret the value of the constant a .

(1)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 6 marks)



11.

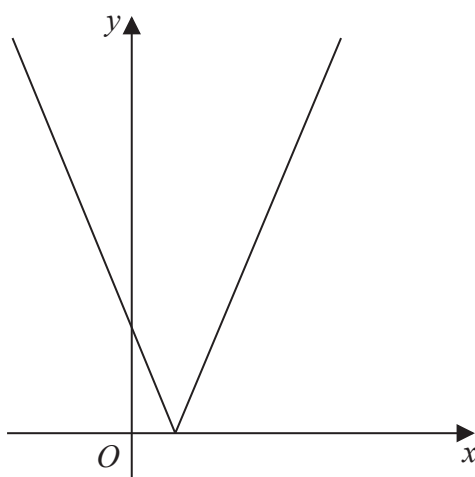


Figure 4

Figure 4 shows a sketch of the graph with equation

$$y = |2x - 3k|$$

where k is a positive constant.

(a) Sketch the graph with equation $y = f(x)$ where

$$f(x) = k - |2x - 3k|$$

stating

- the coordinates of the maximum point
- the coordinates of any points where the graph cuts the coordinate axes

(4)

(b) Find, in terms of k , the set of values of x for which

$$k - |2x - 3k| > x - k$$

giving your answer in set notation.

(4)

(c) Find, in terms of k , the coordinates of the minimum point of the graph with equation

$$y = 3 - 5f\left(\frac{1}{2}x\right)$$

(2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 10 marks)



12. (a) Use the substitution $u = 1 + \sqrt{x}$ to show that

$$\int_0^{16} \frac{x}{1 + \sqrt{x}} \, dx = \int_p^q \frac{2(u-1)^3}{u} \, du$$

where p and q are constants to be found.

(3)

(b) Hence show that

$$\int_0^{16} \frac{x}{1 + \sqrt{x}} \, dx = A - B \ln 5$$

where A and B are constants to be found.

(4)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 7 marks)



13. The curve C has parametric equations

$$x = \sin 2\theta \quad y = \operatorname{cosec}^3 \theta \quad 0 < \theta < \frac{\pi}{2}$$

(a) Find an expression for $\frac{dy}{dx}$ in terms of θ (3)

(b) Hence find the exact value of the gradient of the tangent to C at the point where $y = 8$ (3)



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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 6 marks)



14.

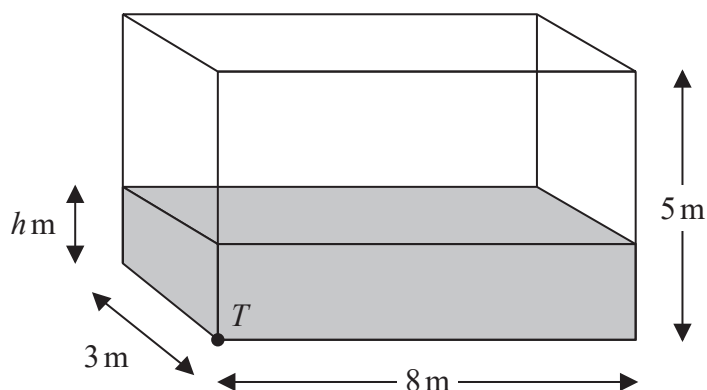


Figure 5

Water flows at a constant rate into a large tank.

The tank is a cuboid, with all sides of negligible thickness.

The base of the tank measures 8 m by 3 m and the height of the tank is 5 m.

There is a tap at a point T at the bottom of the tank, as shown in Figure 5.

At time t minutes after the tap has been opened

- the depth of water in the tank is h metres
- water is flowing into the tank at a constant rate of 0.48 m^3 per minute
- water is modelled as leaving the tank through the tap at a rate of $0.1h \text{ m}^3$ per minute

(a) Show that, according to the model,

$$1200 \frac{dh}{dt} = 24 - 5h \quad (4)$$

Given that when the tap was opened, the depth of water in the tank was 2 m,

(b) show that, according to the model,

$$h = A + Be^{-kt}$$

where A , B and k are constants to be found.

(6)

Given that the tap remains open,

(c) determine, according to the model, whether the tank will ever become full, giving a reason for your answer.

(2)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 12 marks)



15. (a) Express $2\cos\theta - \sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R and the value of α in radians to 3 decimal places.

(3)

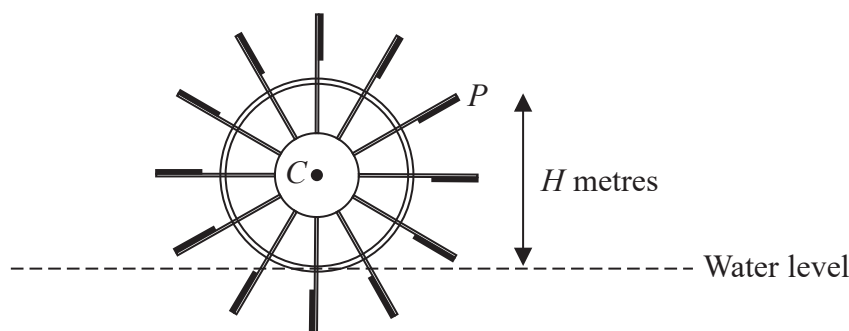


Figure 6

Figure 6 shows the cross-section of a water wheel.

The wheel is free to rotate about a fixed axis through the point C .

The point P is at the end of one of the paddles of the wheel, as shown in Figure 6.

The water level is assumed to be horizontal and of constant height.

The vertical height, H metres, of P above the water level is modelled by the equation

$$H = 3 + 4\cos(0.5t) - 2\sin(0.5t)$$

where t is the time in seconds after the wheel starts rotating.

Using the model, find

- (b) (i) the maximum height of P above the water level,
 (ii) the value of t when this maximum height first occurs, giving your answer to one decimal place.

(3)

In a single revolution of the wheel, P is below the water level for a total of T seconds.

According to the model,

- (c) find the value of T giving your answer to 3 significant figures.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

In reality, the water level may not be of constant height.

- (d) Explain how the equation of the model should be refined to take this into account.

(1)

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 11 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Time 2 hours

Paper
reference

9MA0/01

Mathematics

Advanced

PAPER 1: Pure Mathematics 1

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Q:1/1/1/1/




Pearson

1. The point $P(-2, -5)$ lies on the curve with equation $y = f(x)$, $x \in \mathbb{R}$

Find the point to which P is mapped, when the curve with equation $y = f(x)$ is transformed to the curve with equation

(a) $y = f(x) + 2$ (1)

(b) $y = |f(x)|$ (1)

(c) $y = 3f(x - 2) + 2$ (2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2.

$$f(x) = (x - 4)(x^2 - 3x + k) - 42 \text{ where } k \text{ is a constant}$$

Given that $(x + 2)$ is a factor of $f(x)$, find the value of k .

(3)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 3 marks)



3. A circle has equation

$$x^2 + y^2 - 10x + 16y = 80$$

(a) Find

(i) the coordinates of the centre of the circle,

(ii) the radius of the circle.

(3)

Given that P is the point on the circle that is furthest away from the origin O ,

(b) find the exact length OP

(2)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4. (a) Express $\lim_{\delta x \rightarrow 0} \sum_{x=2.1}^{6.3} \frac{2}{x} \delta x$ as an integral.

(1)

- (b) Hence show that

$$\lim_{\delta x \rightarrow 0} \sum_{x=2.1}^{6.3} \frac{2}{x} \delta x = \ln k$$

where k is a constant to be found.

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 3 marks)



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5. The height, h metres, of a tree, t years after being planted, is modelled by the equation

$$h^2 = at + b \quad 0 \leq t < 25$$

where a and b are constants.

Given that

- the height of the tree was 2.60 m, exactly 2 years after being planted
- the height of the tree was 5.10 m, exactly 10 years after being planted

- (a) find a complete equation for the model, giving the values of a and b to 3 significant figures. (4)

Given that the height of the tree was 7 m, exactly 20 years after being planted

- (b) evaluate the model, giving reasons for your answer. (2)



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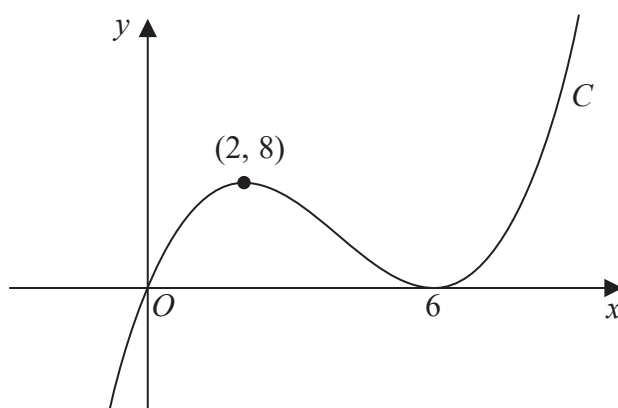


Figure 1

Figure 1 shows a sketch of a curve C with equation $y = f(x)$ where $f(x)$ is a cubic expression in x .

The curve

- passes through the origin
- has a maximum turning point at $(2, 8)$
- has a minimum turning point at $(6, 0)$

(a) Write down the set of values of x for which

$$f'(x) < 0$$

(1)

The line with equation $y = k$, where k is a constant, intersects C at only one point.

(b) Find the set of values of k , giving your answer in set notation.

(2)

(c) Find the equation of C . You may leave your answer in factorised form.

(3)

[illegible]

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7. (i) Given that p and q are integers such that

pq is even

use algebra to prove by contradiction that at least one of p or q is even.

(3)

(ii) Given that x and y are integers such that

- $x < 0$
- $(x + y)^2 < 9x^2 + y^2$

show that $y > 4x$

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 5 marks)



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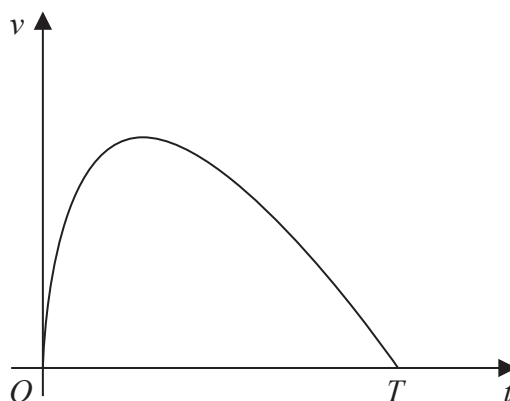


Figure 2

A car stops at two sets of traffic lights.

Figure 2 shows a graph of the speed of the car, $v \text{ ms}^{-1}$, as it travels between the two sets of traffic lights.

The car takes T seconds to travel between the two sets of traffic lights.

The speed of the car is modelled by the equation

$$v = (10 - 0.4t) \ln(t + 1) \quad 0 \leq t \leq T$$

where t seconds is the time after the car leaves the first set of traffic lights.

According to the model,

(a) find the value of T (1)

(b) show that the maximum speed of the car occurs when

$$t = \frac{26}{1 + \ln(t + 1)} - 1 \quad (4)$$

Using the iteration formula

$$t_{n+1} = \frac{26}{1 + \ln(t_n + 1)} - 1$$

with $t_1 = 7$

(c) (i) find the value of t_3 to 3 decimal places,
 (ii) find, by repeated iteration, the time taken for the car to reach maximum speed. (3)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 8 marks)



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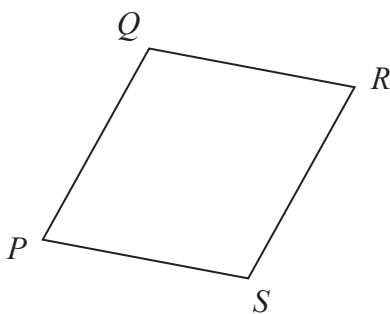


Figure 3

Figure 3 shows a sketch of a parallelogram $PQRS$.

Given that

- $\vec{PQ} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$
- $\vec{QR} = 5\mathbf{i} - 2\mathbf{k}$

(a) show that parallelogram $PQRS$ is a rhombus.

(2)

(b) Find the exact area of the rhombus $PQRS$.

(4)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



11.

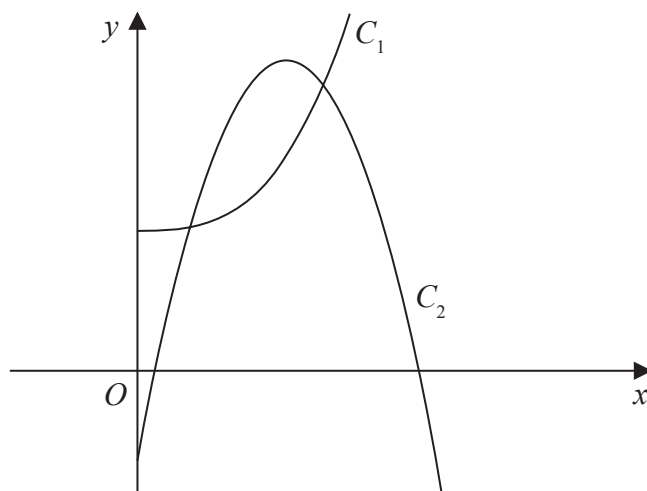
**Figure 4**

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 2x^3 + 10 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 42x - 15x^2 - 7 \quad x > 0$$

- (a) Verify that the curves intersect at $x = \frac{1}{2}$

(2)

The curves intersect again at the point P

- (b) Using algebra and showing all stages of working, find the exact x coordinate of P

(5)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 7 marks)



12.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

Show that

$$\int_1^{e^2} x^3 \ln x \, dx = ae^8 + b$$

where a and b are rational constants to be found.

(5)



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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 5 marks)



DO NOT WRITE IN THIS AREA

Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



14.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Given that

$$2 \sin(x - 60^\circ) = \cos(x - 30^\circ)$$

show that

$$\tan x = 3\sqrt{3} \quad (4)$$

(b) Hence or otherwise solve, for $0 \leq \theta < 180^\circ$

$$2 \sin 2\theta = \cos(2\theta + 30^\circ)$$

giving your answers to one decimal place. (4)

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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 8 marks)



15.

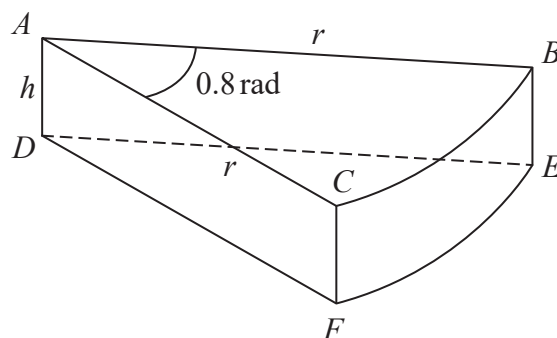


Figure 5

A company makes toys for children.

Figure 5 shows the design for a solid toy that looks like a piece of cheese.

The toy is modelled so that

- face ABC is a sector of a circle with radius r cm and centre A
- angle $BAC = 0.8$ radians
- faces ABC and DEF are congruent
- edges AD , CF and BE are perpendicular to faces ABC and DEF
- edges AD , CF and BE have length h cm

Given that the volume of the toy is 240 cm^3

(a) show that the surface area of the toy, $S \text{ cm}^2$, is given by

$$S = 0.8r^2 + \frac{1680}{r}$$

making your method clear.

(4)

Using algebraic differentiation,

(b) find the value of r for which S has a stationary point.

(4)

(c) Prove, by further differentiation, that this value of r gives the minimum surface area of the toy.

(2)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.

(Total for Question 15 is 10 marks)



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Question 16 continued

Lined area for writing the answer to Question 16.



Question 16 continued

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(Total for Question 16 is 9 marks)

TOTAL FOR PAPER IS 100 MARKS



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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Time 2 hours

Paper
reference

9MA0/02

Mathematics

Advanced

PAPER 2: Pure Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Q:1/1/1/1/




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1. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

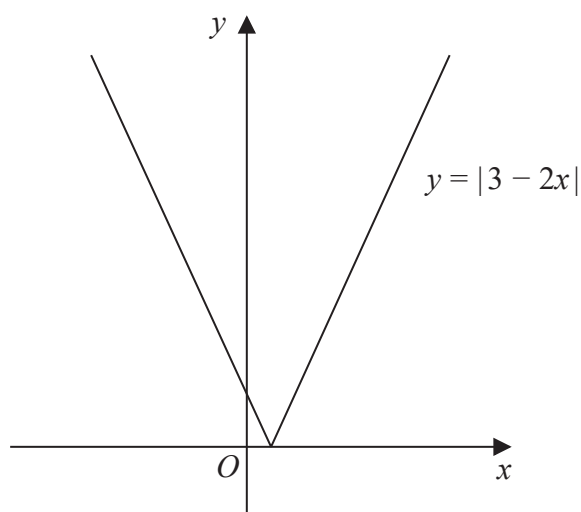


Figure 1

Figure 1 shows a sketch of the graph with equation $y = |3 - 2x|$

Solve

$$|3 - 2x| = 7 + x$$

(4)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. (a) Sketch the curve with equation

$$y = 4^x$$

stating any points of intersection with the coordinate axes.

(2)

- (b) Solve

$$4^x = 100$$

giving your answer to 2 decimal places.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3. A sequence of terms $a_1, a_2, a_3, \dots$ is defined by

$$a_1 = 3$$
$$a_{n+1} = 8 - a_n$$

(a) (i) Show that this sequence is periodic.

(ii) State the order of this periodic sequence.

(2)

(b) Find the value of

$$\sum_{n=1}^{85} a_n$$

(2)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 4 marks)



4. Given that

$$y = 2x^2$$

use differentiation from first principles to show that

$$\frac{dy}{dx} = 4x$$

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 3 marks)



5. The table below shows corresponding values of x and y for $y = \log_3 2x$

The values of y are given to 2 decimal places as appropriate.

x	3	4.5	6	7.5	9
y	1.63	2	2.26	2.46	2.63

- (a) Using the trapezium rule with all the values of y in the table, find an estimate for

$$\int_3^9 \log_3 2x \, dx$$

(3)

Using your answer to part (a) and making your method clear, estimate

(b) (i) $\int_3^9 \log_3 (2x)^{10} \, dx$

(ii) $\int_3^9 \log_3 18x \, dx$

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



6.

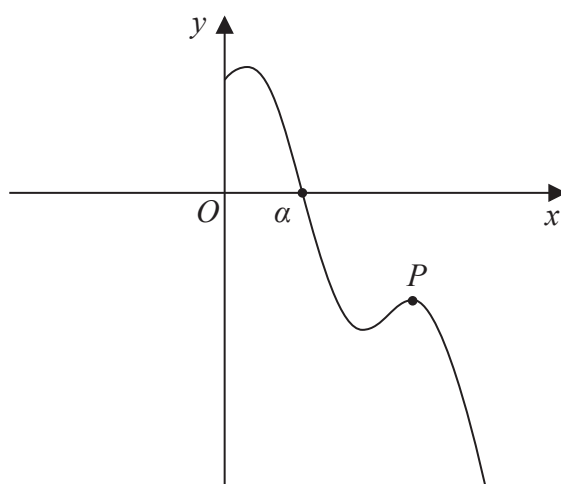


Figure 2

Figure 2 shows a sketch of part of the curve with equation $y = f(x)$ where

$$f(x) = 8 \sin\left(\frac{1}{2}x\right) - 3x + 9 \quad x > 0$$

and x is measured in radians.

The point P , shown in Figure 2, is a local maximum point on the curve.

Using calculus and the sketch in Figure 2,

- (a) find the x coordinate of P , giving your answer to 3 significant figures.

(4)

The curve crosses the x -axis at $x = \alpha$, as shown in Figure 2.

Given that, to 3 decimal places, $f(4) = 4.274$ and $f(5) = -1.212$

- (b) explain why α must lie in the interval $[4, 5]$

(1)

- (c) Taking $x_0 = 5$ as a first approximation to α , apply the Newton-Raphson method once to $f(x)$ to obtain a second approximation to α .

Show your method and give your answer to 3 significant figures.

(2)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 7 marks)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 5 marks)



8.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

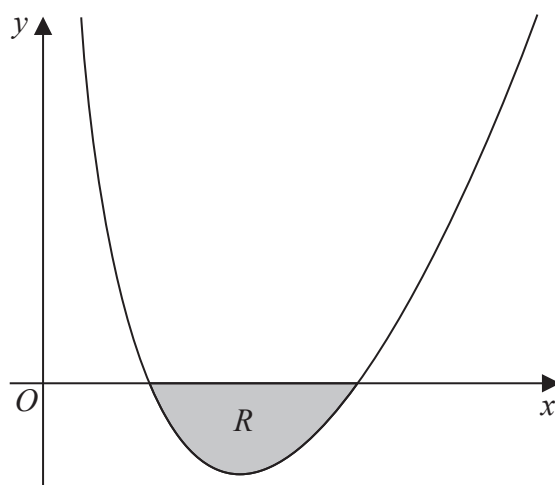


Figure 3

Figure 3 shows a sketch of part of a curve with equation

$$y = \frac{(x-2)(x-4)}{4\sqrt{x}} \quad x > 0$$

The region R , shown shaded in Figure 3, is bounded by the curve and the x -axis.

Find the exact area of R , writing your answer in the form $a\sqrt{2} + b$, where a and b are constants to be found.

(6)

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 6 marks)



9.

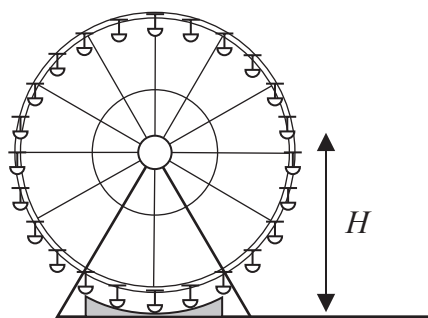


Figure 4

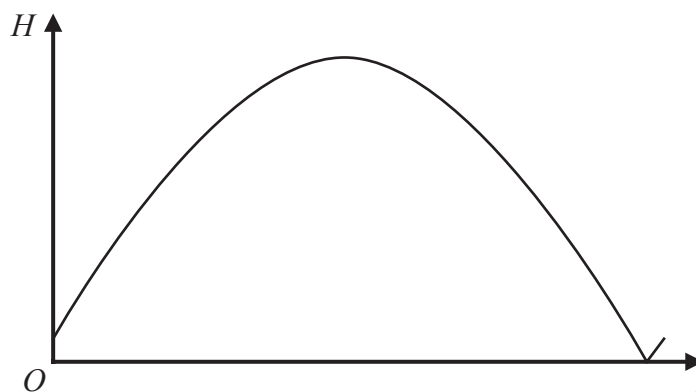


Figure 5

Figure 4 shows a sketch of a Ferris wheel.

The height above the ground, H m, of a passenger on the Ferris wheel, t seconds after the wheel starts turning, is modelled by the equation

$$H = |A \sin(bt + \alpha)|$$

where A , b and α are constants.

Figure 5 shows a sketch of the graph of H against t , for one revolution of the wheel.

Given that

- the maximum height of the passenger above the ground is 50 m
- the passenger is 1 m above the ground when the wheel starts turning
- the wheel takes 720 seconds to complete one revolution

(a) find a complete equation for the model, giving the exact value of A , the exact value of b and the value of α to 3 significant figures.

(4)

(b) Explain why an equation of the form

$$H = |A \sin(bt + \alpha)| + d$$

where d is a positive constant, would be a more appropriate model.

(1)



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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 5 marks)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



11. Prove, using algebra, that

$$n(n^2 + 5)$$

is even for all $n \in \mathbb{N}$.

(4)



DO NOT WRITE IN THIS AREA

Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 4 marks)



12. The function f is defined by

$$f(x) = \frac{e^{3x}}{4x^2 + k}$$

where k is a positive constant.

(a) Show that

$$f'(x) = (12x^2 - 8x + 3k)g(x)$$

where $g(x)$ is a function to be found.

(3)

Given that the curve with equation $y = f(x)$ has at least one stationary point,

(b) find the range of possible values of k .

(3)



DO NOT WRITE IN THIS AREA

Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 6 marks)



13. Relative to a fixed origin O

- the point A has position vector $4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$
- the point B has position vector $4\mathbf{j} + 6\mathbf{k}$
- the point C has position vector $-16\mathbf{i} + p\mathbf{j} + 10\mathbf{k}$

where p is a constant.

Given that A , B and C lie on a straight line,

(a) find the value of p .

(3)

The line segment OB is extended to a point D so that $\vec{CD}$ is parallel to $\vec{OA}$

(b) Find $|\vec{OD}|$, writing your answer as a fully simplified surd.

(3)



DO NOT WRITE IN THIS AREA

Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 6 marks)



14. (a) Express $\frac{3}{(2x-1)(x+1)}$ in partial fractions.

(3)

When chemical A and chemical B are mixed, oxygen is produced.

A scientist mixed these two chemicals and measured the total volume of oxygen produced over a period of time.

The total volume of oxygen produced, $V \text{ m}^3$, t hours after the chemicals were mixed, is modelled by the differential equation

$$\frac{dV}{dt} = \frac{3V}{(2t-1)(t+1)} \quad V \geq 0 \quad t \geq k$$

where k is a constant.

Given that exactly 2 hours after the chemicals were mixed, a total volume of 3 m^3 of oxygen had been produced,

(b) solve the differential equation to show that

$$V = \frac{3(2t-1)}{(t+1)} \quad (5)$$

The scientist noticed that

- there was a **time delay** between the chemicals being mixed and oxygen being produced
- there was a **limit** to the total volume of oxygen produced

Deduce from the model

(c) (i) the **time delay** giving your answer in minutes,

(ii) the **limit** giving your answer in m^3

(2)



DO NOT WRITE IN THIS AREA

Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 10 marks)



15.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

Given that the first three terms of a geometric series are

$$12 \cos \theta \quad 5 + 2 \sin \theta \quad \text{and} \quad 6 \tan \theta$$

(a) show that

$$4 \sin^2 \theta - 52 \sin \theta + 25 = 0 \quad (3)$$

Given that θ is an obtuse angle measured in radians,(b) solve the equation in part (a) to find the exact value of θ (2)

(c) show that the sum to infinity of the series can be expressed in the form

$$k(1 - \sqrt{3})$$

where k is a constant to be found. (5)

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.

(Total for Question 15 is 10 marks)



16.

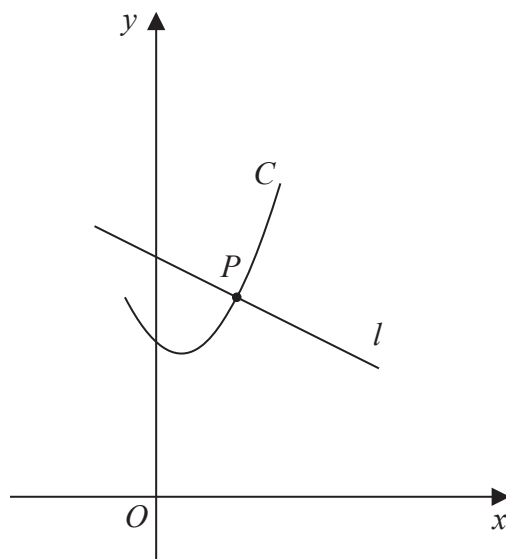


Figure 6

Figure 6 shows a sketch of the curve C with parametric equations

$$x = 2 \tan t + 1 \quad y = 2 \sec^2 t + 3 \quad -\frac{\pi}{4} \leq t \leq \frac{\pi}{3}$$

The line l is the normal to C at the point P where $t = \frac{\pi}{4}$

(a) Using parametric differentiation, show that an equation for l is

$$y = -\frac{1}{2}x + \frac{17}{2} \quad (5)$$

(b) Show that all points on C satisfy the equation

$$y = \frac{1}{2}(x-1)^2 + 5 \quad (2)$$

The straight line with equation

$$y = -\frac{1}{2}x + k \quad \text{where } k \text{ is a constant}$$

intersects C at two distinct points.

(c) Find the range of possible values for k .

(5)



DO NOT WRITE IN THIS AREA

Question 16 continued

Lined area for writing the answer to Question 16.



Question 16 continued

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DO NOT WRITE IN THIS AREA

Question 16 continued

Lined area for writing the answer to Question 16.



Question 16 continued

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(Total for Question 16 is 12 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Tuesday 6 June 2023

Afternoon (Time: 2 hours)

Paper reference **9MA0/01**

Mathematics

Advanced

PAPER 1: Pure Mathematics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Find

$$\int \frac{x^{\frac{1}{2}}(2x-5)}{3} dx$$

writing each term in simplest form.

(4)

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Question 1 continued

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(Total for Question 1 is 4 marks)



2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

$$f(x) = 4x^3 + 5x^2 - 10x + 4a \quad x \in \mathbb{R}$$

where a is a positive constant.

Given $(x - a)$ is a factor of $f(x)$,

(a) show that

$$a(4a^2 + 5a - 6) = 0 \quad (2)$$

(b) Hence

(i) find the value of a

(ii) use algebra to find the exact solutions of the equation

$$f(x) = 3 \quad (4)$$



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 6 marks)



3. Relative to a fixed origin O

- the point A has position vector $5\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$
- the point B has position vector $2\mathbf{i} + 4\mathbf{j} + a\mathbf{k}$

where a is a positive integer.

(a) Show that $|\vec{OA}| = \sqrt{38}$ (1)

(b) Find the smallest value of a for which

$$|\vec{OB}| > |\vec{OA}|$$
 (2)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 3 marks)



4.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$ where $x \in \mathbb{R}$

Given that

- $f'(x) = 2x + \frac{1}{2} \cos x$
 - the curve has a stationary point with x coordinate α
 - α is small
- (a) use the small angle approximation for $\cos x$ to estimate the value of α to 3 decimal places.

(3)

The point $P(0, 3)$ lies on C

- (b) Find the equation of the tangent to the curve at P , giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(2)



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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 5 marks)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



6.

$a = \log_2 x$

$b = \log_2(x + 8)$

Express in terms of a and/or b

$$(a) \log_2 \sqrt{x} \quad (1)$$

$$(b) \log_2(x^2 + 8x) \quad (2)$$

$$(c) \log_2 \left(8 + \frac{64}{x} \right) \quad (3)$$

Give your answer in simplest form.



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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 6 marks)



7. The function f is defined by

$$f(x) = 3 + \sqrt{x - 2} \qquad x \in \mathbb{R} \quad x > 2$$

- (a) State the range of f (1)

- (b) Find f^{-1} (3)

The function g is defined by

$$g(x) = \frac{15}{x-3} \quad x \in \mathbb{R} \quad x \neq 3$$

- (c) Find $gf(6)$ (2)

- (d) Find the exact value of the constant a for which

$$f(a^2 + 2) = g(a) \tag{2}$$

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

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(Total for Question 7 is 8 marks)

8.

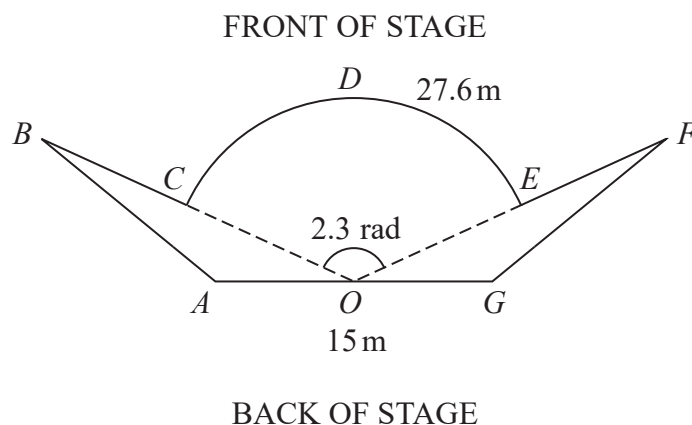
Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the plan view of a stage.

The plan view shows two congruent triangles ABO and GFO joined to a sector $OCDEO$ of a circle, centre O , where

- angle $COE = 2.3$ radians
- arc length $CDE = 27.6$ m
- AOG is a straight line of length 15 m

(a) Show that $OC = 12$ m.

(2)

(b) Show that the size of angle AOB is 0.421 radians correct to 3 decimal places.

(2)

Given that the total length of the front of the stage, $BCDEF$, is 35 m,

(c) find the total area of the stage, giving your answer to the nearest square metre.

(6)

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 10 marks)



DO NOT WRITE IN THIS AREA

Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 7 marks)



10. A circle C has equation

$$x^2 + y^2 + 6kx - 2ky + 7 = 0$$

where k is a constant.

(a) Find in terms of k ,

(i) the coordinates of the centre of C

(ii) the radius of C

(3)

The line with equation $y = 2x - 1$ intersects C at 2 distinct points.

(b) Find the range of possible values of k .

(6)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 9 marks)



11.

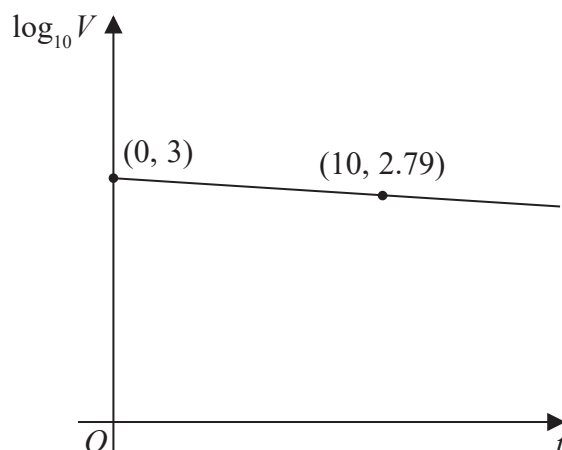


Figure 2

The value, V pounds, of a mobile phone, t months after it was bought, is modelled by

$$V = ab^t$$

where a and b are constants.

Figure 2 shows the linear relationship between $\log_{10} V$ and t .

The line passes through the points $(0, 3)$ and $(10, 2.79)$ as shown.

Using these points,

(a) find the initial value of the phone, (2)

(b) find a complete equation for V in terms of t , giving the exact value of a and giving the value of b to 3 significant figures. (3)

Exactly 2 years after it was bought, the value of the phone was £320

(c) Use this information to evaluate the reliability of the model. (2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 11 is 7 marks)



12.

$$y = \sin x$$

where x is measured in radians.

Use differentiation from first principles to show that

$$\frac{dy}{dx} = \cos x$$

You may

- use without proof the formula for $\sin(A \pm B)$
- assume that as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$

(5)



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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 5 marks)



13. On a roller coaster ride, passengers travel in carriages around a track.

On the ride, carriages complete multiple circuits of the track such that

- the maximum vertical height of a carriage above the ground is 60 m
- a carriage starts a circuit at a vertical height of 2 m above the ground
- the ground is horizontal

The vertical height, H m, of a carriage above the ground, t seconds after the carriage starts the first circuit, is modelled by the equation

$$H = a - b(t - 20)^2$$

where a and b are positive constants.

- (a) Find a complete equation for the model. (3)
- (b) Use the model to determine the height of the carriage above the ground when $t = 40$ (1)

In an alternative model, the vertical height, H m, of a carriage above the ground, t seconds after the carriage starts the first circuit, is given by

$$H = 29 \cos(9t + \alpha)^\circ + \beta \quad 0 \leq \alpha < 360^\circ$$

where α and β are constants.

- (c) Find a complete equation for the alternative model. (2)
- Given that the carriage moves continuously for 2 minutes,
- (d) give a reason why the alternative model would be more appropriate. (1)

Given that the carriage moves continuously for 2 minutes,

(d) give a reason why the alternative model would be more appropriate.



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



14. Prove, using algebra, that

$$(n + 1)^3 - n^3$$

is odd for all $n \in \mathbb{N}$

(4)



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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 4 marks)



15. A curve has equation $y = f(x)$, where

$$f(x) = \frac{7xe^x}{\sqrt{e^{3x} - 2}} \quad x > \ln \sqrt[3]{2}$$

(a) Show that

$$f'(x) = \frac{7e^x(e^{3x}(2-x) + Ax + B)}{2(e^{3x} - 2)^{\frac{3}{2}}}$$

where A and B are constants to be found.

(5)

(b) Hence show that the x coordinates of the turning points of the curve are solutions of the equation

$$x = \frac{2e^{3x} - 4}{e^{3x} + 4}$$

(2)

The equation $x = \frac{2e^{3x} - 4}{e^{3x} + 4}$ has two positive roots α and β where $\beta > \alpha$

A student uses the iteration formula

$$x_{n+1} = \frac{2e^{3x_n} - 4}{e^{3x_n} + 4}$$

in an attempt to find approximations for α and β

Diagram 1 shows a plot of part of the curve with equation $y = \frac{2e^{3x} - 4}{e^{3x} + 4}$ and part of the line with equation $y = x$

Using Diagram 1 on page 42

(c) draw a staircase diagram to show that the iteration formula starting with $x_1 = 1$ can be used to find an approximation for β

(1)

Use the iteration formula with $x_1 = 1$, to find, to 3 decimal places,

(d) (i) the value of x_2

(ii) the value of β

(3)

Using a suitable interval and a suitable function that should be stated

(e) show that $\alpha = 0.432$ to 3 decimal places.

(2)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

Only use the copy of Diagram 1 if you need to redraw your answer to part (c).

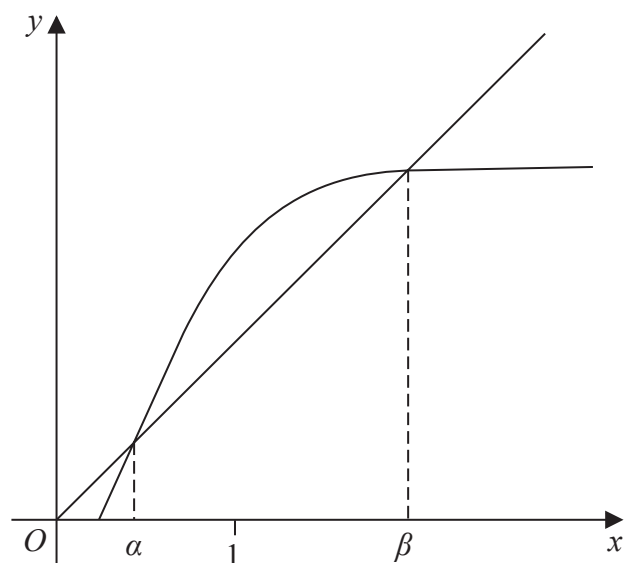
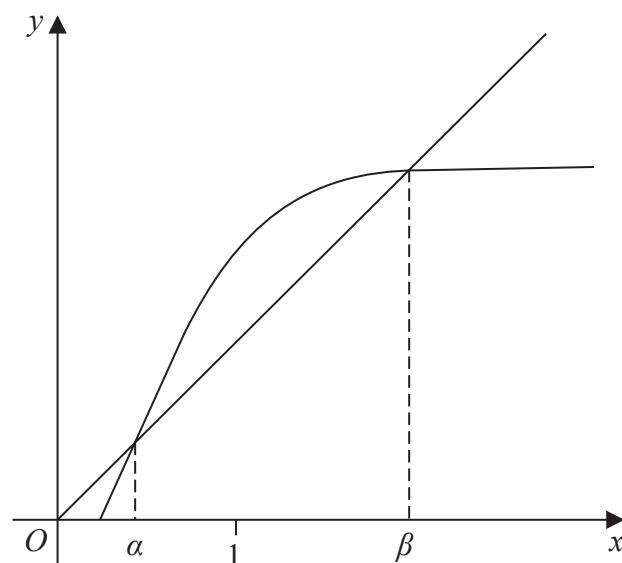


Diagram 1



copy of Diagram 1

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 13 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Tuesday 13 June 2023

Afternoon (Time: 2 hours)	Paper reference	9MA0/02
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Mathematics

Advanced

PAPER 2: Pure Mathematics 2

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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$$f(x) = x^3 + 2x^2 - 8x + 5$$

(b) (i) Solve $f''(x) = 0$

(ii) Hence find the range of values of x for which $f(x)$ is concave. (2)

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DO NOT WRITE IN THIS AREA

Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 4 marks)



2. A sequence $u_1, u_2, u_3 \dots$ is defined by

$$u_1 = 35$$

$$u_{n+1} = u_n + 7 \cos\left(\frac{n\pi}{2}\right) - 5(-1)^n$$

(a) (i) Show that $u_2 = 40$

(ii) Find the value of u_3 and the value of u_4

(3)

Given that the sequence is periodic with order 4

(b) (i) write down the value of u_5

(ii) find the value of $\sum_{r=1}^{25} u_r$

(3)



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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 6 marks)



3. Given that

$$\log_2(x+3) + \log_2(x+10) = 2 + 2\log_2 x$$

(a) show that

$$3x^2 - 13x - 30 = 0 \quad (3)$$

(b) (i) Write down the roots of the equation

$$3x^2 - 13x - 30 = 0$$

(ii) Hence state which of the roots in part (b)(i) is not a solution of

$$\log_2(x+3) + \log_2(x+10) = 2 + 2\log_2 x$$

giving a reason for your answer.

(2)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 5 marks)



- (3)

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 4 marks)



- 5.** The curve C has equation $y = f(x)$

The curve

- passes through the point $P(3, -10)$
- has a turning point at P

Given that

$$\frac{dy}{dx} = 2x^3 - 9x^2 + 5x + k$$

where k is a constant,

- (a) show that $k = 12$ **(2)**
- (b) Hence find the coordinates of the point where C crosses the y -axis. **(3)**



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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 5 marks)



6. Relative to a fixed origin O ,

- A is the point with position vector $12\mathbf{i}$
- B is the point with position vector $16\mathbf{j}$
- C is the point with position vector $(50\mathbf{i} + 136\mathbf{j})$
- D is the point with position vector $(22\mathbf{i} + 24\mathbf{j})$

(a) Show that AD is parallel to BC .

(2)

Points A , B , C and D are used to model the vertices of a running track in the shape of a quadrilateral.

Runners complete one lap by running along all four sides of the track.

The lengths of the sides are measured in metres.

Given that a particular runner takes exactly 5 minutes to complete 2 laps,

(b) calculate the average speed of this runner, giving the answer in kilometres per hour.

(4)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

A curve has equation

$$x^3 + 2xy + 3y^2 = 47$$

- (a) Find $\frac{dy}{dx}$ in terms of x and y

(4)

The point $P(-2, 5)$ lies on the curve.

- (b) Find the equation of the normal to the curve at P , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)



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Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 7 marks)



8. (a) Express $2\cos\theta + 8\sin\theta$ in the form $R\cos(\theta - \alpha)$, where R and α are constants,
 $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R and give the value of α in radians to 3 decimal places.

(3)

The first three terms of an arithmetic sequence are

$$\cos x \quad \cos x + \sin x \quad \cos x + 2\sin x \quad x \neq n\pi$$

Given that S_9 represents the sum of the first 9 terms of this sequence as x varies,

- (b) (i) find the exact maximum value of S_9

- (ii) deduce the smallest positive value of x at which this maximum value of S_9 occurs.

(3)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 6 marks)



9. The curve C has parametric equations

$$x = t^2 + 6t - 16 \quad y = 6 \ln(t + 3) \quad t > -3$$

(a) Show that a Cartesian equation for C is

$$y = A \ln(x + B) \quad x > -B$$

where A and B are integers to be found.

(3)

The curve C cuts the y -axis at the point P

(b) Show that the equation of the tangent to C at P can be written in the form

$$ax + by = c \ln 5$$

where a , b and c are integers to be found.

(4)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 7 marks)



10. $f(x) = \frac{3kx - 18}{(x + 4)(x - 2)}$ where k is a positive constant

(a) Express $f(x)$ in partial fractions in terms of k .

(3)

(b) Hence find the exact value of k for which

$$\int_{-3}^1 f(x) \, dx = 21$$

(4)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 7 marks)



11.

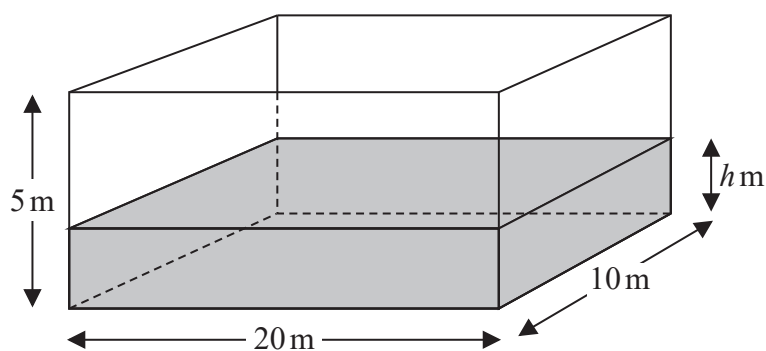


Figure 1

A tank in the shape of a cuboid is being filled with water.

The base of the tank measures 20 m by 10 m and the height of the tank is 5 m, as shown in Figure 1.

At time t minutes after water started flowing into the tank the height of the water was h m and the volume of water in the tank was $V \text{ m}^3$

In a model of this situation

- the sides of the tank have negligible thickness
- the rate of change of V is inversely proportional to the square root of h

(a) Show that

$$\frac{dh}{dt} = \frac{\lambda}{\sqrt{h}}$$

where λ is a constant.

(3)

Given that

- initially the height of the water in the tank was 1.44 m
- exactly 8 minutes after water started flowing into the tank the height of the water was 3.24 m

(b) use the model to find an equation linking h with t , giving your answer in the form

$$h^{\frac{3}{2}} = At + B$$

where A and B are constants to be found.

(5)

(c) Hence find the time taken, from when water started flowing into the tank, for the tank to be completely full.

(2)



DO NOT WRITE IN THIS AREA

Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 10 marks)



12.

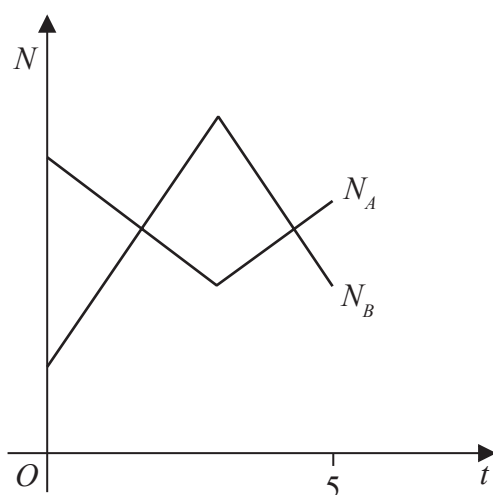


Figure 2

The number of subscribers to two different music streaming companies is being monitored.

The number of subscribers, N_A , in thousands, to **company A** is modelled by the equation

$$N_A = |t - 3| + 4 \quad t \geq 0$$

where t is the time in years since monitoring began.

The number of subscribers, N_B , in thousands, to **company B** is modelled by the equation

$$N_B = 8 - |2t - 6| \quad t \geq 0$$

where t is the time in years since monitoring began.

Figure 2 shows a sketch of the graph of N_A and the graph of N_B over a 5-year period.

Use the equations of the models to answer parts (a), (b), (c) and (d).

- (a) Find the initial difference between the number of subscribers to **company A** and the number of subscribers to **company B**.

(2)

When $t = T$ **company A** reduced its subscription prices and the number of subscribers increased.

- (b) Suggest a value for T , giving a reason for your answer.

(2)

- (c) Find the range of values of t for which $N_A > N_B$ giving your answer in set notation.

(5)

- (d) State a limitation of the model used for **company B**.

(1)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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DO NOT WRITE IN THIS AREA

Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 10 marks)



13.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

- (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$(3 + x)^{-2}$$

writing each term in simplest form.

(4)

- (b) Using the answer to part (a) and using algebraic integration, estimate the value of

$$\int_{0.2}^{0.4} \frac{6x}{(3+x)^2} dx$$

giving your answer to 4 significant figures.

(4)

- (c) Find, using algebraic integration, the exact value of

$$\int_{0.2}^{0.4} \frac{6x}{(3+x)^2} dx$$

giving your answer in the form $a \ln b + c$, where a , b and c are constants to be found.

(5)



DO NOT WRITE IN THIS AREA

Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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DO NOT WRITE IN THIS AREA

Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 13 marks)



14.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that the equation

$$2 \tan \theta (8 \cos \theta + 23 \sin^2 \theta) = 8 \sin 2\theta (1 + \tan^2 \theta)$$

may be written as

$$\sin 2\theta (A \cos^2 \theta + B \cos \theta + C) = 0$$

where A , B and C are constants to be found.

(3)

(b) Hence, solve for $360^\circ \leq x \leq 540^\circ$

$$2 \tan x (8 \cos x + 23 \sin^2 x) = 8 \sin 2x (1 + \tan^2 x) \quad x \in \mathbb{R} \quad x \neq 450^\circ$$

(4)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 7 marks)



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Question 15 continued

Handwriting practice area with horizontal lines.



Question 15 continued

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(Total for Question 15 is 3 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Tuesday 4 June 2024

Afternoon (Time: 2 hours)

Paper reference **9MA0/01**

Mathematics

Advanced

PAPER 1: Pure Mathematics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

$$g(x) = 3x^3 - 20x^2 + (k + 17)x + k$$

where k is a constant.

Given that $(x - 3)$ is a factor of $g(x)$, find the value of k .

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 3 marks)



2. (a) Find, in ascending powers of x , the first four terms of the binomial expansion of

$$(1 - 9x)^{\frac{1}{2}}$$

giving each term in simplest form.

(3)

- (b) Give a reason why $x = -\frac{2}{9}$ should **not** be used in the expansion to find an approximation to $\sqrt{3}$

(1)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3.

$$f(x) = x + \tan\left(\frac{1}{2}x\right) \quad \pi < x < \frac{3\pi}{2}$$

Given that the equation $f(x) = 0$ has a single root α

(a) show that α lies in the interval $[3.6, 3.7]$ (2)

(b) Find $f'(x)$ (2)

(c) Using 3.7 as a first approximation for α , apply the Newton–Raphson method once to obtain a second approximation for α . Give your answer to 3 decimal places. (2)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 6 marks)



4. Given that $y = x^2$, use differentiation from first principles to show that $\frac{dy}{dx} = 2x$ (3)

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 3 marks)



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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 6 marks)



6.

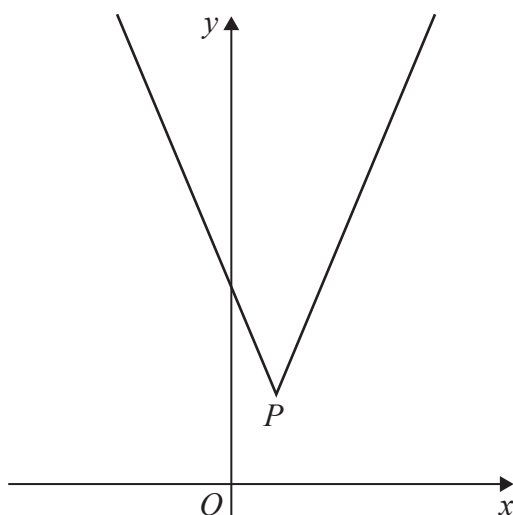


Figure 1

Figure 1 shows a sketch of the graph with equation

$$y = 3|x - 2| + 5$$

The vertex of the graph is at the point P , shown in Figure 1.

- (a) Find the coordinates of P .

(2)

- (b) Solve the equation

$$16 - 4x = 3|x - 2| + 5$$

(2)

A line l has equation $y = kx + 4$ where k is a constant.

Given that l intersects $y = 3|x - 2| + 5$ at 2 distinct points,

- (c) find the range of values of k .

(2)

[illegible]

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 6 marks)



7.

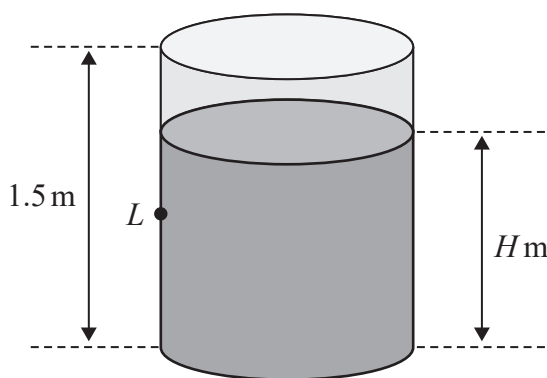


Diagram not drawn to scale.

Figure 2

Figure 2 shows a cylindrical tank of height 1.5 m.

Initially the tank is full of water.

The water starts to leak from a small hole, at a point L , in the side of the tank.

While the tank is leaking, the depth, H metres, of the water in the tank is modelled by the differential equation

$$\frac{dH}{dt} = -0.12e^{-0.2t}$$

where t hours is the time after the leak starts.

Using the model,

(a) show that

$$H = Ae^{-0.2t} + B$$

where A and B are constants to be found,

(3)

(b) find the time taken for the depth of the water to decrease to 1.2 m. Give your answer in hours and minutes, to the nearest minute.

(3)

In the long term, the water level in the tank falls to the same height as the hole.

(c) Find, according to the model, the height of the hole from the bottom of the tank.

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



$$\begin{aligned} f(x) &= 4 - 3x^2 & x \in \mathbb{R} \\ g(x) &= \frac{5}{2x-9} & x \in \mathbb{R}, \ x \neq \frac{9}{2} \end{aligned}$$

- The function h is defined by

$$h(x) = 2x^2 - 6x + k \quad x \in \mathbb{R}$$

where k is a constant.

- (d) Find the range of values of k for which the equation

$$f(x) = h(x)$$

has no real solutions. (3)

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Question 8 continued

Handwriting practice area with horizontal lines.



Question 8 continued

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DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 11 marks)



9. The first 3 terms of a geometric sequence are

$$3^{4k-5} \quad 9^{7-2k} \quad 3^{2(k-1)}$$

where k is a constant.

(a) Using algebra and making your reasoning clear, prove that $k = \frac{5}{2}$ (3)

(b) Hence find the sum to infinity of the geometric sequence. (3)



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Question 9 continued

Lined area for writing answers.

(Total for Question 9 is 6 marks)



10.

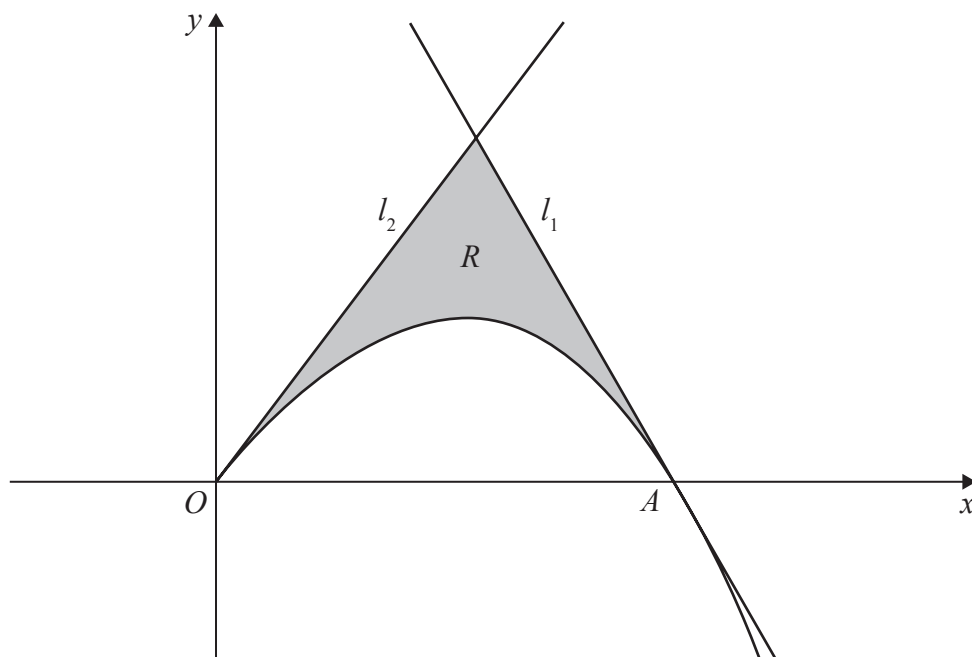


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve with equation

$$y = 8x - x^{\frac{5}{2}} \quad x \geq 0$$

The curve crosses the x -axis at the point A .

(a) Verify that the x coordinate of A is 4

(1)

The line l_1 is the tangent to the curve at A .

(b) Use calculus to show that an equation of line l_1 is

$$12x + y = 48$$

(3)

The line l_2 has equation $y = 8x$

The region R , shown shaded in Figure 3, is bounded by the curve, the line l_1 and the line l_2

(c) Use algebraic integration to find the exact area of R .

(5)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 9 marks)



11.

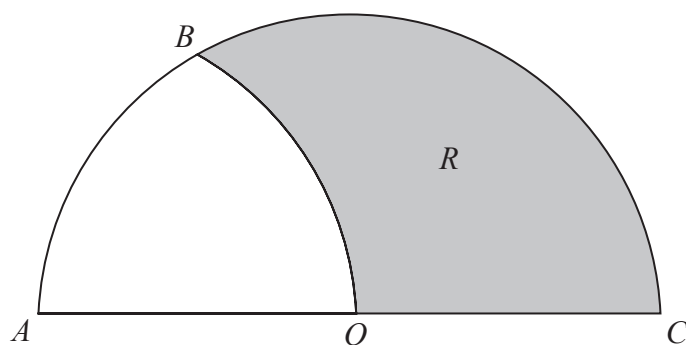


Figure 4

Figure 4 shows the design of a badge.

The shape $ABCOA$ is a semicircle with centre O and diameter 10 cm.

OB is the arc of a circle with centre A and radius 5 cm.

The region R , shown shaded in Figure 4, is bounded by the arc OB , the arc BC and the line OC .

Find the exact area of R .

Give your answer in the form $(a\sqrt{3} + b\pi)\text{cm}^2$, where a and b are rational numbers.

(4)



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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 4 marks)



- 12.** (a) Express $140 \cos \theta - 480 \sin \theta$ in the form $K \cos(\theta + \alpha)$

where $K > 0$ and $0 < \alpha < 90^\circ$

State the value of K and give the value of α , in degrees, to 2 decimal places.

(3)

A scientist studies the number of rabbits and the number of foxes in a wood for one year.

The number of rabbits, R , is modelled by the equation

$$R = A + 140 \cos(30t)^\circ - 480 \sin(30t)^\circ$$

where t months is the time after the start of the year and A is a constant.

Given that, during the year, the maximum number of rabbits in the wood is 1500

- (b) (i) find a complete equation for this model.
- (ii) Hence write down the minimum number of rabbits in the wood during the year according to the model.
- (2)**

The actual number of rabbits in the wood is at its minimum value in the middle of April.

- (c) Use this information to comment on the model for the number of rabbits. (2)

The number of foxes, F , in the wood during the same year is modelled by the equation

$$F = 100 + 70 \sin(30t + 70)^\circ$$

The number of foxes is at its minimum value after T months.

- (d) Find, according to the models, the number of **rabbits** in the wood at time T months. (4)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 11 marks)



13. (a) Given that a is a positive constant, use the substitution $x = a \sin^2 \theta$ to show that

$$\int_0^a x^{\frac{1}{2}} \sqrt{a-x} \, dx = \frac{1}{2} a^2 \int_0^{\frac{\pi}{2}} \sin^2 2\theta \, d\theta \quad (4)$$

(b) Hence use algebraic integration to show that

$$\int_0^a x^{\frac{1}{2}} \sqrt{a-x} \, dx = k\pi a^2$$

where k is a constant to be found.

(4)



DO NOT WRITE IN THIS AREA

Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Handwriting practice area with horizontal lines.

(Total for Question 13 is 8 marks)



14. A balloon is being inflated.

In a simple model,

- the balloon is modelled as a sphere
- the rate of increase of the radius of the balloon is inversely proportional to the square root of the radius of the balloon

At time t seconds, the radius of the balloon is r cm.

(a) Write down a differential equation to model this situation.

(1)

At the instant when $t = 10$

- the radius is 16 cm
- the radius is increasing at a rate of 0.9 cm s^{-1}

(b) Solve the differential equation to show that

$$r^{\frac{3}{2}} = 5.4t + 10$$

(5)

(c) Hence find the radius of the balloon when $t = 20$

Give your answer to the nearest millimetre.

(2)

(d) Suggest a limitation of the model.

(1)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 9 marks)



15. (i) Show that $k^2 - 4k + 5$ is positive for all real values of k .

(2)

(ii) A student was asked to prove by contradiction that

“There are no positive integers x and y such that $(3x + 2y)(2x - 5y) = 28$ ”

The start of the student’s proof is shown below.

Assume that positive integers x and y exist such that
 $(3x + 2y)(2x - 5y) = 28$

If $3x + 2y = 14$ and $2x - 5y = 2$

$$\left. \begin{array}{l} 3x + 2y = 14 \\ 2x - 5y = 2 \end{array} \right\} \Rightarrow x = \frac{74}{19}, y = \frac{22}{19} \text{ Not integers}$$

Show the calculations and statements needed to complete the proof.

(4)



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Question 15 continued

Handwriting practice area with horizontal lines.



Question 15 continued

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(Total for Question 15 is 6 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Tuesday 11 June 2024

Afternoon (Time: 2 hours)

Paper reference **9MA0/02**

Mathematics

Advanced

PAPER 2: Pure Mathematics 2

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

1.

$$y = 4x^3 - 7x^2 + 5x - 10$$

(a) Find in simplest form

(i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$

(3)

(b) Hence find the exact value of x when $\frac{d^2y}{dx^2} = 0$

(2)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 5 marks)



2. Jamie takes out an interest-free loan of £8100

Jamie makes a payment every month to pay back the loan.

Jamie repays £400 in month 1, £390 in month 2, £380 in month 3, and so on, so that the amounts repaid each month form an arithmetic sequence.

(a) Show that Jamie repays £290 in month 12 (1)

After Jamie's N th payment, the loan is completely paid back.

(b) Show that $N^2 - 81N + 1620 = 0$ (2)

(c) Hence find the value of N . (2)



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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 5 marks)



3. The point $P(3, -2)$ lies on the curve with equation $y = f(x)$, $x \in \mathbb{R}$

Find the coordinates of the point to which P is mapped when the curve with equation $y = f(x)$ is transformed to the curve with equation

(i) $y = f(x - 2)$

(ii) $y = f(2x)$

(iii) $y = 3f(-x) + 5$

(4)



DO NOT WRITE IN THIS AREA

Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 4 marks)



4. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = ku_n - 5$$

$$u_1 = 6$$

where k is a positive constant.

Given that $u_3 = -1$

- (a) show that

$$6k^2 - 5k - 4 = 0 \quad (2)$$

- (b) Hence

- (i) find the value of k ,

- (ii) find the value of $\sum_{r=1}^3 u_r$ (3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



5. Given that θ is small and in radians, use the small angle approximations to find an approximate numerical value of

$$\frac{\theta \tan 2\theta}{1 - \cos 3\theta}$$

(3)

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DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 3 marks)



6.

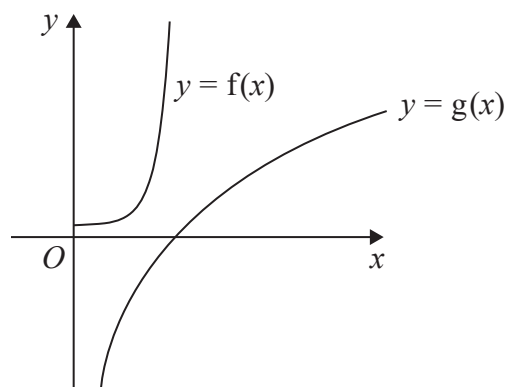


Figure 1

Figure 1 shows a sketch of the curves with equations $y = f(x)$ and $y = g(x)$ where

$$f(x) = e^{4x^2 - 1} \quad x > 0$$

$$g(x) = 8 \ln x \quad x > 0$$

(a) Find

(i) $f'(x)$

(ii) $g'(x)$

(2)

Given that $f'(x) = g'(x)$ at $x = \alpha$

(b) show that α satisfies the equation

$$4x^2 + 2 \ln x - 1 = 0$$

(2)

The iterative formula

$$x_{n+1} = \sqrt{\frac{1 - 2 \ln x_n}{4}}$$

is used with $x_1 = 0.6$ to find an approximate value for α

(c) Calculate, giving each answer to 4 decimal places,

(i) the value of x_2

(ii) the value of α

(3)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 7 marks)



7.

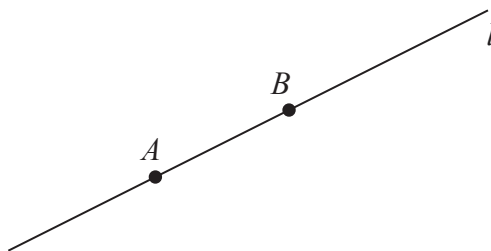


Figure 2

Figure 2 shows a sketch of the straight line l .

Line l passes through the points A and B .

Relative to a fixed origin O

- the point A has position vector $2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$
- the point B has position vector $5\mathbf{i} + 6\mathbf{j} + 8\mathbf{k}$

(a) Find $\vec{AB}$

(1)

Given that a point P lies on l such that

$$|\vec{AP}| = 2|\vec{BP}|$$

(b) find the possible position vectors of P .

(4)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 5 marks)



8.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Prove that

$$\frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1} \equiv 2 \tan \theta \sec \theta \quad \theta \neq (90n)^\circ, n \in \mathbb{Z} \quad (3)$$

(b) Hence solve, for $0 < x < 90^\circ$, the equation

$$\frac{1}{\operatorname{cosec} 2x - 1} + \frac{1}{\operatorname{cosec} 2x + 1} = \cot 2x \sec 2x$$

Give each answer, in degrees, to one decimal place.

(4)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 7 marks)



9.

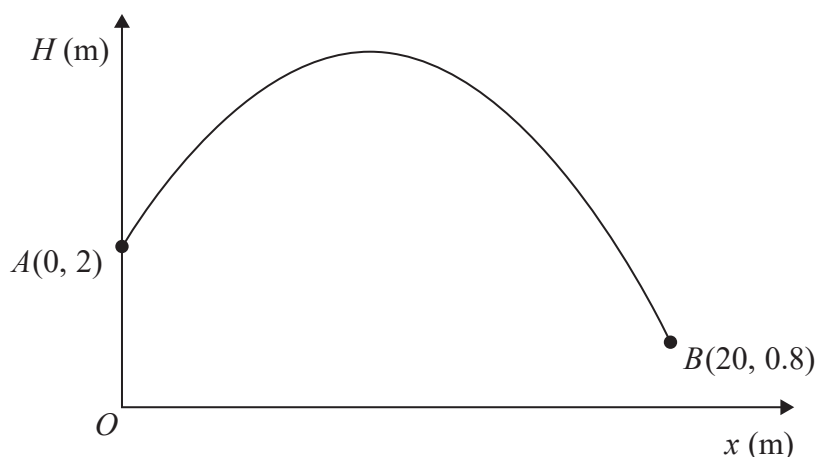


Figure 3

The graph in Figure 3 shows the path of a small ball.

The ball travels in a vertical plane above horizontal ground.

The ball is thrown from the point represented by A and caught at the point represented by B .

The height, H metres, of the ball above the ground has been plotted against the horizontal distance, x metres, measured from the point where the ball was thrown.

With respect to a fixed origin O , the point A has coordinates $(0, 2)$ and the point B has coordinates $(20, 0.8)$, as shown in Figure 3.

The ball reaches its maximum height when $x = 9$

A quadratic function, linking H with x , is used to model the path of the ball.

(a) Find H in terms of x . (4)

(b) Give one limitation of the model. (1)

Chandra is standing directly under the path of the ball at a point 16 m horizontally from O .

Chandra can catch the ball if the ball is less than 2.5 m above the ground.

(c) Use the model to determine if Chandra can catch the ball. (2)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 7 marks)



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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 6 marks)



11.

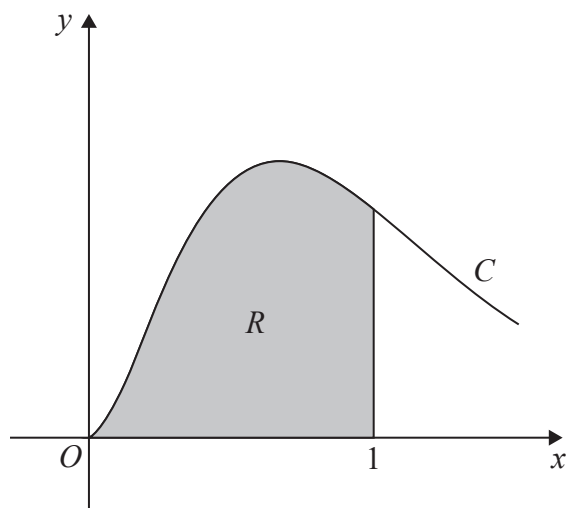


Figure 5

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 5 shows a sketch of part of the curve C with equation

$$y = 8x^2e^{-3x} \quad x \geq 0$$

The finite region R , shown shaded in Figure 5, is bounded by

- the curve C
- the line with equation $x = 1$
- the x -axis

Find the exact area of R , giving your answer in the form

$$A + Be^{-3}$$

where A and B are rational numbers to be found.

(5)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 5 marks)



12. (a) Express $\frac{1}{V(25 - V)}$ in partial fractions. (2)

The volume, V microlitres, of a plant cell t hours after the plant is watered is modelled by the differential equation

$$\frac{dV}{dt} = \frac{1}{10}V(25 - V)$$

The plant cell has an initial volume of 20 microlitres.

- (b) Find, according to the model, the time taken, in minutes, for the volume of the plant cell to reach 24 microlitres. (5)
- (c) Show that

$$V = \frac{A}{e^{-kt} + B}$$

where A , B and k are constants to be found. (3)

The model predicts that there is an upper limit, L microlitres, on the volume of the plant cell.

- (d) Find the value of L , giving a reason for your answer. (2)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 12 marks)



13. The world human population, P billions, is modelled by the equation

$$P = ab^t$$

where a and b are constants and t is the number of years after 2004

Using the estimated population figures for the years from 2004 to 2007, a graph is plotted of $\log_{10} P$ against t .

The points lie approximately on a straight line with

- gradient 0.0054
- intercept 0.81 on the $\log_{10} P$ axis

(a) Estimate, to 3 decimal places, the value of a and the value of b .

(4)

In the context of the model,

- (b) (i) interpret the value of the constant a ,
(ii) interpret the value of the constant b .

(2)

(c) Use the model to estimate the world human population in 2030

(2)

(d) Comment on the reliability of the answer to part (c).

(1)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Handwriting practice area with horizontal lines.

(Total for Question 13 is 9 marks)



14. The circle C_1 has equation

$$x^2 + y^2 - 6x + 14y + 33 = 0$$

(a) Find

- (i) the coordinates of the centre of C_1
- (ii) the radius of C_1

(3)

A different circle C_2

- has centre with coordinates $(-6, -8)$
- has radius k , where k is a constant

Given that C_1 and C_2 intersect at 2 distinct points,

(b) find the range of values of k , writing your answer in set notation.

(5)



DO NOT WRITE IN THIS AREA

Question 14 continued

Handwriting practice area with horizontal lines.



Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 8 marks)



15. The curve C has equation

$$(x + y)^3 = 3x^2 - 3y - 2$$

- (a) Find an expression for $\frac{dy}{dx}$ in terms of x and y .

(5)

The point $P(1, 0)$ lies on C .

- (b) Show that the normal to C at P has equation

$$y = -2x + 2$$

(2)

- (c) Prove that the normal to C at P does **not** meet C again.

You should use algebra for your proof and make your reasoning clear.

(5)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 12 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Wednesday 4 June 2025

Afternoon (Time: 2 hours)

Paper reference **9MA0/01**

Mathematics

Advanced

PAPER 1: Pure Mathematics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

1. The point $P(6, -4)$ lies on the curve with equation $y = f(x)$, $x \in \mathbb{R}$

Find the point to which P is mapped when the curve with equation $y = f(x)$ is transformed to the curve with equation

(a) $y = f(x + 2)$ (1)

(b) $y = f^{-1}(x)$ (1)

(c) $y = 2|f(x)| - 3$ (2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. The circle C has equation

$$(x + 3)^2 + (y - 4)^2 = 24$$

(a) (i) State the coordinates of the centre of C .

(ii) Find the radius of C writing your answer as a fully simplified surd.

(3)

(b) Determine, giving a reason, whether or not the origin lies inside the circle C .

(2)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. The first three terms of an arithmetic sequence are

$$6k, 10 \text{ and } 2k$$

where k is a constant.

- (a) Find the value of k .

(2)

- (b) Hence find the value of the sum of the first 50 terms of this sequence.

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4. Given that

- $f(x) = 2x^3 + 3x^2 - 16x + 16$
- $f(-4) = 0$

(a) write $f(x)$ in the form

$$(x + a)Q(x)$$

where a is a constant and $Q(x)$ is a quadratic expression.

(3)

(b) Hence prove that -4 is the only real root of the equation

$$f(x) = 0$$

(Solutions relying on calculator technology are not acceptable.)

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



5. (a) Express $\lim_{\delta x \rightarrow 0} \sum_{x=1.44}^{2.89} \frac{2}{\sqrt{x}} \delta x$ as an integral.

(1)

- (b) Hence show that

$$\lim_{\delta x \rightarrow 0} \sum_{x=1.44}^{2.89} \frac{2}{\sqrt{x}} \delta x = k$$

where k is an integer to be found.

(Solutions relying on calculator technology are not acceptable.)

(2)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 3 marks)



6. A scientist is monitoring the flight of sea birds after they leave their nests on a cliff.

The height above the sea, h metres, of one of the birds is modelled by the equation

$$h = A - Bt^{1.5} \quad h \geq 0 \quad 0 \leq t \leq T$$

where t seconds is the time after the bird leaves its nest and A , B and T are positive constants.

Given that

- the bird was 17.6m above the sea exactly 4 seconds after leaving its nest
- the bird was 11.9m above the sea exactly 9 seconds after leaving its nest

(a) find a complete equation for the model.

(4)

Find, according to the model,

(b) the height of the bird's nest above the sea,

(1)

(c) the limitation on the value of T .

(2)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

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(Total for Question 6 is 7 marks)



7.

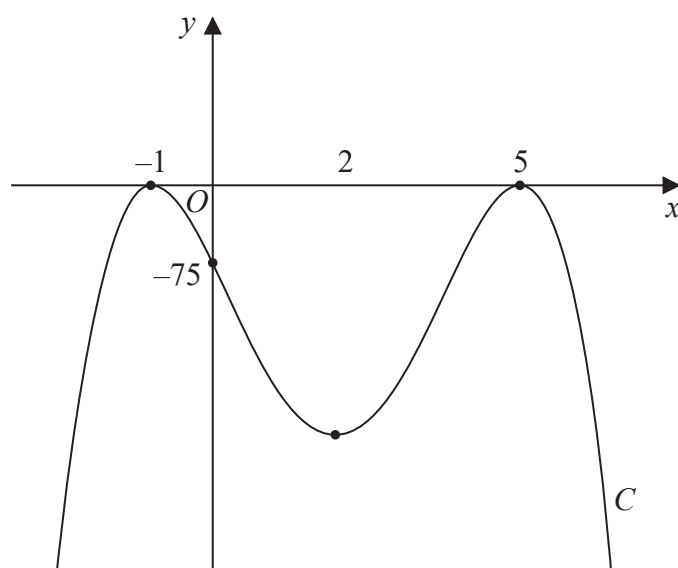


Figure 1

Figure 1 shows a sketch of a curve C with equation $y = f(x)$, where $f(x)$ is a quartic expression in x .

The curve

- has maximum turning points at $(-1, 0)$ and $(5, 0)$
- crosses the y -axis at $(0, -75)$
- has a minimum turning point at $x = 2$

(a) Find the set of values of x for which

$$f'(x) \geq 0$$

writing your answer in set notation.

(2)

(b) Find the equation of C . You may leave your answer in factorised form.

(3)

The curve C_1 has equation $y = f(x) + k$, where k is a constant.

Given that the graph of C_1 intersects the x -axis at exactly four places,

(c) find the range of possible values for k .

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 7 marks)



8. A student was asked to prove that the square of any number can be expressed in the form $5n$ or $5n \pm 1$ where $n \in \mathbb{N}$

The start of the student's proof is shown in the box below.

Let $m = 5k$

Consider $m^2 = (5k)^2 = 25k^2 = 5 \times 5k^2 = 5n$ where $n = 5k^2$ ✓

Let $m = 5k + 1$

Consider $m^2 = (5k + 1)^2$

$$= 25k^2 + 10k + 1 = 5(5k^2 + 2k) + 1 = 5n + 1 \quad \text{where } n = 5k^2 + 2k \quad \checkmark$$

Let $m = 5k + 2$

Consider $m^2 = (5k + 2)^2$

$$= 25k^2 + 10k + 4 = 5(5k^2 + 2k + 1) - 1 = 5n - 1 \quad \text{where } n = 5k^2 + 2k + 1 \quad \checkmark$$

- (a) Identify and correct an algebraic error in the box above.

(1)

- (b) Show the calculations and statements that are required to complete the proof.

(4)



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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 5 marks)



9.

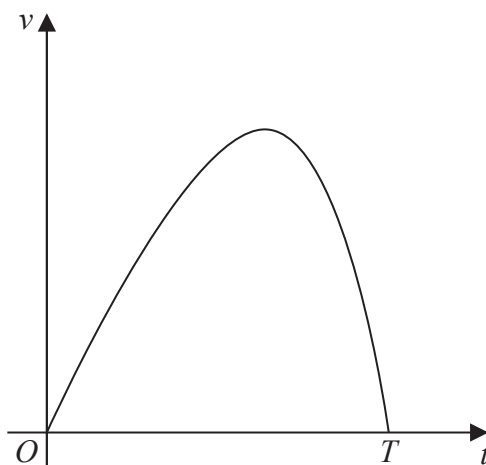


Figure 2

A racing car is driven along a straight road.

Figure 2 shows a graph of the speed of the car as it travels along the road.

The car starts from rest and is driven for T seconds before stopping.

The speed of the car is modelled by the equation

$$v = 15t - te^{0.2t} \quad 0 \leq t \leq T$$

where t seconds is the time after the car starts to move.

According to the model,

(a) find the value of T , giving your answer to one decimal place, (2)

(b) show that the maximum speed of the car occurs when

$$t = 5 \ln \left(\frac{75}{t + 5} \right) \quad (4)$$

Using the iteration formula

$$t_{n+1} = 5 \ln \left(\frac{75}{t_n + 5} \right) \quad \text{with } t_1 = 8$$

(c) (i) find the value of t_3 to 3 decimal places,
 (ii) find, by repeated iteration, the time taken for the car to reach maximum speed. (3)



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Question 9 continued

Handwriting practice area with horizontal lines.



Question 9 continued

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DO NOT WRITE IN THIS AREA

Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 9 marks)



10. Given that

• $\vec{PQ} = 2\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}$

• $\vec{QR} = 6\mathbf{i} + 6\mathbf{k}$

(a) find $\vec{PR}$.

(2)

(b) Hence show that triangle PQR is both right-angled and isosceles.

(4)



DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 6 marks)



11.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The value of a car, £ V , is modelled by the equation

$$V = 1500 + Ae^{-kt}$$

where A and k are positive constants and t is the age of the car in years.

Given that

- the initial value of the car was £20 000
 - the value of the car was £12 000 when it was 2.5 years old
- (a) find a complete equation for the model, giving the exact value of A and the value of k to 3 significant figures.
- (b) Show that the rate of change in the value of the car can be expressed in the form
- $$-k(V - 1500)$$
- (c) State a limitation of this model.

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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 8 marks)



12.

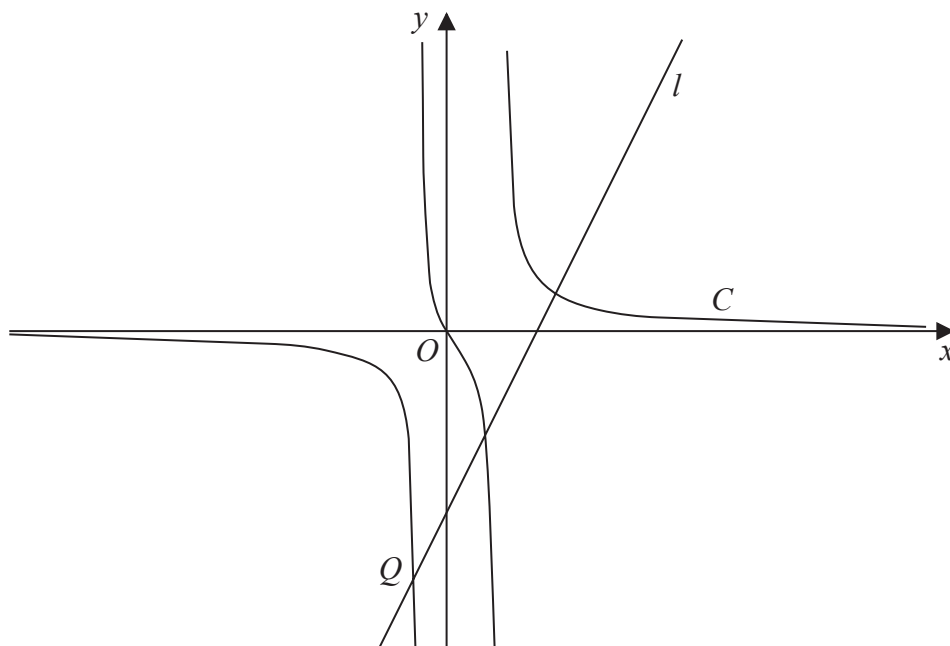


Figure 3

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{15x}{(2x+3)(x-3)} \quad x \neq -\frac{3}{2} \quad x \neq 3$$

and the straight line l with equation

$$y = 2x - 10$$

(a) Verify that C and l intersect where $x = 6$

(2)

The curve and line also intersect at the point Q shown in Figure 3.

(b) Show that the x coordinate of Q is a solution of

$$4x^3 - 26x^2 - 3x + 90 = 0$$

(2)

(c) Using algebra and showing all stages of working, find the exact x coordinate of Q .

(4)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 8 marks)



13.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

Show that

$$\int_0^2 \frac{x}{(2x+1)^3} dx = \frac{2}{25}$$

(5)



DO NOT WRITE IN THIS AREA

Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 5 marks)



14.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Show that

$$\sin(x + 30^\circ) + \sqrt{3} \cos(x + 30^\circ) \equiv 2 \cos x \quad (3)$$

(b) Hence solve, for $0 \leq \theta < 180^\circ$

$$\sin(\theta + 30^\circ) + \sqrt{3} \cos(\theta + 30^\circ) = 3 \sin 2\theta$$

giving your answers to one decimal place, where appropriate. (4)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 7 marks)



15.

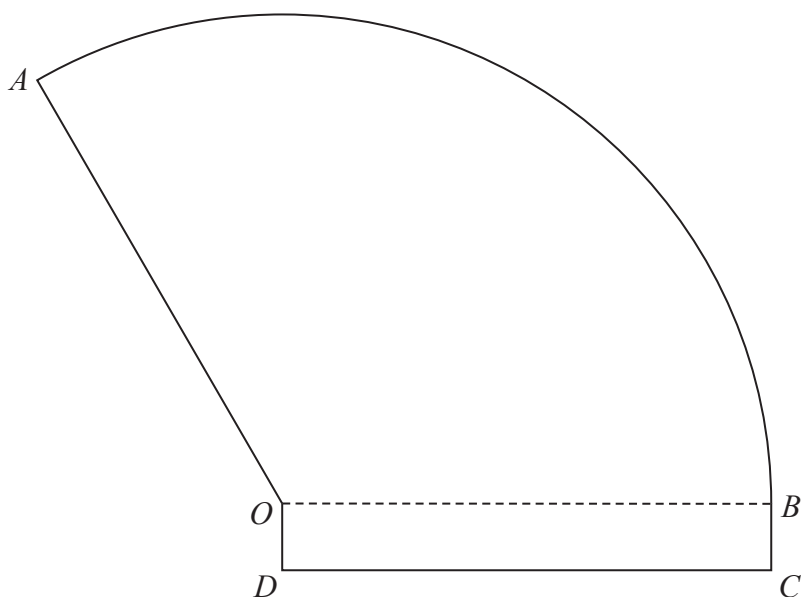


Figure 4

Figure 4 shows the plan view for the design of a stage.

The shape of this design consists of a sector of a circle AOB joined to a rectangle $OBCD$.

Given that

- the radius of the sector is r metres and angle AOB is θ radians
- the length and width of the rectangle are r metres and $\frac{1}{10}r$ metres respectively
- the total area of the stage is 240 m^2

(a) show that the perimeter of the stage, P metres, is given by

$$P = 2r + \frac{480}{r}$$

You must make your method clear.

(4)

Using algebraic differentiation,

(b) find the value of r for which P has a stationary value.

(3)

(c) Prove, by further differentiation, that this value of r gives the minimum perimeter of the stage.

(2)



DO NOT WRITE IN THIS AREA

Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Handwriting practice area with horizontal lines.

(Total for Question 15 is 9 marks)



16.

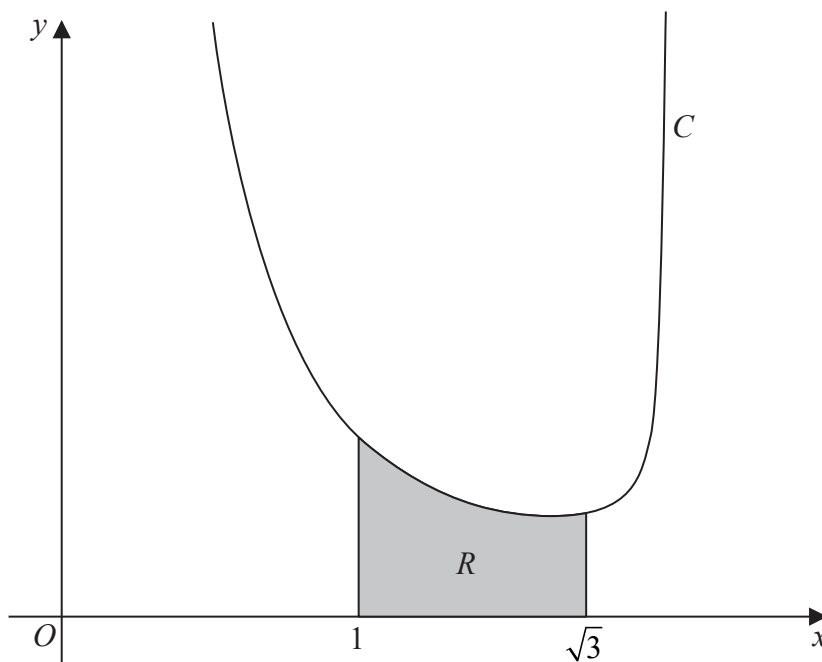


Figure 5

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{1}{x^2 \sqrt{4 - x^2}} \quad 0 < x < 2$$

The region R , shown shaded in Figure 5, is bounded by C , the line with equation $x = 1$, the x -axis and the line with equation $x = \sqrt{3}$

- (a) Use the substitution $x = 2 \sin u$ to show that the area of R is given by

$$\int_a^b k \operatorname{cosec}^2 u \, du$$

where a , b and k are constants to be found.

(4)

- (b) Hence, using algebraic integration, find the exact area of R .
Give your answer in simplest form.

(3)



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Question 16 continued

Handwriting practice area with horizontal lines.



Question 16 continued

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(Total for Question 16 is 7 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Thursday 12 June 2025

Afternoon (Time: 2 hours)

Paper reference **9MA0/02**

Mathematics

Advanced

PAPER 2: Pure Mathematics 2

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

Factorise completely

$$2x^3 - 24x^2 + 40x$$

(3)



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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 3 marks)



2. Find

$$\int \left(x^4 - 6x^{\frac{1}{2}} - 3 \right) dx$$

giving the answer in simplest form.

(4)



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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 4 marks)



3. Given that

$$3^x = 7^y$$

find the exact value of $\frac{x}{y}$

(2)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 2 marks)



4.

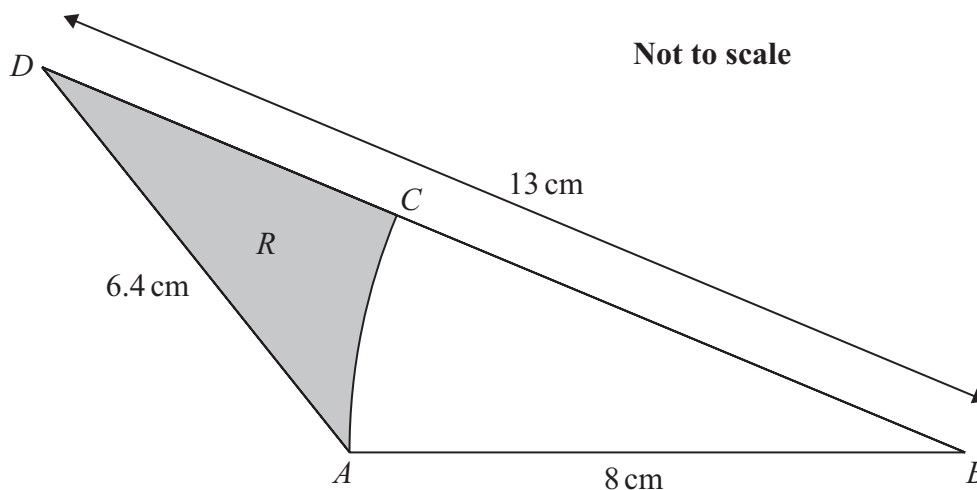


Figure 1

The shape $ABCD$, shown in Figure 1, consists of a triangle ABD containing a sector ABC of a circle with centre B .

Given that

- $AD = 6.4$ cm
- $BD = 13$ cm
- $BA = BC = 8$ cm

(a) show that angle $ABC = 0.394$ radians to 3 significant figures.

(2)

The region R , shown shaded in Figure 1, is bounded by the line CD , the line DA and the arc AC .

(b) Find the area of R , giving the answer in cm^2 to 3 significant figures.
You must make your method clear.

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 5 marks)





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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



6.

In this question you must show detailed reasoning.

- (a) Given that x is small and in radians, use the small angle approximation for $\cos \theta$ to show that

$$1 - \cos^2(2x) \approx 4x^2 - 4x^4 \quad (2)$$

- (b) Given that x is small and in radians, use

- the answer to part (a)
- the small angle approximations for $\sin \theta$ and $\tan \theta$

to show that

$$\frac{1 - \cos^2(2x)}{\sin\left(\frac{x}{3}\right)\tan\left(\frac{x}{2}\right)} \approx a + bx^2$$

where a and b are constants to be found. (2)

- (c) Hence, given that x is **very** small, deduce an approximate value for

$$\frac{1 - \cos^2(2x)}{\sin\left(\frac{x}{3}\right)\tan\left(\frac{x}{2}\right)}$$

giving a reason for your answer. (2)

[illegible]

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7. Given that

$$\frac{3x^3 - 8x^2 - 6x - 11}{(x+1)(x-3)} \equiv Ax + B + \frac{C}{x+1} + \frac{D}{x-3} \quad x \in \mathbb{R} \quad x \neq -1, 3$$

find the value of each of the constants A , B , C and D .

(4)



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Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 4 marks)



8.

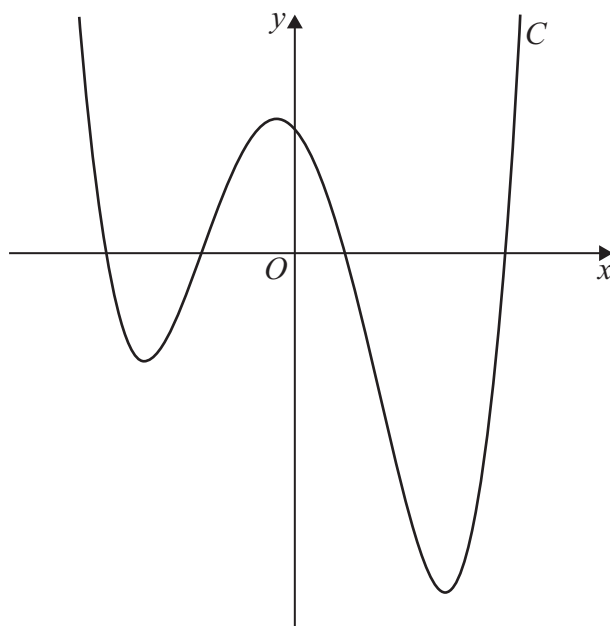


Figure 2

$$f(x) = x^4 + \frac{1}{3}x^3 - 8x^2 + ax + \frac{17}{3}$$

where a is a constant.

Figure 2 shows a sketch of the curve C with equation $y = f(x)$

Given that C has a local maximum at $x = -\frac{1}{4}$

(a) show that $a = -4$ (4)

(b) find the exact y coordinate of the local maximum. (1)

The equation $f(x) = k$, where k is a constant, has 4 distinct solutions.

(c) Using algebra and showing all stages of your working, find the range of values of k .
Give the answer using set notation.

(Solutions relying on calculator technology are not acceptable.)

(4)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 9 marks)



9. In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

A new type of car is released for sale.

The total number of this type of car sold, N , in a particular region, t months after the cars were released for sale, is modelled by the equation

$$N = 5000 - 5000e^{-0.075t} \quad t \geq 0$$

Use the equation of the model to answer parts (a), (b), (c) and (d).

- (a) Find the total number of cars sold in the first 3 months. (2)

Given that $N = 3000$ when $t = T$

- (b) find the value of T giving the answer to 2 decimal places. (3)

- (c) Find the rate of increase in the total number of cars sold when $t = 3$, giving the answer to 3 significant figures. (2)

After a marketing campaign, the total number of cars sold is expected to rise and have an upper limit of 6500

- (d) Using this information, suggest **one** refinement to the model. (1)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 8 marks)





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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 9 marks)



11. A company is trying to determine the most profitable selling price for a new toy.

Given that

- if the selling price of each toy is £30, the company expects to sell 1500 toys in one year
- if the selling price of each toy is £50, the company expects to sell 300 toys in one year

Using a **linear** model, with y being the expected number of toys sold in one year and x pounds being the selling price of the toy,

- (a) find an equation for y in terms of x . (3)

Given that

- the cost of making each toy is £10
 - the company has additional costs of £8 000 per year
- (b) show that, according to the model, the yearly profit, P , in **thousands** of pounds, is given by

$$P = -0.06x^2 + 3.9x - 41 \quad (3)$$

Use the model given in part (b) to answer parts (c), (d) and (e).

Given that the company wishes to make a profit on sales of the toy,

- (c) find the range of possible selling prices of the toy. (2)
- (d) Hence, or otherwise, deduce the selling price of the toy that maximises the profit. (1)

In one particular year, the company sold the toy for £35 and made £21 750 profit.

- (e) Use this information to evaluate the suitability of the model. (2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 11 marks)



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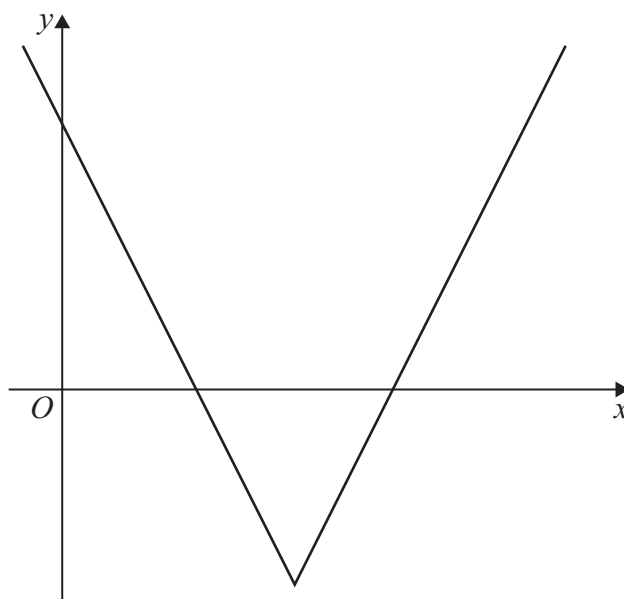


Figure 3

Figure 3 shows a sketch of the graph with equation $y = f(x)$ where

$$f(x) = 4|x - 3| - 5 \quad x \in \mathbb{R}$$

Given that a is a constant and $|a| = 1$

- (a) find the possible values of $f(a)$ (2)

The function g is defined by

$$g(x) = 2x + 17 \quad x \in \mathbb{R}$$

- (b) Find the range of $g \circ f(x)$ (2)

The function h is defined by

$$h(x) = kx \quad x \in \mathbb{R}$$

where k is a constant.

Given that the equation $f(x) = h(x)$ has no solutions,

- (c) find the range of values of k . (4)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 8 marks)



13. Given that for all values of k , where $0 < |k| < 1$, the equation

$$\sin(nx) = k \quad n \in \mathbb{N}$$

has exactly 6 solutions in the interval $0 \leq x < 2\pi$

(i) deduce the value of n

(1)

(ii) deduce the number of solutions of the equation

$$\sin^2(nx) = k^2$$

in the interval $0 \leq x < 5\pi$, justifying your answer.

(2)



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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 3 marks)



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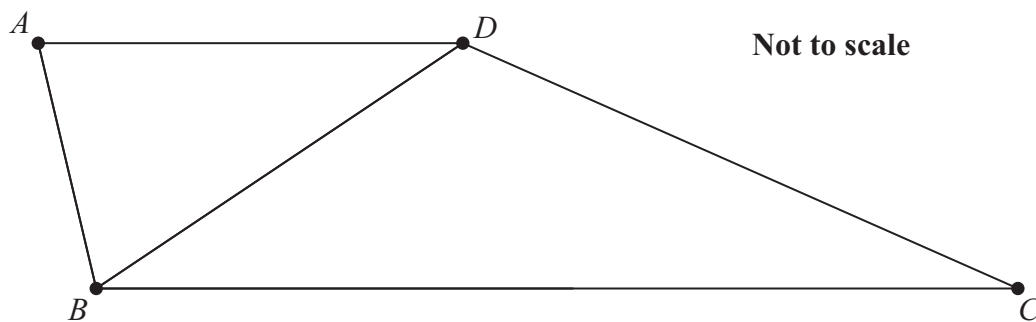


Figure 4

In this question you must show detailed reasoning.

Figure 4 shows a trapezium $ABCD$ where AD is parallel to BC

Given that

- $\vec{AB} = 2\mathbf{a} + 3\mathbf{b}$
- $\vec{BC} = 15\mathbf{a} - 5\mathbf{b}$
- $\vec{DB} = -4\mathbf{a} + k\mathbf{b}$ where k is an integer

(a) show that $k = 5$

(3)

Given also that

- the point N lies on BC such that $BN : NC = 1 : 4$
- AN intersects BD at X

(b) find $BX : XD$

(5)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 8 marks)



15.

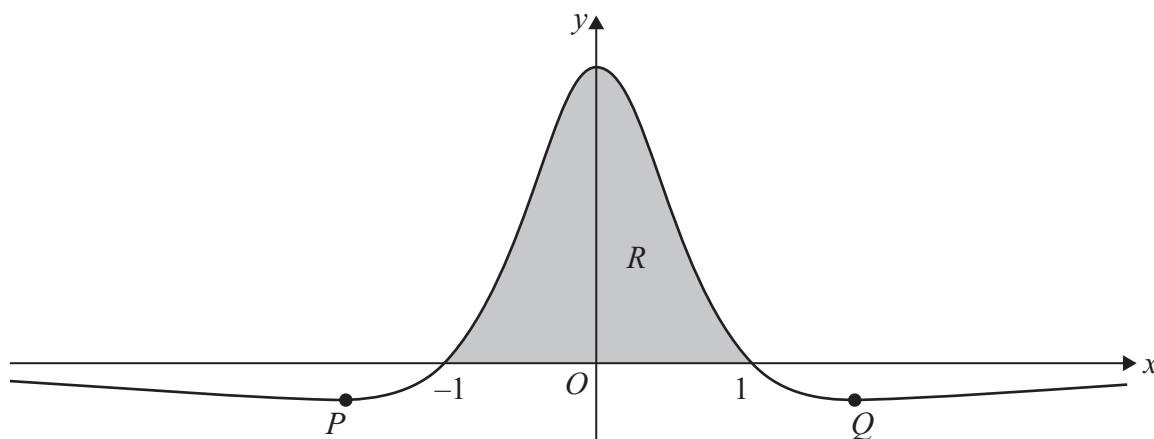


Figure 5

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Figure 5 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = \frac{1 - x^2}{(1 + x^2)^2}$$

The curve

- intersects the x -axis at -1 and 1
- has minimum turning points at P and Q

as shown in Figure 5.

(a) Use calculus to find the exact coordinates of P .

(5)

(b) Using the substitution $x = \tan \theta$ show that

$$\int_{-1}^1 f(x) dx = \int_{\alpha}^{\beta} \cos 2\theta d\theta$$

where α and β are constants to be found.

(5)

The finite region R , shown shaded in Figure 5, is bounded by the x -axis and the curve.

(c) Use algebraic integration to find the area of R .

(3)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Handwriting practice area with horizontal lines.



Question 15 continued

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(Total for Question 15 is 13 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Mock Paper Set 1			
(Time: 2 hours)		Paper Reference 9MA0/01	
Mathematics Advanced Paper 1: Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

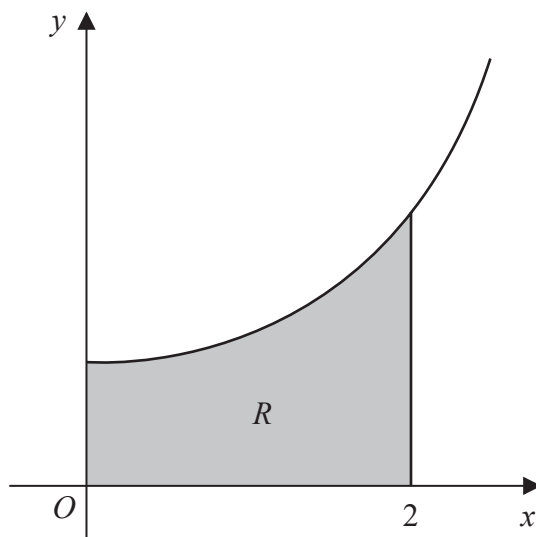


Figure 1

Figure 1 shows part of the curve with equation $y = e^{\frac{1}{5}x^2}$ for $x \geq 0$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the y -axis, the x -axis, and the line with equation $x = 2$

The table below shows corresponding values of x and y for $y = e^{\frac{1}{5}x^2}$

x	0	0.5	1	1.5	2
y	1	$e^{0.05}$	$e^{0.2}$	$e^{0.45}$	$e^{0.8}$

- (a) Use the trapezium rule, with all the values of y in the table, to find an estimate for the area of R , giving your answer to 2 decimal places.

(3)

- (b) Use your answer to part (a) to deduce an estimate for

(i) $\int_0^2 \left(4 + e^{\frac{1}{5}x^2}\right) dx$

(ii) $\int_1^3 e^{\frac{1}{5}(x-1)^2} dx$

giving your answers to 2 decimal places.

(2)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 5 marks)



2.

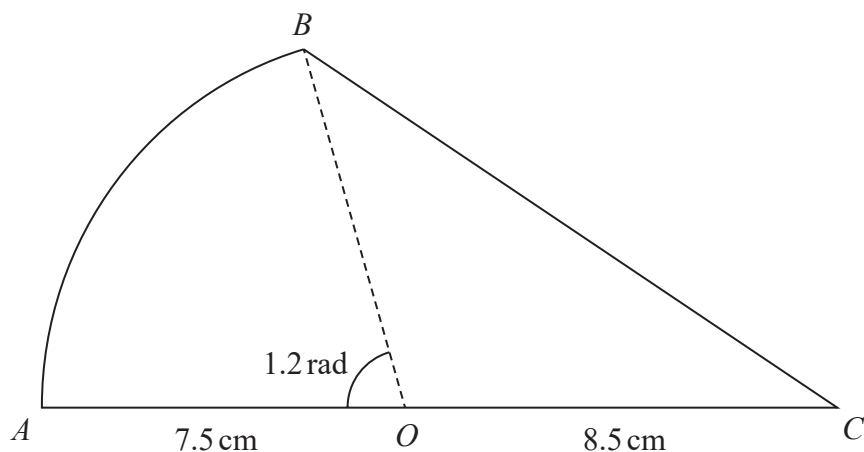


Figure 2

The shape $AOCBA$, shown in Figure 2, consists of a sector AOB of a circle centre O joined to a triangle BOC .

The points A , O and C lie on a straight line with $AO = 7.5$ cm and $OC = 8.5$ cm.

The size of angle AOB is 1.2 radians.

Find, in cm, the perimeter of the shape $AOCBA$, giving your answer to one decimal place.

(5)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. The equation

$$3x^2 + k = 5x + 2 \quad k \in \mathbb{R}$$

where k is a constant, has no real roots.

Find the range of possible values for k .

(4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 10 marks)



5. A curve has equation

$$y = 4x^2 - 5x$$

The curve passes through the point $P(2, 6)$

Use differentiation from first principles to find the value of the gradient of the curve at P .

(5)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



6.

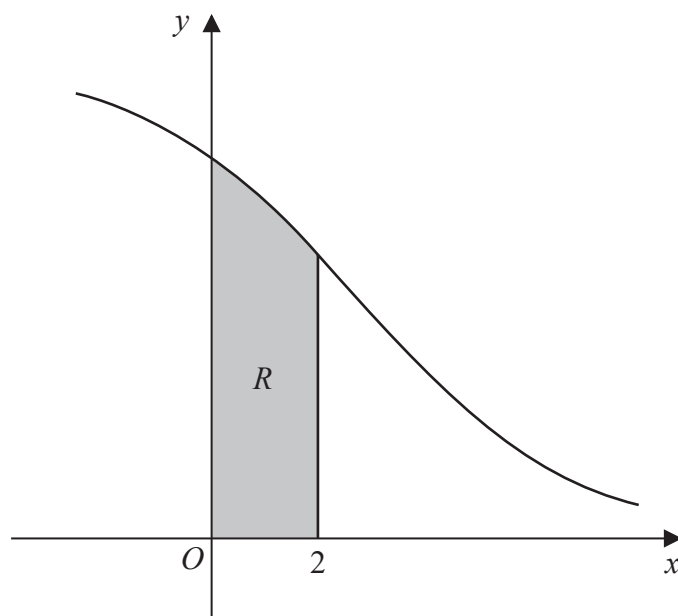


Figure 3

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{6}{e^{\frac{1}{2}x} + 4} \quad x \in \mathbb{R}$$

The finite region R , shown shaded in Figure 3, is bounded by the curve, the y -axis, the x -axis, and the line with equation $x = 2$

- (a) Use the substitution $u = e^{\frac{1}{2}x}$ to show that the area of R can be given by

$$\int_a^b \frac{12}{u(u+4)} du$$

where a and b are constants to be found.

(3)

- (b) Hence use algebraic integration to show that the exact area of R is $3 \ln \left(\frac{5e}{e+4} \right)$

(5)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 8 marks)



7. (a) Express $3 \sin \theta - 4 \cos \theta$ in the form $R \sin(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$.
State the value of R and give the value of α to 2 decimal places. (3)

The temperature in a greenhouse, $G^\circ\text{C}$, is modelled by the equation

$$G = 17 + 3 \sin(15t)^\circ - 4 \cos(15t)^\circ \quad 0 \leq t \leq 17$$

where t is the time in hours after 5 a.m.

- (b) Find, according to this model,
- (i) the maximum temperature in the greenhouse, (1)
- (ii) the time, after midday, when the temperature in the greenhouse is 20°C .
Give your answer to the nearest minute.
(Solutions based entirely on graphical or numerical methods are not acceptable.) (4)



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Question 7 continued

Lined area for writing the answer to Question 7.



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Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8. (i) Show that $y^2 - 4y + 7$ is positive for all real values of y . (2)

(ii) Bobby claims that

$$e^{3x} \geq e^{2x} \quad x \in \mathbb{R}$$

Determine whether Bobby's claim is always true, sometimes true or never true, justifying your answer.

(iii) Elsa claims that

‘for $n \in \mathbb{Z}^+$, if n^2 is even, then n must be even’

Use proof by contradiction to show that Elsa's claim is true.

(iv) Ying claims that

‘the sum of two different irrational numbers is irrational’

Determine whether Ying's claim is always true, sometimes true or never true, justifying your answer.



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 8 marks)



9. (a) Show that

$$\frac{\sin x}{1 - \cos x} + \frac{1 - \cos x}{\sin x} \equiv k \operatorname{cosec} x \quad x \neq n\pi, \quad n \in \mathbb{Z}$$

where k is a constant to be found.

(4)

(b) Hence explain why the equation

$$\frac{\sin x}{1 - \cos x} + \frac{1 - \cos x}{\sin x} = 1.6$$

has no real solutions.

(1)



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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 5 marks)



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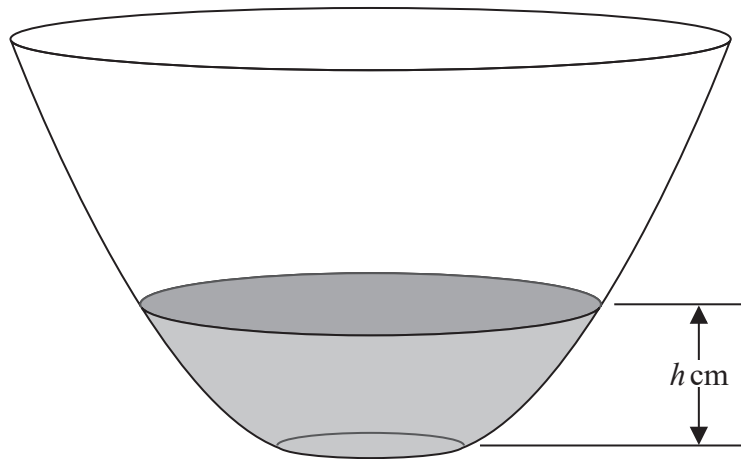


Figure 4

Figure 4 shows a bowl with a circular cross-section.

Initially the bowl is empty. Water begins to flow into the bowl.

At time t seconds after the water begins to flow into the bowl, the height of the water in the bowl is h cm.

The volume of water, $V \text{ cm}^3$, in the bowl is modelled as

$$V = 4\pi h(h + 6) \quad 0 \leq h \leq 25$$

The water flows into the bowl at a constant rate of $80\pi \text{ cm}^3 \text{ s}^{-1}$

- (a) Show that, according to the model, it takes 36 seconds to fill the bowl with water from empty to a height of 24 cm. (1)
- (b) Find, according to the model, the rate of change of the height of the water, in cm s^{-1} , when $t = 8$ (8)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 9 marks)



11. (i) Given that

$$y = a^x$$

where a is a positive constant, show that

$$\frac{dy}{dx} = a^x \ln a \quad (2)$$

(ii) Given that

$$x = 2 \tan y \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$

show that

$$\frac{dy}{dx} = \frac{k}{4 + x^2} \quad (4)$$

where k is a constant to be found.



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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 6 marks)



12.

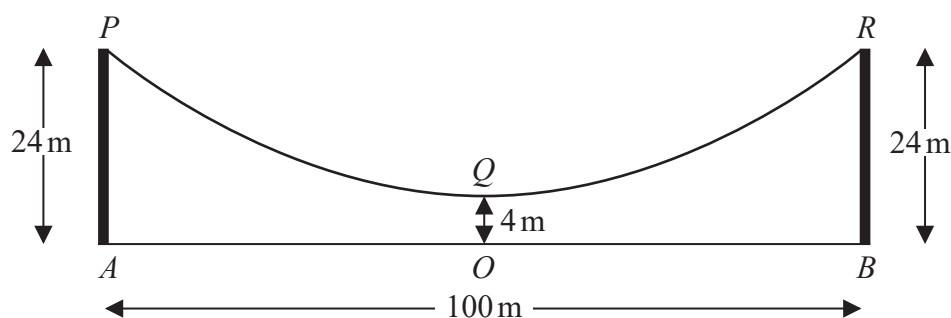


Figure 5

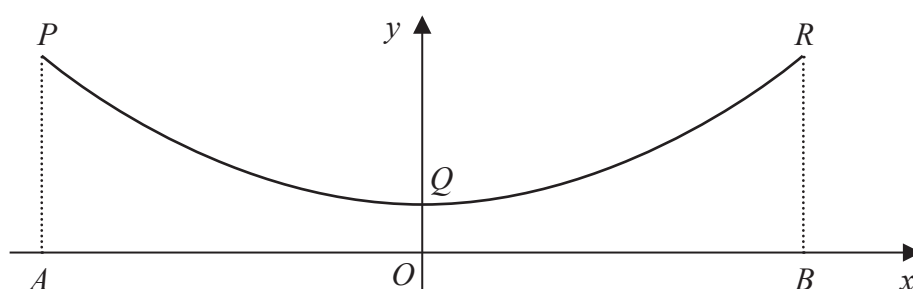


Figure 6

A suspension bridge cable PQR hangs between the tops of two vertical towers, AP and BR , as shown in Figure 5.

A walkway AOB runs between the bases of the towers, directly under the cable.

The towers are 100m apart and each tower is 24m high.

At the point O , midway between the towers, the cable is 4m above the walkway.

The points P , Q , R , A , O and B are assumed to lie in the same vertical plane and AOB is assumed to be horizontal.

Figure 6 shows a symmetric quadratic curve PQR used to model this cable.

Given that O is the origin,

- (a) find an equation for the curve PQR . (3)

Lee can safely inspect the cable up to a height of 12m above the walkway.

A defect is reported on the cable at a location 19m horizontally **from one of the towers**.

- (b) Determine whether, according to the model, Lee can safely inspect this defect. (2)

- (c) Give a reason why this model may not be suitable to determine whether Lee can safely inspect this defect. (1)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 6 marks)



13. Given that p is a positive constant,

(a) show that

$$\sum_{n=1}^{11} \ln(p^n) = k \ln p$$

where k is a constant to be found,

(2)

(b) show that

$$\sum_{n=1}^{11} \ln(8p^n) = 33 \ln(2p^2)$$

(2)

(c) Hence find the set of values of p for which

$$\sum_{n=1}^{11} \ln(8p^n) < 0$$

giving your answer in set notation.

(2)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 6 marks)



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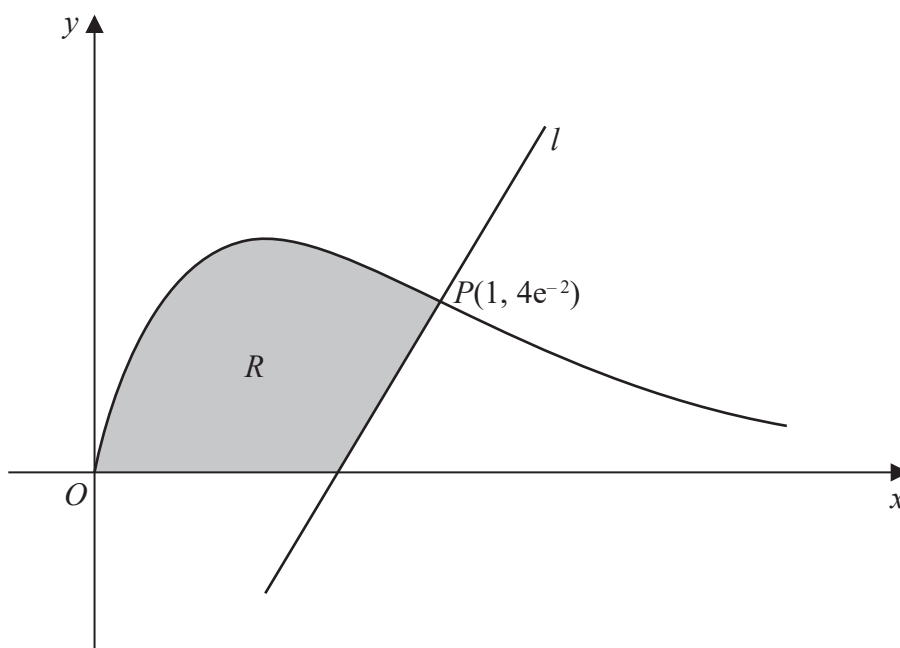


Figure 7

Figure 7 shows a sketch of the curve with equation

$$y = 4xe^{-2x} \quad x \geq 0$$

The line l is the normal to the curve at the point $P(1, 4e^{-2})$

The finite region R , shown shaded in Figure 7, is bounded by the curve, the line l , and the x -axis.

Find the exact value of the area of R .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(10)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 10 marks)



15. Relative to a fixed origin O , the points A and B are such that

$$\vec{OA} = \begin{pmatrix} -3 \\ 2 \\ 7 \end{pmatrix} \text{ and } \vec{OB} = \begin{pmatrix} 3 \\ -1 \\ p \end{pmatrix}, \text{ where } p \text{ is a constant}$$

and the points C and D are such that

$$\vec{BC} = \begin{pmatrix} 0 \\ 6 \\ -7 \end{pmatrix} \text{ and } \vec{AD} = \begin{pmatrix} 2 \\ 5 \\ -4 \end{pmatrix}$$

(a) Find the position vector of the point D .

(1)

Given that $ABCD$ is a trapezium,

(b) find the value of p .

(4)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	
		Candidate Number	
Mock Paper Set 1			
(Time: 2 hours)		Paper Reference 9MA0/02	
Mathematics Advanced Paper 2: Pure Mathematics 2			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. (a) Given that θ is small and in radians, show that the equation

$$\cos \theta - \sin \left(\frac{1}{2} \theta \right) + 2 \tan \theta = \frac{11}{10} \quad (\text{I})$$

can be written as

$$5\theta^2 - 15\theta + 1 \approx 0 \quad (3)$$

The solutions of the equation

$$5\theta^2 - 15\theta + 1 = 0$$

are 0.068 and 2.932, correct to 3 decimal places.

- (b) Comment on the validity of each of these values as approximate solutions to equation (I).

(1)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. A curve has parametric equations

$$x = 6t + 1 \quad y = 5 - \frac{4}{3t} \quad t \neq 0$$

Show that the Cartesian equation of the curve can be expressed in the form

$$y = \frac{ax + b}{x - 1} \quad x \neq k$$

where a , b and k are constants to be found.

(3)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 3 marks)



3. A curve has equation

$$y = x^2 + kx + 14 - \frac{8}{(x - 5)}$$

where k is a constant.

Given that the curve has a stationary point P , where $x = 3$

- (a) show that $k = -8$ (4)
- (b) Determine the nature of the stationary point P , giving a reason for your answer. (2)
- (c) Show that the curve has a point of inflection where $x = 7$ (2)

The curve passes through the points $(4.5, 14.25)$ and $(5.5, -15.75)$

Jane uses this information to write down the following

As there is a change of sign, the curve cuts the x -axis in the interval $(4.5, 5.5)$

- (d) Explain the error in Jane's reasoning. (1)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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(Total for Question 3 is 9 marks)



4.

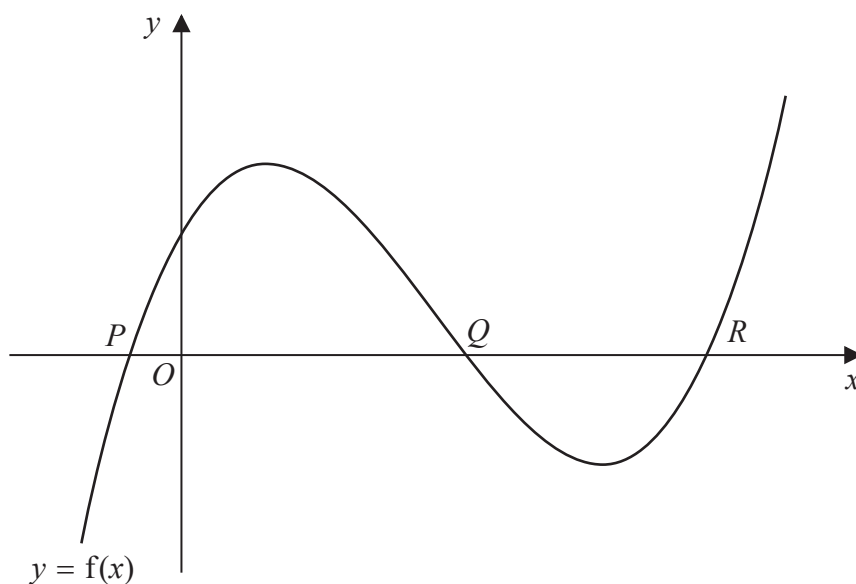


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = x^3 - 6x^2 + 7x + 2 \quad x \in \mathbb{R}$$

The curve cuts the x -axis at the points P , Q and R , as shown in Figure 1.

The coordinates of Q are $(2, 0)$

(a) Write $f(x)$ as a product of two algebraic factors.

(2)

(b) Find, giving your answer in simplest form,

(i) the exact x coordinate of P ,

(ii) the exact x coordinate of R .

(2)

(c) Deduce the number of real solutions, for $-\pi \leq \theta \leq 12\pi$, to the equation

$$\sin^3 \theta - 6 \sin^2 \theta + 7 \sin \theta + 2 = 0$$

justifying your answer.

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



5.

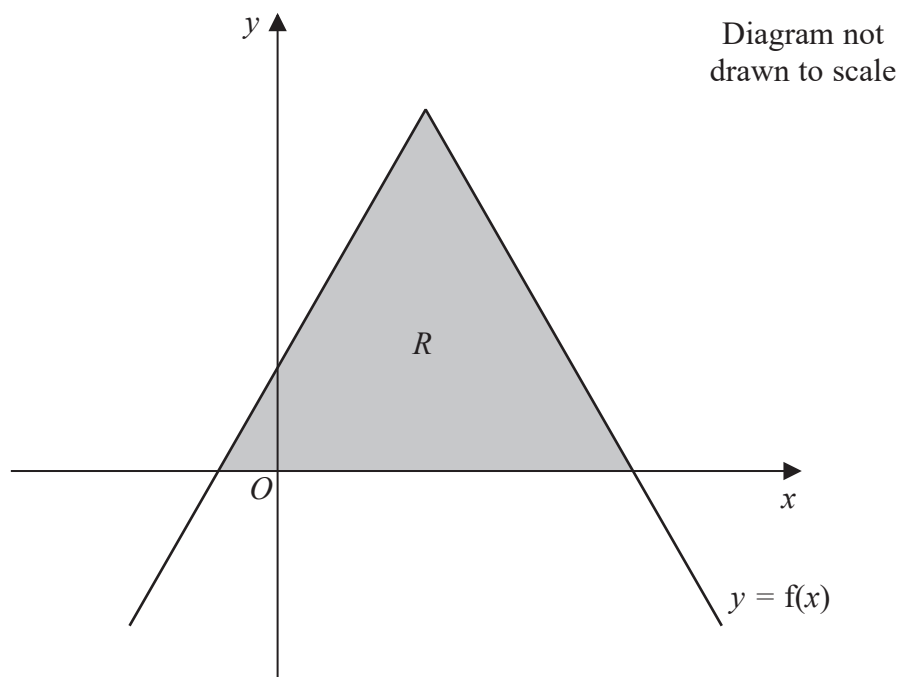
**Figure 2**

Figure 2 shows part of a graph with equation $y = f(x)$, where

$$f(x) = 7 - |3x - 5| \quad x \in \mathbb{R}$$

The finite region R , shown shaded in Figure 2, is bounded by the graph with equation $y = f(x)$ and the x -axis.

(a) Find the area of R , giving your answer in simplest form.

(4)

The equation

$$7 - |3x - 5| = k$$

where k is a constant, has two distinct real solutions.

(b) Write down the range of possible values for k .

(1)

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



6.

$$f(x) = (2 + kx)^{-4} \quad \text{where } k \text{ is a positive constant}$$

The binomial expansion of $f(x)$, in ascending powers of x , up to and including the term in x^2 , is

$$\frac{1}{16} + Ax + \frac{125}{32}x^2$$

where A is a constant.

(a) Find the value of A , giving your answer in simplest form.

(5)

(b) Determine, giving a reason for your answer, whether the binomial expansion for $f(x)$

is valid when $x = \frac{1}{10}$

(1)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7.

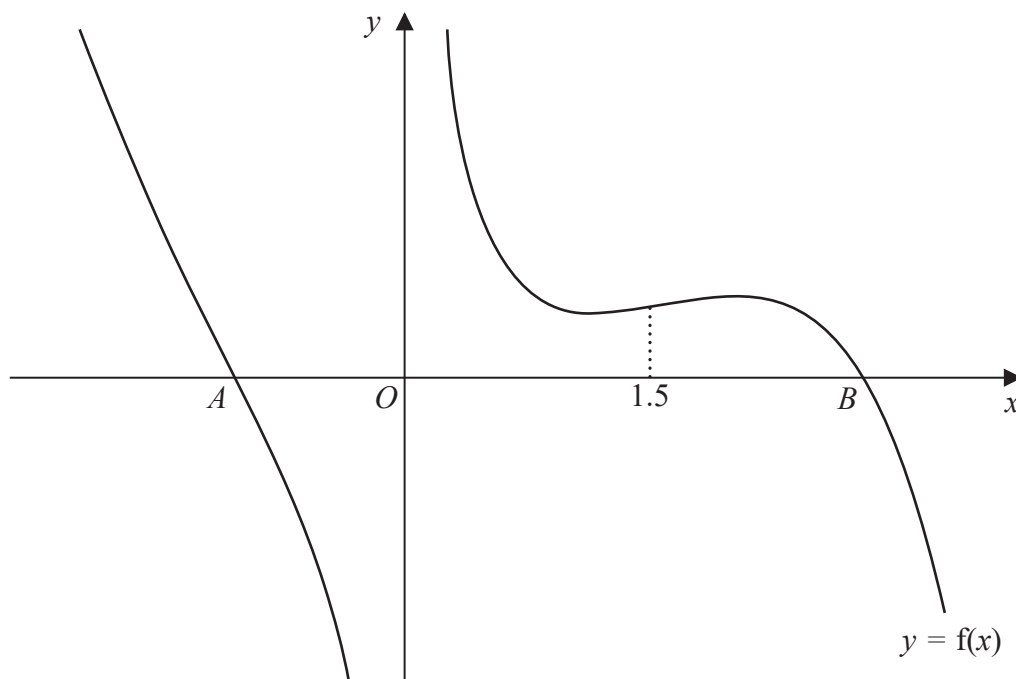
**Figure 3**

Figure 3 shows a plot of part of the curve with equation $y = f(x)$, where

$$f(x) = \frac{2}{x} - e^x + 2x^2 \quad x \in \mathbb{R}, x \neq 0$$

The curve cuts the x -axis at the point A , where $x = \alpha$, and at the point B , where $x = \beta$, as shown in Figure 3.

(a) Show that α lies between -1.5 and -1

(2)

(b) The iterative formula

$$x_{n+1} = -\sqrt{\left(\frac{1}{2}e^{x_n} - \frac{1}{x_n}\right)} \quad n \in \mathbb{N}$$

with $x_1 = -1$ can be used to estimate the value of α .

(i) Find the value of x_3 to 4 decimal places.

(ii) Find the value of α correct to 2 decimal places.

(2)



[If you need to rework your answer to part (d) turn over for a spare copy of Figure 3]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 7 continued

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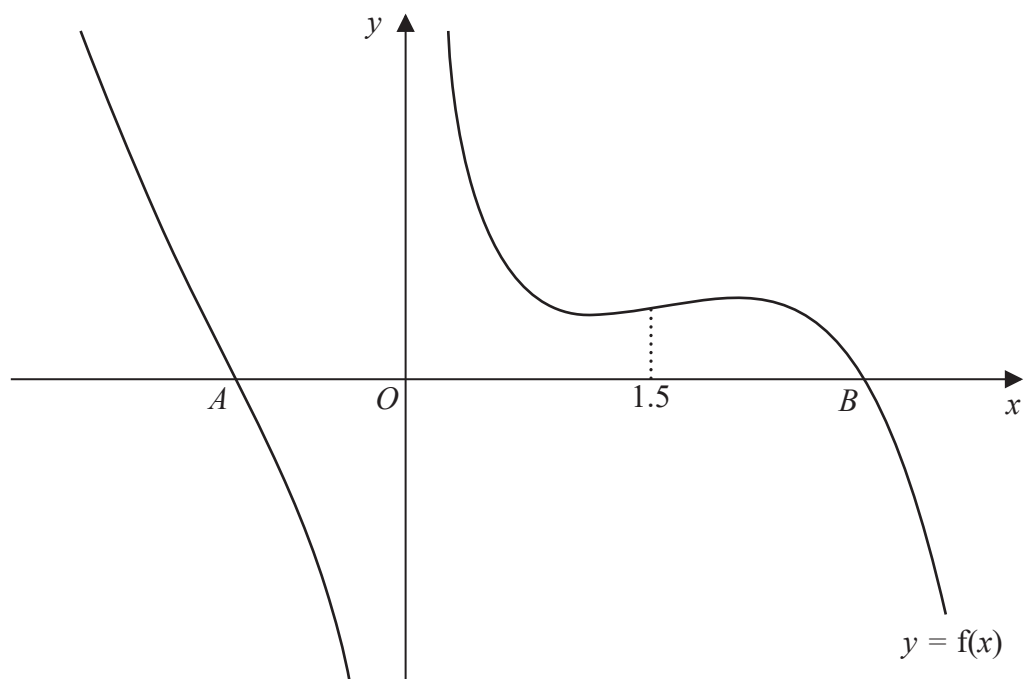
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Question 7 continued

Only use this spare copy of Figure 3 if you have to rework your answer to part (d).



Spare copy of Figure 3

(Total for Question 7 is 8 marks)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 7 marks)



9. The amount of antibiotic, y milligrams, in a patient's bloodstream, t hours after the antibiotic was first given, is modelled by the equation

$$y = ab^t$$

where a and b are constants.

- (a) Show that this equation can be written in the form

$$\log_{10} y = t \log_{10} b + c$$

expressing the constant c in terms of a .

(2)

A doctor measures the amount of antibiotic in the patient's bloodstream at regular intervals for the first 5 hours after the antibiotic was first given.

She plots a graph of $\log_{10} y$ against t and finds that the points on the graph lie close to a straight line passing through the point $(0, 2.23)$ with gradient -0.076

- (b) Estimate, to 2 significant figures, the value of a and the value of b .

(2)

With reference to this model,

- (c) (i) give a practical interpretation of the value of the constant a ,

- (ii) give a practical interpretation of the value of the constant b .

(2)

- (d) Use the model to estimate the time taken, after the antibiotic was first given, for the amount of antibiotic in the patient's bloodstream to fall to 30 milligrams. Give your answer, in hours, correct to one decimal place.

(2)

- (e) Comment on the reliability of your estimate in part (d).

(1)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 9 marks)



10. A biologist conducted an experiment to investigate the growth of mould on a slice of bread.

The biologist measured the surface area of bread, $A \text{ cm}^2$, covered by mould at times, t days, after the start of the experiment.

Initially 9.00 cm^2 of the bread was covered by mould and 6 days later, 56.25 cm^2 of the bread was covered by mould.

In the biologist's model, the rate of increase of the surface area of bread covered by mould, at any time t days, is proportional to the square root of that area.

By forming and solving a differential equation,

(a) show that the biologist's model leads to the equation

$$A = \left(\frac{3}{4}t + 3 \right)^2 \quad (6)$$

The biologist's full set of results are shown in the table below.

t (days)	0	6	12	18	24	30
$A \text{ (cm}^2\text{)}$	9.00	56.25	143.78	271.19	334.81	337.33

Table 1

Use the last four measurements from Table 1 to

(b) (i) evaluate the biologist's model,

(ii) suggest a possible explanation of the results.

(3)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 9 marks)



11.

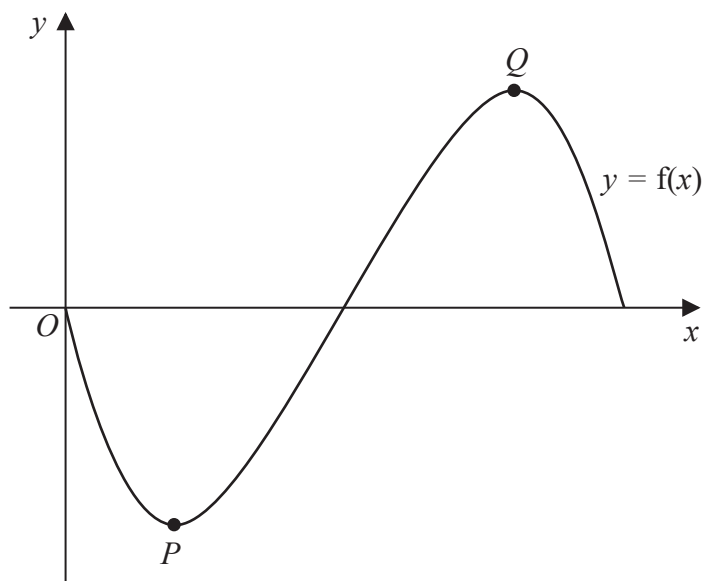


Figure 5

Figure 5 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \frac{\sin 2x}{-3 + \cos 2x} \quad 0 \leq x \leq \pi$$

The curve has a minimum turning point at P and a maximum turning point at Q , as shown in Figure 5.

(a) Show that the x coordinate of P and the x coordinate of Q are solutions of the equation

$$\cos 2x = \frac{1}{3} \quad (4)$$

(b) Hence find, to 2 decimal places, the x coordinate of the maximum turning point on the curve with equation

(i) $y = f(3x) + 5 \quad 0 \leq x \leq \frac{\pi}{3}$

(ii) $y = -f\left(\frac{1}{4}x\right) \quad 0 \leq x \leq 4\pi \quad (4)$



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 8 marks)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 7 marks)



13.

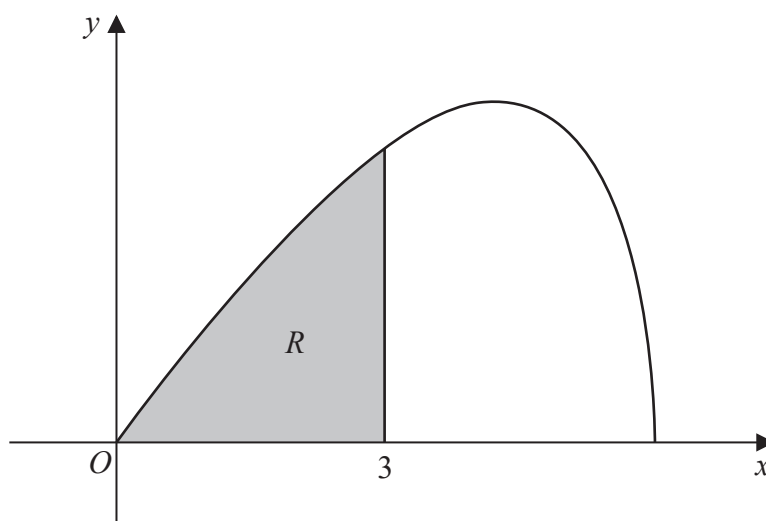


Figure 6

Figure 6 shows the curve with parametric equations

$$x = 6 \cos t \quad y = 5 \sin 2t \quad 0 \leq t \leq \frac{\pi}{2}$$

The finite region R , shown shaded in Figure 6, is bounded by the curve, the x -axis, and the line with equation $x = 3$

Use calculus to show that the area of R is $20 - \frac{15}{2}\sqrt{3}$

(7)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



14.

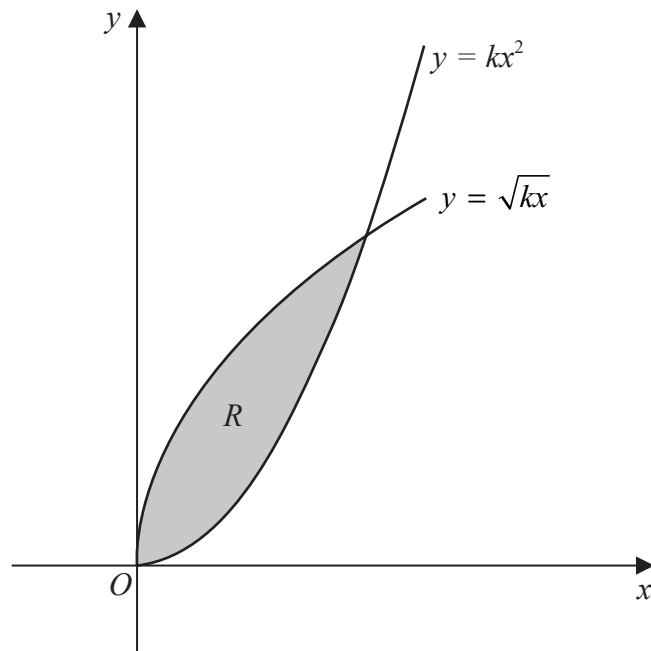


Figure 7

Figure 7 shows the curves with equations

$$y = kx^2 \quad x \geq 0$$

$$y = \sqrt{kx} \quad x \geq 0$$

where k is a positive constant.

The finite region R , shown shaded in Figure 7, is bounded by the two curves.

Show that, for all values of k , the area of R is $\frac{1}{3}$

(5)

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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 5 marks)



15. A sequence of numbers $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = k - \frac{3k}{a_n} \quad n \in \mathbb{Z}^+$$

where k is a constant.

The sequence is periodic of order 3

Given that $a_2 = 2$

(a) show that $k^2 + k - 12 = 0$

(3)

Given that $a_1 \neq a_2$

(b) find the value of $\sum_{r=1}^{121} a_r$

(4)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 7 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper – Set 2			
(Time: 2 hours)		Paper Reference 9MA0/01	
Mathematics			
Advanced			
Paper 1: Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Weed is completely covering the surface of a pond.

Fish are introduced into the pond in an effort to control the weed.

The surface area of the pond, $A \text{ m}^2$, covered by the weed, t days after the fish are introduced is modelled by the equation

$$A = 105 - 12e^{0.08t} \quad t \in \mathbb{R}, t \geq 0$$

According to the model,

- (a) state the surface area of the pond covered by the weed at the start of the investigation, (1)
- (b) find the time taken in days, to one decimal place, for the surface area of the pond covered by the weed to fall to 40 m^2 (3)

Stuart wants to predict the surface area of the pond covered by the weed 30 days after the fish are introduced.

- (c) Explain why he should not use this model. (2)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\sqrt{1+4x}$$

giving each coefficient in its simplest form.

(4)

The expansion can be used to find an approximation for $\sqrt{26}$

- (b) Explain why $x = \frac{25}{4}$ **should not** be used in the expansion to find an approximation for $\sqrt{26}$

(1)

- (c) Explain how you could use $x = \frac{1}{100}$ in the expansion to find an approximation for $\sqrt{26}$

There is no need to carry out the calculation.

(2)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 7 marks)



3. The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = \frac{4}{2 - u_n} \quad u_1 = 1$$

(a) Show that this sequence is periodic, stating the period.

(3)

(b) Hence find $\sum_{n=1}^{50} u_n$

(2)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4.

$$h(x) = \frac{4x^3 - 19x^2 + 28x - 4}{(x-2)^2} \quad x > 2$$

- (a) Write $h(x)$ in the form $Ax + B + \frac{C}{(x-2)^2}$ where A , B and C are constants to be found.

(3)

- (b) Hence find $\int h(x) dx$

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



5. Relative to a fixed origin O ,

- the point A has position vector $-2\mathbf{i} + 3\mathbf{j}$
- the point B has position vector $3\mathbf{i} + p\mathbf{j}$, where p is constant
- the point C has position vector $q\mathbf{i} + 7\mathbf{j}$, where q is constant

Given that $|\overrightarrow{AB}| = 5\sqrt{2}$

(a) find the possible values of p .

(3)

Given that the angle between $\overrightarrow{AC}$ and the unit vector $\mathbf{i}$ is $\frac{\pi}{3}$ radians,

(b) find the exact value of q .

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



6. (a) Use the identity for $\tan(A + B)$ to show that

$$\tan 3\theta \equiv \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \quad (4)$$

Given $\tan \beta = \sqrt{6}$

- (b) use the answer to part (a) to show that

$$\tan 3\beta = k\sqrt{6}$$

where k is a constant to be found.

(2)



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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 6 marks)



7.

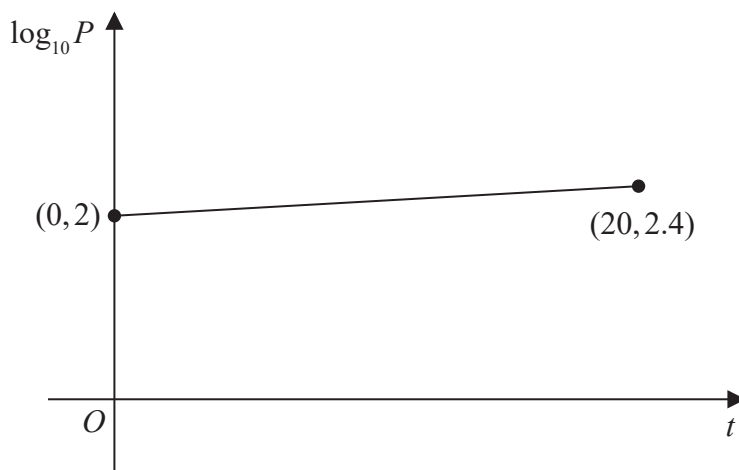


Figure 1

Red squirrels were introduced into a large wood in Northumberland on 1st June 1996.

Scientists counted the number of red squirrels in the wood, P , on 1st June each year for t years after 1996.

The scientists found that over time the number of red squirrels can be modelled by the formula

$$P = ab^t$$

where a and b are constants.

The line l , shown in Figure 1, illustrates the linear relationship between $\log_{10} P$ and t over a period of 20 years.

Using the information given on the graph and using the model,

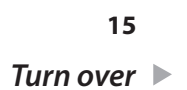
- (a) find an equation for l , (2)
- (b) find the initial number of red squirrels that were introduced into the wood, (2)
- (c) find a complete equation for the model giving the value of b to 4 significant figures. (2)

On 1st June 2019 there were found to be 198 red squirrels in the wood.

- (d) (i) Use this information to show that the model is not valid on 1st June 2019.
- (ii) Suggest a reason for the model not being valid at this time. (3)



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Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 9 marks)



8.

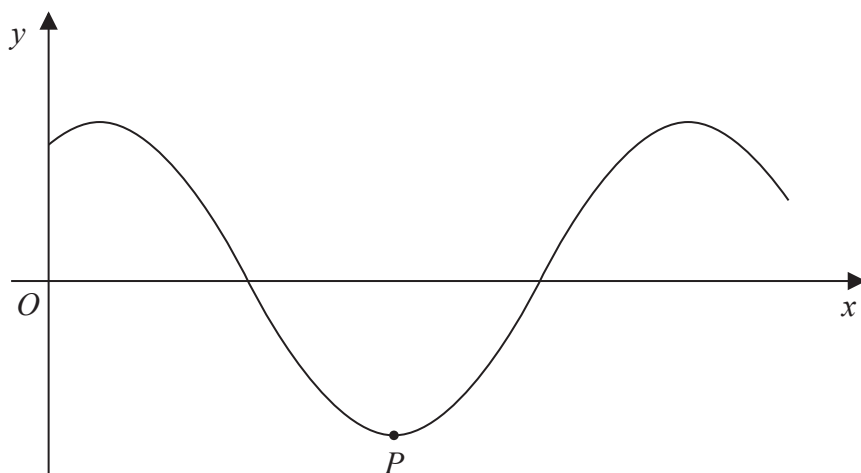


Figure 2

Figure 2 shows a sketch of part of the curve with equation

$$y = 5 \cos(x - 30)^\circ \quad x \geq 0$$

The point P on the curve is the minimum point with the smallest positive x coordinate.

- (a) State the coordinates of P .

(2)

- (b) Solve, for $0 \leq x < 360$, the equation

$$5 \cos(x - 30)^\circ = 4 \sin x^\circ$$

giving your answers to one decimal place.

(4)

- (c) Deduce, giving reasons for your answer, the **number of roots** of the equation

$$5 \cos(2x - 30)^\circ = 4 \sin 2x^\circ \text{ for } 0 \leq x < 360$$

(2)

[illegible]

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 8 marks)



9.

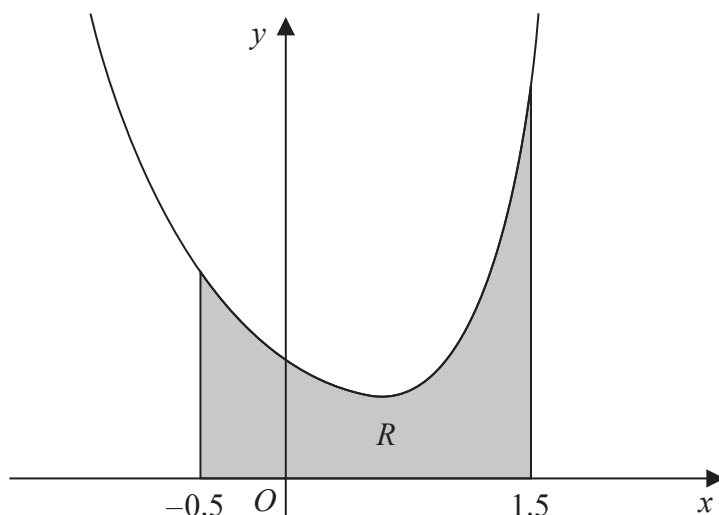
**Figure 3**

Figure 3 shows a sketch of the curve with equation $y = 2x^2 - x$

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line with equation $x = -0.5$, the x -axis and the line with equation $x = 1.5$

- (a) Use the trapezium rule with four strips of equal width to find an estimate for the area of R .
Show your working and give your answer to two decimal places.

(4)

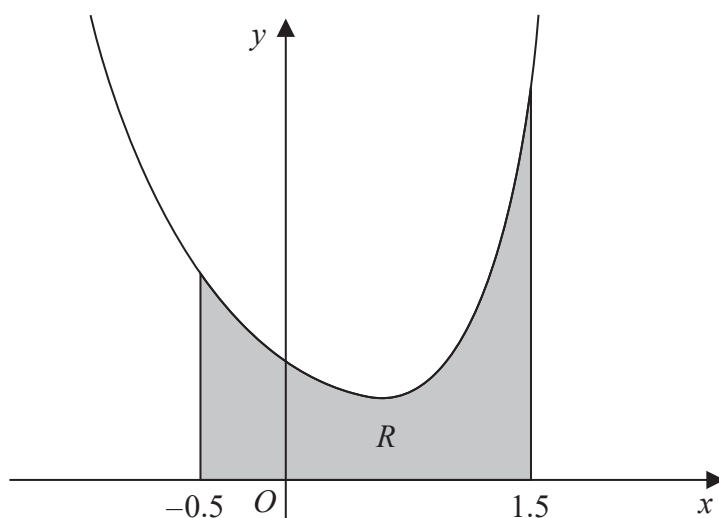
A copy of Figure 3, called Diagram 1, is drawn below.

- (b) Explain, with the aid of Diagram 1, whether your answer in part (a) is an underestimate or overestimate of the true value for the area of R .

(1)

Using your answer to part (a) and showing your working,

- (c) estimate the value of $\int_{-0.5}^{1.5} (2x^2 + 1 + 2x) dx$

(3)**Diagram 1**

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Question 9 continued

Handwriting practice area with horizontal lines.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 8 marks)



10. (a) Sketch the graph with equation

$$y = |2x - 9| + 3$$

stating the coordinates of the vertex and any points where the graph crosses the coordinate axes.

(3)

(b) Solve the equation

$$3x + 1 = |2x - 9| + 3$$

(2)

A straight line l has equation $y = kx + 1$, where k is a constant.

Given that l does not meet or intersect the graph with equation $y = |2x - 9| + 3$

(c) find the range of possible values of k .

(3)



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Question 10 continued

Handwriting practice area with 25 horizontal lines.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 6 marks)



12. Using algebraic integration and making your method clear, show that

$$\int_1^5 \frac{4x+9}{x+3} dx = a + \ln b$$

where a and b are constants to be found.



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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 5 marks)



13.

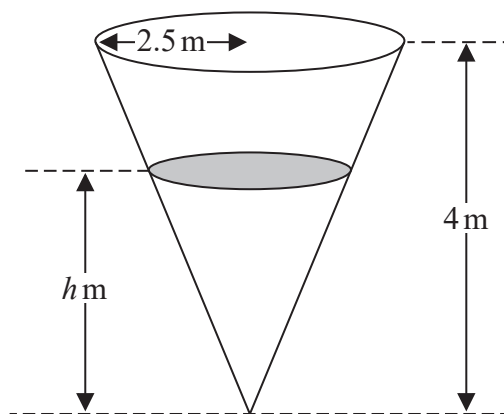
Diagram not
drawn to scale

Figure 4

[The volume of a cone of base radius r and height h is $\frac{1}{3}\pi r^2 h$]

Figure 4 shows a container in the shape of an inverted right circular cone which contains some water.

The cone has an internal base radius of 2.5 m and a vertical height of 4 m.

At time t seconds

- the height of the water is h m
- the volume of the water is V m³
- the water is modelled as leaking from a hole at the bottom of the container at a rate of

$$\left(\frac{\pi}{512}\sqrt{h}\right)m^3s^{-1}$$

(a) Show that, while the water is leaking

$$h^{\frac{3}{2}}\frac{dh}{dt} = -\frac{1}{200} \quad (5)$$

Given that the container was initially full of water,

(b) find an equation, in terms of h and t , to model this situation. (3)

It takes approximately 43 minutes for the container to empty.

(c) Use this information to comment on the suitability of this model. (3)



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Question 13 continued

Handwriting practice area with horizontal lines.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 11 marks)



14.

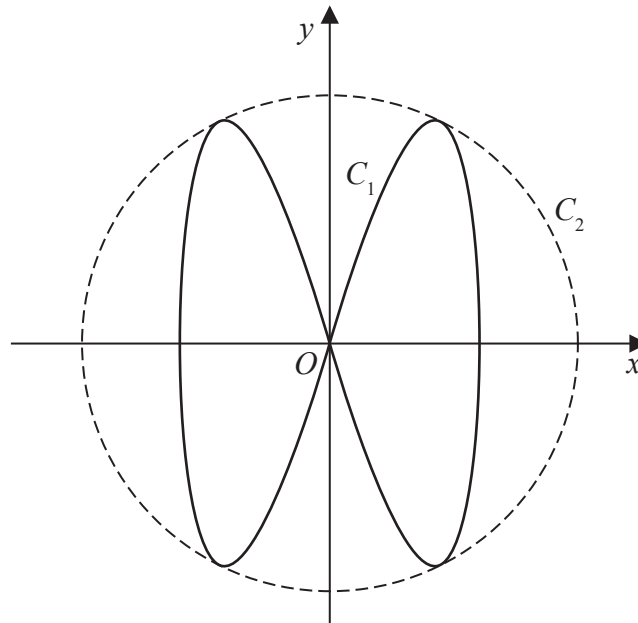


Figure 5

Figure 5 shows a sketch of the curve C_1 with parametric equations

$$x = 2\sin t, \quad y = 3\sin 2t \quad 0 \leq t < 2\pi$$

(a) Show that the Cartesian equation of C_1 can be expressed in the form

$$y^2 = kx^2(4 - x^2)$$

where k is a constant to be found.

(4)

The circle C_2 with centre O touches C_1 at four points as shown in Figure 5.

(b) Find the radius of this circle.

(5)



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Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 9 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number 	Candidate Number
Mock Paper – Set 2			
(Time: 2 hours)		Paper Reference 9MA0/02	
Mathematics Advanced Paper 2: Pure Mathematics 2			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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1. (a) Given that

$$2\log(3-x) - \log(21-2x) = 0$$

show that

$$x^2 - 4x - 12 = 0 \quad (3)$$

- (b) (i) Write down the roots of $x^2 - 4x - 12 = 0$
- (ii) Which of these two roots is **not** a solution of

$$2\log(3-x) - \log(21-2x) = 0$$

giving a reason for your answer. (2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2.

$$f(x) = 3x^4 + 2x^2 - 12x + 8$$

Given that $y = f(x)$ has a single turning point at $x = \alpha$,

(a) show that α is a solution of the equation

$$x = \sqrt[3]{1 - \frac{x}{3}} \quad (3)$$

The iterative formula

$$x_{n+1} = \sqrt[3]{1 - \frac{x_n}{3}}$$

is used with $x_1 = 1$ to find an approximate value for α .

(b) Calculate the value of x_2 and the value of x_5 , giving each answer to 4 decimal places.

(3)

(c) Using a suitable interval and a suitable function that should be stated, show that to 3 decimal places α is 0.889

(2)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 8 marks)



3.

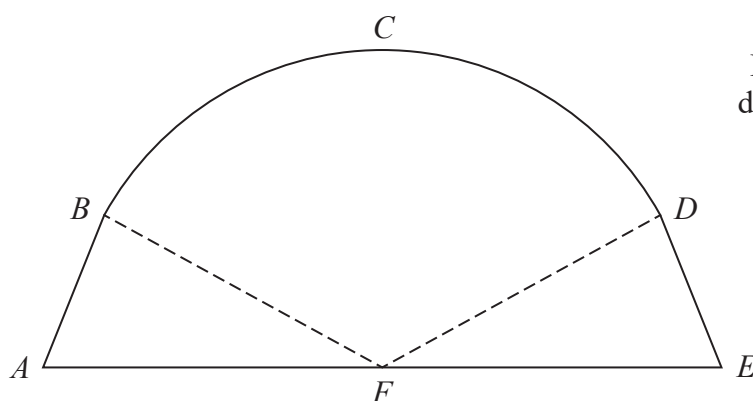


Figure 1

Figure 1 shows the plan view of a design for a stage at a concert.

The stage is modelled as a sector $BCDF$, of a circle centre F , joined to two congruent triangles ABF and EDF .

Given that

AFE is a straight line

$$AF = FE = 10.7 \text{ m}$$

$$BF = FD = 9.2 \text{ m}$$

angle $BFD = 1.82$ radians

find

- (a) the perimeter of the stage, in metres, to one decimal place,

- (b) the area of the stage, in m^2 , to one decimal place. (4)



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(Total for Question 3 is 9 marks)



4.

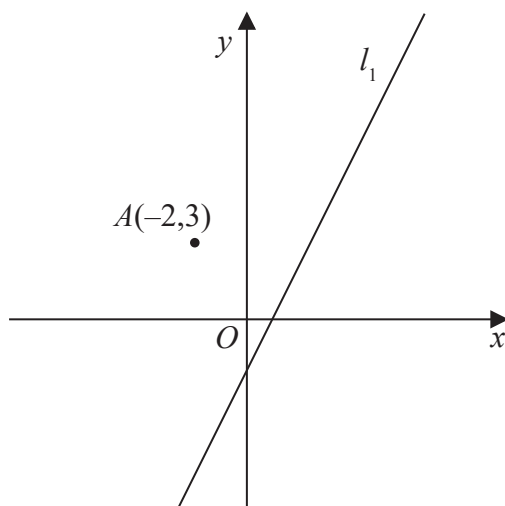


Figure 2

Figure 2 is a sketch showing the line l_1 with equation $y = 2x - 1$ and the point A with coordinates $(-2, 3)$.

The line l_2 passes through A and is perpendicular to l_1

- (a) Find the equation of l_2 writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The point B and the point C lie on l_1 such that ABC is an isosceles triangle with $AB = AC = 2\sqrt{13}$

- (b) Show that the x coordinates of points B and C satisfy the equation

$$5x^2 - 12x - 32 = 0$$

(4)

Given that B lies in the 3rd quadrant

- (c) find, using algebra and showing your working, the coordinates of B .

(4)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 11 marks)



5. Vectors $\vec{AB}$ and $\vec{BC}$ are given by

$$\vec{AB} = \begin{pmatrix} 2p \\ q \\ 4 \end{pmatrix} \quad \vec{BC} = \begin{pmatrix} q \\ -3p \\ 2 \end{pmatrix} \quad \text{where } p \text{ and } q \text{ are constants}$$

Given that $\vec{AC}$ is parallel to $\begin{pmatrix} 3 \\ -4 \\ 3 \end{pmatrix}$ find the value of p and the value of q . (5)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



6. A small company which makes batteries for electric cars has a 10-year plan for growth.
- In year 1 the company will make 2 600 batteries
 - In year 10 the company aims to make 12 000 batteries

In order to calculate the number of batteries it will need to make each year, from year 2 to year 9, the company considers two models, Model *A* and Model *B*.

In Model *A* the number of batteries made will increase by the same **number** each year.

- (a) Using Model A , determine the number of batteries the company will make in year 2 (3)

In Model *B* the number of batteries made will increase by the same **percentage** each year.

- (b) Using Model *B*, determine the number of batteries the company will make in year 2.

Give your answer to the nearest 10 batteries. (3)

Sam calculates the total number of batteries made from year 1 to year 10 inclusive using each of the two models.

- (c) Calculate the difference between the two totals, giving your answer to the nearest 100 batteries.
- (3)**

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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7.

$$g(x) = 4x^3 + ax^2 + 4x + b$$

where a and b are constants.

Given that

- $(2x + 1)$ is a factor of $g(x)$
- the curve with equation $y = g(x)$ has a point of inflection at $x = \frac{1}{6}$

(a) find the value of a and the value of b

(5)

(b) Show that there are no stationary points on the curve with equation $y = g(x)$.

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 7 marks)



8. (a) Sketch the curve with equation

$$y = k - \frac{1}{2x} \quad \text{where } k \text{ is a positive constant}$$

State, in terms of k , the coordinates of any points of intersection with the coordinate axes and the equation of the horizontal asymptote.

(3)

The straight line l has equation $y = 2x + 3$

Given that l cuts the curve in two distinct places,

- (b) find the range of values of k , writing your answer in set notation.

(6)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 9 marks)



9.

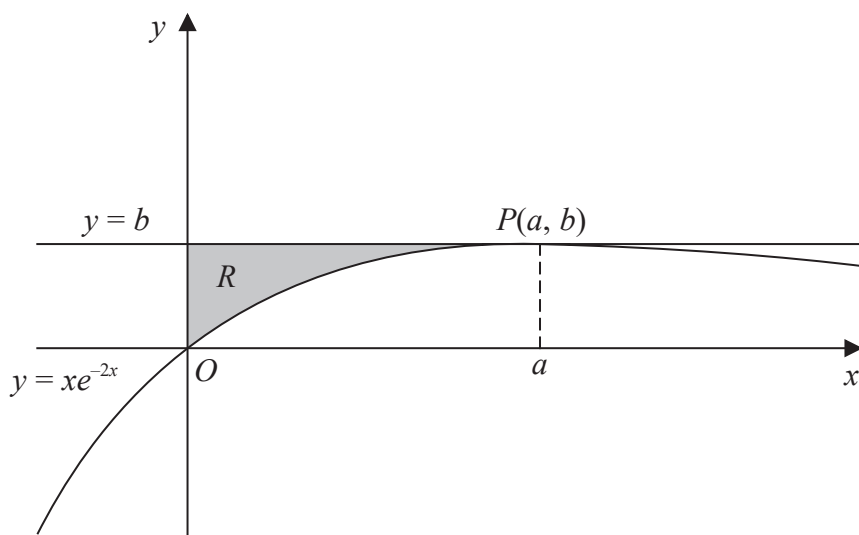


Figure 3

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve with equation

$$y = xe^{-2x}$$

The point $P(a, b)$ is the turning point of the curve.

(a) Find the value of a and the exact value of b

(4)

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line with equation $y = b$ and the y -axis.

(b) Find the exact area of R .

(5)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 9 marks)



10.

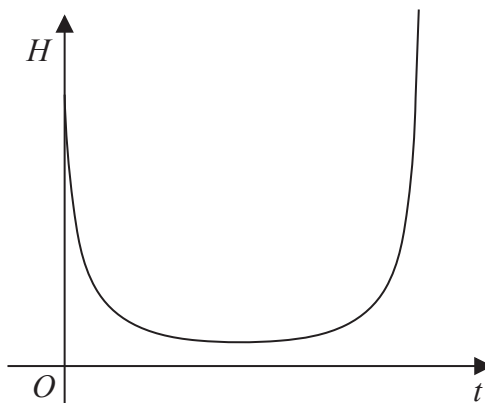


Figure 4

A scientist is studying the flight of seabirds in a colony.

She models the height above sea level, H metres, of one of the birds in the colony by the equation

$$H = \frac{140}{A + 45\sin 2t^\circ - 28\cos 2t^\circ} \quad 0 \leq t \leq T$$

where t seconds is the time after the bird leaves its nest and A and T are constants.

Figure 4 is a sketch showing the graph of H against t .

Given that this seabird's nest is 20 m above sea level,

- (a) find a complete equation for H . (3)

Given that

$$45\sin 2t^\circ - 28\cos 2t^\circ \equiv 53\sin(2t - \alpha)^\circ \quad 0 < \alpha < 90$$

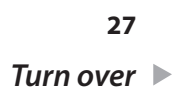
- (b) find the value of α to one decimal place. (2)

Find, according to this model,

- (c) the minimum height of the seabird above sea level giving your answer to the nearest cm, (2)
- (d) the limitation on the value of T . (2)



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Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 9 marks)



11.

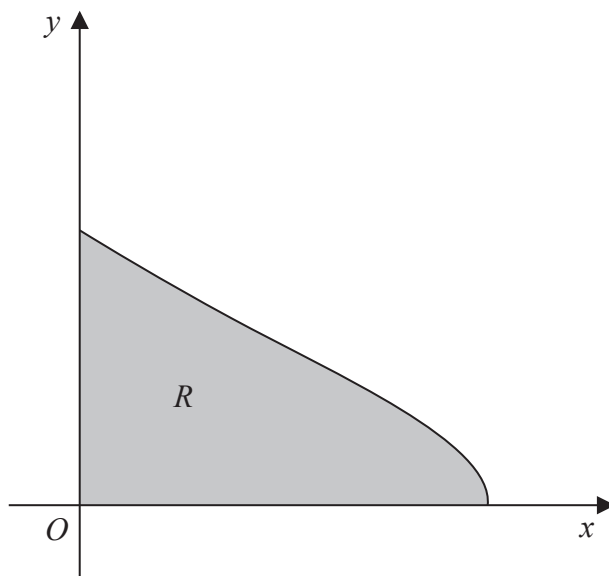
**Figure 5**

Figure 5 shows a sketch of the curve with parametric equations

$$x = 3 \cos 2t, \quad y = 2 \tan t \quad 0 \leq t \leq \frac{\pi}{4}$$

The region R , shown shaded in Figure 5, is bounded by the curve, the x -axis and the y -axis.

(a) Show that the area of R is given by

$$\int_0^{\frac{\pi}{4}} 24 \sin^2 t \, dt \quad (4)$$

(b) Hence, using algebraic integration, find the exact area of R .

(3)



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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 7 marks)



12.

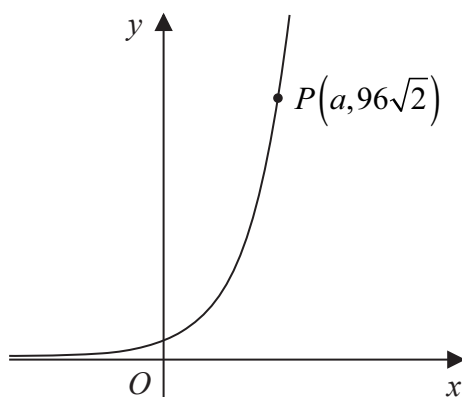


Figure 6

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 6 shows a sketch of part of the curve with equation

$$y = 3 \times 2^{2x}$$

The point $P(a, 96\sqrt{2})$ lies on the curve.

(a) Find the exact value of a .

(3)

The curve with equation $y = 3 \times 2^{2x}$ meets the curve with equation $y = 6^{3-x}$ at the point Q .

(b) Show that the x coordinate of Q is

$$\frac{3 + 2\log_2 3}{3 + \log_2 3}$$

(5)



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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 8 marks)



13. Prove by contradiction that there are no positive integers a and b with a odd such that

$$a + 2b = \sqrt{8ab}$$

(4)

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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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(Total for Question 13 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper Set 3			
Time: 2 hours		Paper Reference 9MA0/01	
Mathematics			
Advanced			
Paper 1: Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Curve C has equation

$$y = x^3 - 7x^2 + 5x + 4$$

(a) Find $\frac{dy}{dx}$

(2)

The point $P(2, -6)$ lies on C

(b) Find the equation of the tangent to C at P

Give your answer in the form $y = mx + c$ where m and c are integers to be found.

(3)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2.

$$f(x) = 3x^3 - 7x^2 + 7x - 10$$

(a) Use the factor theorem to show that $(x - 2)$ is a factor of $f(x)$ (2)

(b) Find the values of the constants a , b and c such that

$$f(x) \equiv (x-2)(ax^2 + bx + c) \quad (3)$$

(c) Using your answer to part (b) show that the equation $f(x) = 0$ has only one real root. (2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3.

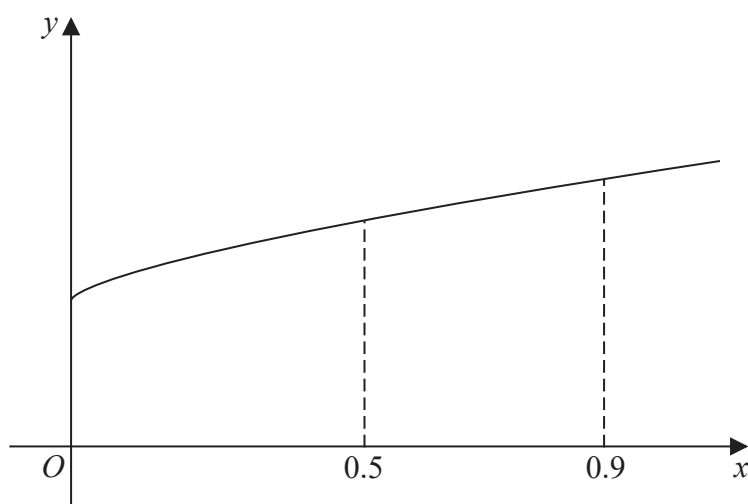
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$

The table below shows corresponding values of x and y for this curve between $x = 0.5$ and $x = 0.9$

The values of y are given to 4 significant figures.

x	0.5	0.6	0.7	0.8	0.9
y	1.632	1.711	1.786	1.859	1.930

(a) Use the trapezium rule, with all the values of y in the table, to find an estimate for

$$\int_{0.5}^{0.9} f(x) dx$$

Give your answer to 3 significant figures.

(3)

(b) Using your answer to part (a), deduce an estimate for

$$\int_{0.5}^{0.9} (3f(x) + 2) dx$$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



4.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Express as an integral

$$\lim_{\delta x \rightarrow 0} \sum_{x=4}^{12} (1 + 2x)^{\frac{1}{2}} \delta x \quad (1)$$

(b) Using your answer to part (a) show that

$$\lim_{\delta x \rightarrow 0} \sum_{x=4}^{12} (1 + 2x)^{\frac{1}{2}} \delta x = \frac{98}{3} \quad (3)$$



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 4 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



6.

$$f(x) = \frac{10}{\sqrt{4-3x}}$$

- (a) Show that the first 4 terms in the binomial expansion of $f(x)$, in ascending powers of x , are

$$A + Bx + Cx^2 + \frac{675}{1024}x^3$$

where A , B and C are constants to be found. Give each constant in simplest form.

(4)

Given that this expansion is valid for $|x| < k$

- (b) state the largest value of k .

(1)

By substituting $x = \frac{1}{3}$ into $f(x)$ and into the answer for part (a),

- (c) find an approximation for $\sqrt{3}$

Give your answer in the form $\frac{a}{b}$ where a and b are integers to be found.

(2)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 7 marks)



7. Curve C has equation

$$y = (x^2 - 5x + 8)e^{x^2} \quad x \in \mathbb{R}$$

(a) Show that

$$\frac{dy}{dx} = (2x^3 - 10x^2 + 18x - 5)e^{x^2} \quad (3)$$

Given that

- C has only one stationary point
- the stationary point has x coordinate α
- $\frac{dy}{dx} \approx -0.5$ at $x = 0.3$
- $\frac{dy}{dx} \approx 0.9$ at $x = 0.4$

(b) explain why $0.3 < \alpha < 0.4$ (1)

(c) Show that α is a solution of the equation

$$x = \frac{5(2x^2 + 1)}{2(x^2 + 9)} \quad (3)$$

(d) Using the iteration formula

$$x_{n+1} = \frac{5(2x_n^2 + 1)}{2(x_n^2 + 9)} \quad \text{with } x_1 = 0.3$$

find

- the value of x_3 to 4 decimal places,
 - the value of α to 4 decimal places.
- (3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that

$$\frac{1 - \cos 2\theta}{\sin^2 2\theta} \equiv k \sec^2 \theta \quad \theta \neq \frac{n\pi}{2} \quad n \in \mathbb{Z}$$

where k is a constant to be found.

(3)

(b) Hence solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\frac{1 - \cos 2x}{\sin^2 2x} = (1 + 2 \tan x)^2$$

Give your answers to 3 significant figures where appropriate.

(4)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 7 marks)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



10. (a) Sketch the graph with equation

$$y = |3x - 2a|$$

where a is a positive constant.

State the coordinates of each point where the graph cuts or meets the coordinate axes.

(2)

(b) Solve, in terms of a , the inequality

$$|3x - 2a| \leq x + a$$

(4)

Given that $|3x - 2a| \leq x + a$

(c) find, in terms of a , the range of possible values of $g(x)$, where

$$g(x) = 5a - \left| \frac{1}{2}a - x \right|$$

(3)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 9 marks)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 10 marks)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 9 marks)



13. Given that

$$y^2 - x^2 = 8$$

show that

$$\frac{d^2y}{dx^2} = \frac{8}{y^3}$$

(5)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 5 marks)



14. (i) Prove that the sum of the squares of 2 consecutive odd integers is always 2 more than a multiple of 8

(3)

(ii) Use proof by contradiction to show that $\log_2 5$ is irrational.

(4)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 7 marks)

(TOTAL FOR PAPER IS 100 MARKS)



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper Set 3			
Time: 2 hours		Paper Reference 9MA0/02	
Mathematics			
Advanced			
Paper 2: Pure Mathematics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Find

$$\int (x^4 - 6x^2 + 7) dx$$

giving your answer in simplest form.

(3)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 3 marks)



2. The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = k$$
$$u_{n+1} = 3u_n - 2$$

where k is a constant.

(a) Find, in simplest form in terms of k ,

(i) u_2

(ii) u_3

(2)

Given that $\sum_{r=1}^4 u_r = 44$

(b) find the value of k .

(3)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. A student was asked to solve the simultaneous equations

$$\begin{aligned}x + y &= 9 \\x^2 - 3xy + 2y^2 &= 0\end{aligned}$$

The student's solution is shown below:

line 1: $x + y = 9 \Rightarrow y = 9 - x$

line 2: $x^2 - 3xy + 2y^2 = 0 \Rightarrow x^2 - 3x(9 - x) + 2(9 - x)^2 = 0$

line 3: $x^2 - 27x - 3x^2 + 162 - 36x + 2x^2 = 0$

line 4: $63x = 162$

line 5: $x = \frac{162}{63} \Rightarrow y = 9 - \frac{162}{63} = \frac{45}{7}$

- (a) Identify the error in line 3 of the solution.

(1)

- (b) Using algebra and showing all your working, solve the simultaneous equations.

(4)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4. Curve C has equation

$$y = (x + k)(2 - x)$$

where k is a constant and $k > 2$

- (a) Sketch C , showing the coordinates of any points of intersection with the coordinate axes. (3)
- (b) Find, in simplest form in terms of k , the coordinates of the stationary point of C . (3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



6.

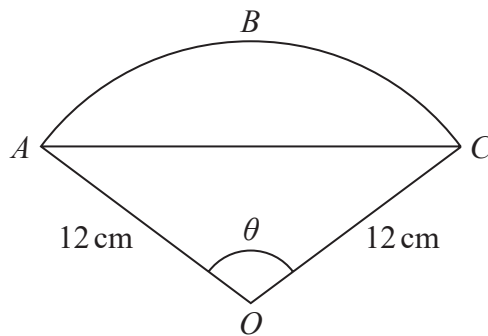


Diagram **NOT**
accurately drawn

Figure 1

Figure 1 shows a sector $OABCO$ of a circle centre O .

Given that

- $OA = OC = 12$ cm
- angle $AOC = \theta$ radians
- area triangle OAC : area segment $ABC = 3 : 1$

(a) show that

$$3\theta - 4 \sin \theta = 0 \quad (2)$$

(b) Taking 1.2 as a first approximation to θ , apply the Newton-Raphson method once to

$$f(\theta) = 3\theta - 4 \sin \theta$$

to find a second approximation to θ

Give your answer to 3 decimal places.

(3)

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 5 marks)



7. A ball is released from rest from a height of 5 m and bounces repeatedly on horizontal ground.

After hitting the ground for the first time, the ball rises to a maximum height of 3 m.

In a model for the motion of the ball

- the maximum height after each bounce is 60% of the previous maximum height
- the motion takes place in a vertical line

(a) Using the model

- show that the maximum height after the 3rd bounce is 1.08 m,
- find the total distance the ball travels from release to when the ball hits the ground for the 5th time.

(3)

According to the model, after the ball is released, there is a limit, D metres, to the total distance the ball will travel.

(b) Find the value of D

(2)

With reference to the model,

(c) give a reason why, in reality, the ball will not travel D metres in total.

(1)

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 6 marks)



8.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Express $3 \cos x + \sin x$ in the form $R \cos(x - \alpha)$ where

- R and α are
- $R > 0$
- $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R and the value of α in radians to 3 decimal places.

(3)

The temperature, $\theta^{\circ}\text{C}$, inside a rabbit hole on a particular day is modelled by the equation

$$\theta = 6.5 + 3 \cos\left(\frac{\pi t}{13} - 4\right) + \sin\left(\frac{\pi t}{13} - 4\right) \quad 0 \leq t < 24$$

where t is the number of hours after midnight.

Using the equation of the model and your answer to part (a)

(b) (i) deduce the minimum value of θ during this day,

(ii) find the time of day when this minimum value occurs, giving your answer to the nearest minute.

(4)

(c) Find the rate of temperature increase in the rabbit hole at midday.

(2)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 9 marks)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 7 marks)



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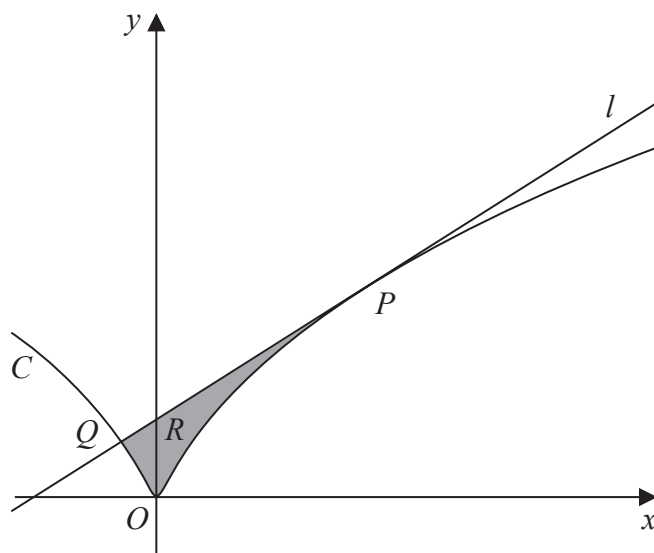


Figure 2

The curve C shown in Figure 2 has parametric equations

$$x = t^3 + 3t \quad y = 3t^2 \quad -2 < t < 4$$

The point P lies on C where $t = 3$

- (a) Write down the coordinates of P (1)

The line l is the tangent to C at P as shown in Figure 2.

- (b) Use calculus to show that an equation for l is
- $$3x - 5y + 27 = 0 \quad (3)$$

The line l meets C again at the point Q

- (c) Using algebra and showing all stages of your working, find the coordinates of Q (3)

The finite region R , shown shaded in Figure 2, is bounded by the curve C and the line l

- (d) Using algebraic integration, find the exact area of R (5)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 12 marks)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 8 marks)



12. Given that

$$y = a^x$$

where a is a positive constant

(a) prove that

$$\frac{dy}{dx} = a^x \ln a \quad (3)$$

(b) Hence show that

$$\int_1^2 4^x dx = k(\ln 2)^n$$

where k and n are integers to be found.

(3)



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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 6 marks)



13. The resting metabolic rate, R ml of oxygen consumed per hour, of a particular species of mammal is modelled by the formula,

$$R = aM^b$$

where

- M grams is the mass of the mammal
- a and b are constants

- (a) Show that this relationship can be written in the form

$$\log_{10} R = b \log_{10} M + \log_{10} a \quad (2)$$

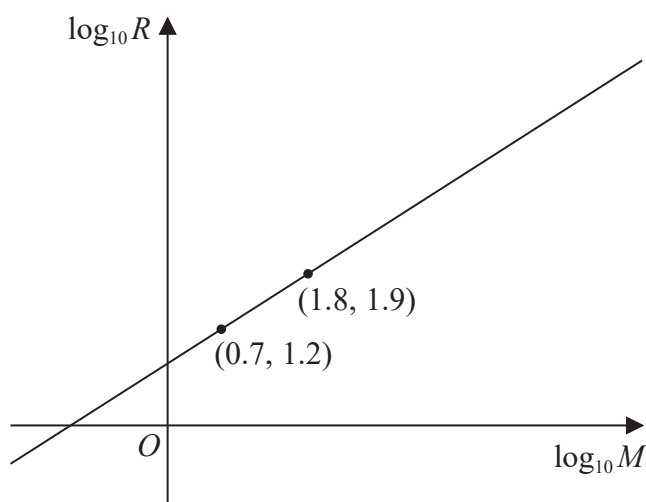


Figure 3

A student gathers data for R and M and plots a graph of $\log_{10} R$ against $\log_{10} M$

The graph is a straight line passing through points $(0.7, 1.2)$ and $(1.8, 1.9)$ as shown in Figure 3.

- (b) Using this information, find a complete equation for the model.
Write your answer in the form

$$R = aM^b$$

giving the value of each of a and b to 3 significant figures. (3)

- (c) With reference to the model, interpret the value of the constant a (1)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 6 marks)



14. (a) Use the substitution $u = 1 + \sin^2 x$ to show that

$$\int_0^{\frac{\pi}{6}} \frac{8 \tan x}{1 + \sin^2 x} dx = \int_p^q \frac{4}{u(2-u)} du$$

where p and q are constants to be found.

(5)

(b) Hence, using algebraic integration, show that

$$\int_0^{\frac{\pi}{6}} \frac{8 \tan x}{1 + \sin^2 x} dx = \ln A$$

where A is a rational number to be found.

(6)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 11 marks)



15.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

The first 3 terms of an arithmetic sequence are

$$\ln 3 \quad \ln(3^k - 1) \quad \ln(3^k + 5)$$

Find the exact value of the constant k .

(5)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 5 marks)

(TOTAL FOR PAPER IS 100 MARKS)



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Other names

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Candidate Number

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Pearson Edexcel Level 3 GCE

Time 2 hours

Paper
reference

9MA0/01

Mathematics

Advanced

PAPER 1: Pure Mathematics 1

Mock set 4

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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- Check your answers if you have time at the end.

Turn over ►

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1. The point A has coordinates $(-3, -2)$
The point B has coordinates $(7, 5)$
The straight line l_1 passes through A and B .

(a) Find the gradient of l_1 giving your answer in simplest form. (2)

(b) Find an equation for l_1 giving your answer in the form $ax + by + c = 0$,
where a , b and c are integers to be found. (2)

Given that the line l_2 is perpendicular to l_1

(c) write down the gradient of l_2 (1)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2. The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = pu_n + q$$

where p and q are constants.

Given that $u_1 = 4$

(a) write down u_2 in terms of p and q .

(1)

Given also that

- $u_2 = 10$
- $\sum_{r=1}^3 u_r = 42$

(b) find the value of p and the value of q .

(3)



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Question 2 continued

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(Total for Question 2 is 4 marks)



3. The functions f and g are defined by

$$\begin{aligned} f(x) &= e^{3x} + 2 & x &\in \mathbb{R} \\ g(x) &= \ln x & x &\in \mathbb{R} \quad x > 0 \end{aligned}$$

- (a) State the range of f (1)
- (b) Find $fg(x)$, giving your answer in simplest form. (2)
- (c) Find f^{-1} (3)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 6 marks)



4. Using the rules of logarithms, solve the equation

$$\log_2(x+2) + \log_2(x+3) - 2\log_2 x = 1$$

(4)

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 4 marks)



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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 7 marks)



6. Prove by contradiction that for all positive real values of x

$$x + \frac{4}{x} \geq 4$$

(4)

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 4 marks)



7. Relative to a fixed origin O

- the point A has position vector $8\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$
- the point B has position vector $t\mathbf{i} + 2t\mathbf{j} + 5t\mathbf{k}$

where t is a non-zero constant.

(a) Show that $\left| \overrightarrow{AB} \right|^2 = 30t^2 - 24t + 77$ (3)

(b) Hence find

- (i) the value of t when $\left| \overrightarrow{AB} \right|$ takes its minimum value,
- (ii) the minimum value of $\left| \overrightarrow{AB} \right|$, giving your answer as a simplified surd.
- (4)**

Given that

- $\left| \overrightarrow{AB} \right|$ takes its minimum value
- the point C lies on AB such that C divides AB in the ratio 5 : 3

(c) find the coordinates of C (2)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 9 marks)



8.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**(a) Given that $\cos 2\theta + \cos \theta + 1 \neq 0$, show that

$$\frac{\sin 2\theta + \sin \theta}{\cos 2\theta + \cos \theta + 1} \equiv \tan \theta \quad (3)$$

(b) Hence solve, for $0 < x < \pi$

$$\frac{\sin 2x + \sin x}{\cos 2x + \cos x + 1} = 2 \cos x$$

Give your answers to 3 decimal places.

(5)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 8 marks)



9. A kettle contains a quantity of water.
The water is cooling down.
The temperature, $\theta^{\circ}\text{C}$, of the water at time t minutes after cooling began is modelled by the equation

$$\theta = 21 + Ae^{-kt}$$

where A and k are constants.

Given that

- at the instant cooling began, the temperature of the water was 80°C
 - 40 minutes after cooling began, the temperature of the water was 33°C
- (a) find a complete equation for the model. Give the exact value of A and the value of k to 3 significant figures.
- (4)**

Given that 20 minutes after cooling began, the temperature of the water was 48°C

- (b) evaluate the model. (2)
- (c) Find, according to the model, the rate of the decrease in the temperature of the water 20 minutes after cooling began.

(Solutions relying entirely on calculator technology are not acceptable.)

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Question 9 continued

Handwriting practice area with 20 horizontal lines.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 9 marks)



10. A geometric series has

- first term $4 + k$
- second term $8 - 2k$

where k is a positive constant.

(a) Explain why the series does **not** converge when $k = 20$

(1)

Given that the series does converge,

(b) find the range of possible values of k .

(4)



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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 5 marks)



11. The global yearly energy output from onshore wind farms was first measured in 1996. The equation

$$E = ab^t \quad \text{where } a \text{ and } b \text{ are constants}$$

models the global yearly energy output, E gigawatts (GW), from onshore wind farms t years after 1996.

Given that the global yearly energy output from onshore wind farms was

- 74GW in 2006
- 198GW in 2010

- (a) (i) find the value of b to 3 decimal places,
(ii) find the value of a to one decimal place.

(4)

- (b) With reference to the model

- (i) interpret the value of a ,
- (ii) interpret the value of b .

(2)

Using the model,

- (c) find the predicted global yearly energy output from onshore wind farms in 2025.

(2)

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 8 marks)



12. (a) Find

$$\int x^2 e^{x^3} dx$$

(2)

(b) Using the substitution $u = x^3$ or otherwise, show by integration that

$$\int x^8 e^{x^3} dx = \frac{1}{3} e^{x^3} (x^6 + Ax^3 + B) + c$$

where A and B are constants to be found and c is an arbitrary constant.

(5)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 7 marks)



13. (a) Express $\frac{1}{x(100-x)}$ in partial fractions.

(2)

A scientist is studying the spread of a fungus in a field.
The field has an area of 100 m^2

The area of the field affected by the fungus, $x\text{ m}^2$, is modelled by the differential equation

$$500 \frac{dx}{dt} = x(100 - x)$$

where t days is the time after the start of the study.

Given that at the start of the study 5 m^2 of the field was affected by the fungus,

- (b) solve the differential equation to show that

$$x = \frac{A}{1 + Be^{-\frac{1}{5}t}}$$

where A and B are constants to be found.

(6)

- (c) Hence find the area of the field affected by the fungus 10 days after the start of the study.

(2)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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[illegible]

14.

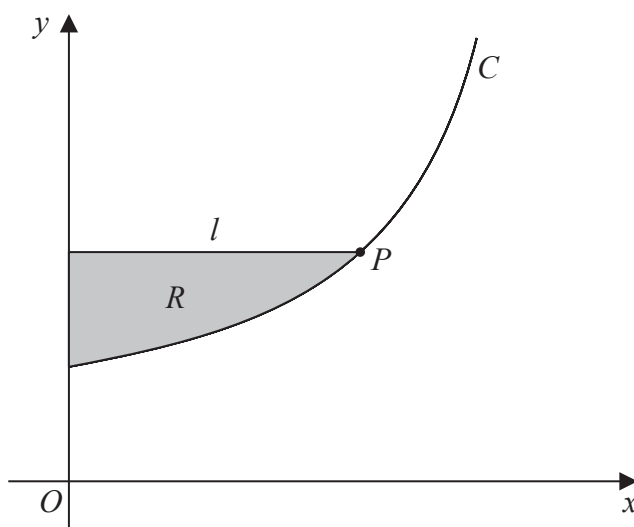


Figure 1

Figure 1 shows a sketch of the curve C with parametric equations

$$x = \frac{1}{2} \theta \sin \theta \quad y = \frac{1}{3} \sec \theta \quad 0 \leq \theta \leq \frac{5\pi}{12}$$

The point P lies on C where $\theta = \frac{\pi}{3}$

(a) Find, in simplest form, the exact Cartesian coordinates of P

(2)

The line l meets C at P and is parallel to the x -axis.

The region R shown shaded in Figure 1 is bounded by C , the y -axis and l .

(b) Show that the area of R is given by

$$A\pi\sqrt{3} - B \int_0^{\frac{\pi}{3}} (\tan \theta + \theta) d\theta$$

where A and B are rational numbers to be found.

(5)

(c) Hence, use algebraic integration to find the exact area of R

(3)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 10 marks)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Time 2 hours

Paper
reference

9MA0/02

Mathematics

Advanced

PAPER 2: Pure Mathematics 2

Mock set 4

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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1.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**(a) Find the values of x that satisfy

$$(i) \quad 2x - 10 < 3(5 - x) \quad (2)$$

$$(ii) \quad x^2 - 11x + 24 \leq 0 \quad (2)$$

(b) Hence find the values of x that satisfy both

$$2x - 10 < 3(5 - x) \quad \text{and} \quad x^2 - 11x + 24 \leq 0 \quad (1)$$



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 5 marks)



2.

$$f(x) = x^3 - 5x^2 + \frac{3}{x} - 4 \quad x > 0$$

(a) Find $f'(x)$.

(2)

Given that $f(0.5) = 0.875$ and $f(0.6) = -0.584$ (b) explain why the equation $f(x) = 0$ has a root in the interval $[0.5, 0.6]$

(1)

Given that this root is α and using 0.5 as a first approximation to α (c) apply the Newton-Raphson method once to $f(x)$ to find a second approximation to α .
Give your answer to 3 decimal places.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3.

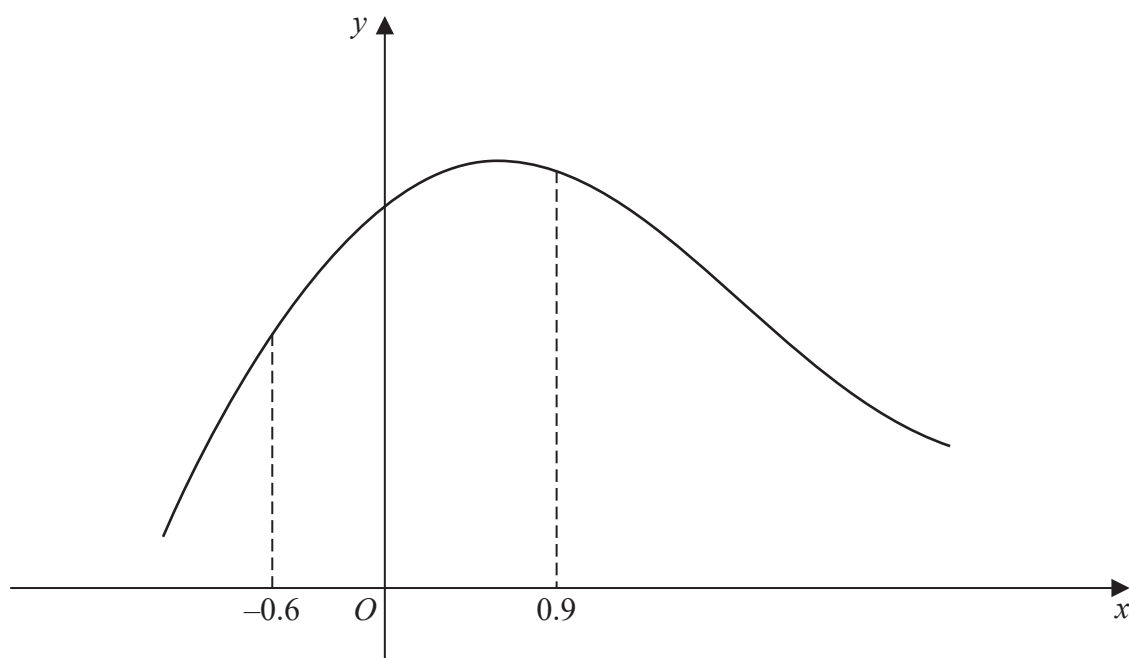
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$.

The table shows corresponding values of x and y for this curve between $x = -0.6$ and $x = 0.9$

x	-0.6	-0.3	0	0.3	0.6	0.9
y	1.811	2.342	2.718	2.941	3.011	2.944

- (a) Use the trapezium rule, with all the values of y in the table, to find an estimate for

$$\int_{-0.6}^{0.9} f(x) \, dx$$

Give your answer to 3 significant figures.

(3)

- (b) State whether the answer to part (a) is an underestimate or overestimate of

$$\int_{-0.6}^{0.9} f(x) \, dx$$

justifying your answer.

(1)

- (c) Using the answer to part (a), find an estimate for

$$\int_{-0.6}^{0.9} (8 - 2f(x)) \, dx$$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 7 marks)



$$f(x) = px^3 - 7x^2 - x + q$$

where p and q are constants.

Given that $(2x + 1)$ is a factor of $f(x)$

(a) show that

$$8q - p = 10 \tag{2}$$

Given that $(x - 1)$ is also a factor of $f(x)$

(b) find the value of p and the value of q . (2)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 4 marks)



5. Given that θ is small and in radians, use the small angle approximations to find an approximate value of

$$\frac{\cos 9\theta - 1}{\theta \sin \theta}$$

(3)

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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 3 marks)



6. The curve C has equation

$$y = ax^3 + bx^2 + 12x + 2$$

where a and b are constants.

(a) Find, in terms of a and b ,

(i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$

(3)

Given that

- the point $P\left(\frac{3}{2}, \frac{13}{2}\right)$ lies on C
- $\frac{d^2y}{dx^2} = 0$ at P

(b) find the value of a and the value of b .

(2)

(c) Show that P is a point of inflection.

(2)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 7 marks)



7.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Show that

$$\lim_{\delta x \rightarrow 0} \sum_{x=2}^3 \frac{(x-3)^2}{x^{\frac{3}{2}}} \delta x = p\sqrt{2} + q\sqrt{3}$$

where p and q are constants to be found.

(5)



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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 5 marks)



- 8.** Lina has started a business selling candles.
In the first week of trading, Lina sold 150 candles.

In a model for future trading, Lina predicts that the number of candles sold each week will increase by a fixed number, d , so that

- $150 + d$ candles will be sold in week 2
- $150 + 2d$ candles will be sold in week 3

and so on.

- (a) Use the model with $d = 10$ to find
- (i) the number of candles sold in week 30
 - (ii) the **total** number of candles sold in the first 30 weeks of trading.
- (4)**

Lina must sell at least 15 000 candles in the first year of trading to make a profit. Given that Lina's business does make a profit in the first year,

- (b) find, according to the model, the smallest possible value of d . (2)
- (c) State a limitation of the model. (1)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 7 marks)



9.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

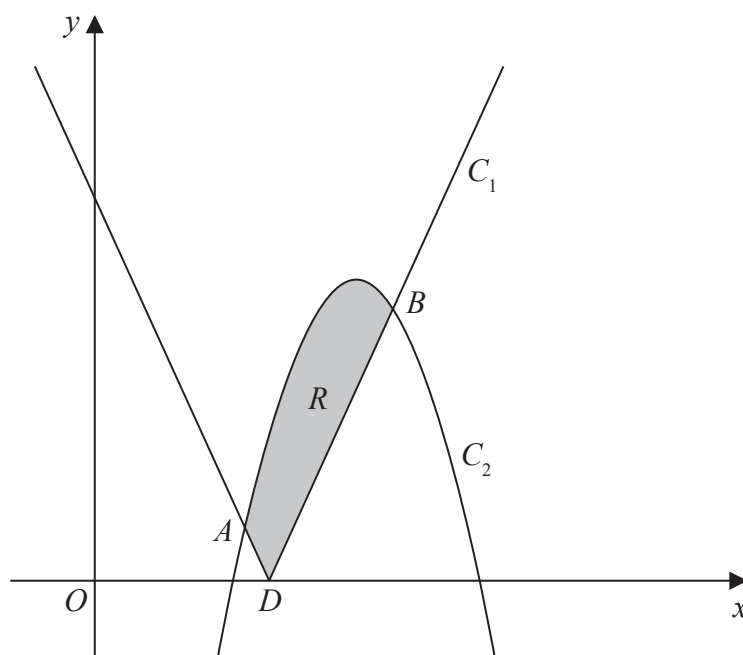


Figure 2

In Figure 2

- the graph C_1 has equation $y = |7 - 3x|$
- the curve C_2 has equation $y = -2x^2 + 14x - 19$
- C_1 intersects C_2 at the points A and B

Given that C_1 meets the x -axis at the point D , as shown in Figure 2,(a) find the x coordinate of D . (1)(b) Verify that the x coordinate of A is 2 (1)(c) Using algebra and showing your working, find the coordinates of B . (3)The region R , shown shaded in Figure 2, is bounded by C_1 and C_2 (d) Use algebraic integration to find the area of R . (5)

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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 10 marks)



10. The circle C_1 has Cartesian equation

$$x^2 + y^2 = 10x + k \quad x \in \mathbb{R} \quad y \in \mathbb{R}$$

where k is a constant.

The curve C_2 has parametric equations

$$x = t^2 \quad y = 2t \quad t \in \mathbb{R}$$

The curves C_1 and C_2 intersect at 4 distinct points.

Find the range of possible values for k , giving your answer in set notation.

(6)



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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 6 marks)



11. Prove, using algebra, that for all positive integers n that are **not** divisible by 3

$$n^2 - 1$$

is divisible by 3

(4)



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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 4 marks)



12.

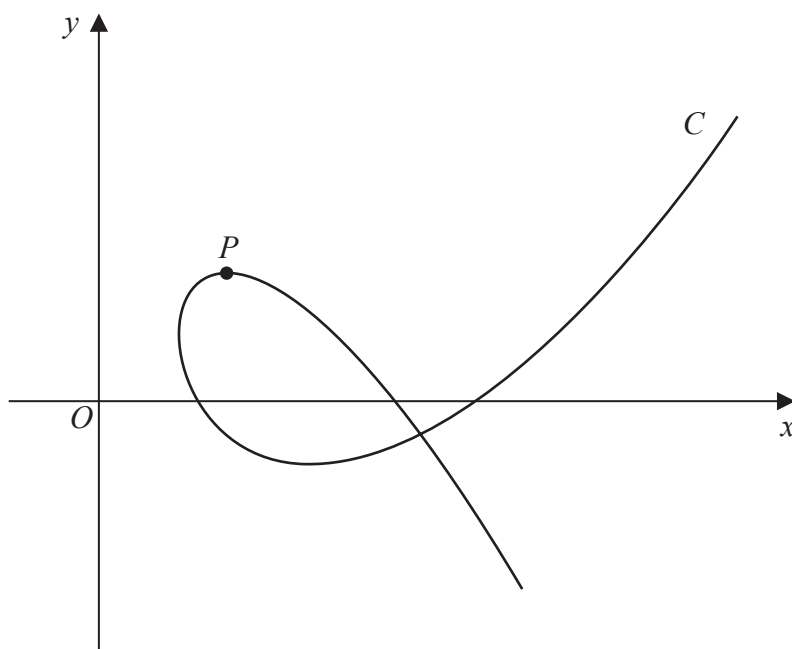


Figure 3

Figure 3 shows a sketch of the curve C with parametric equations

$$x = t^2 + t + 4 \quad y = t^3 - 12t + 5 \quad -5 \leq t \leq 5$$

- (a) Find $\frac{dy}{dx}$ in terms of t .

(2)

Given that the point P lies on C such that the tangent to C at P is parallel to the x -axis,

- (b) find the coordinates of P .

(2)

Given that this tangent meets C again at the point Q ,

- (c) find, using algebra, the coordinates of Q .

(3)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 7 marks)



- 13.** (a) Express $\sin x + \frac{1}{2} \cos x$ in the form $R \sin(x + \alpha)$ where R and α are constants,
 $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R and the value of α in radians to 3 decimal places.

(3)

The temperature, $\theta^{\circ}\text{C}$, of the water in a fish tank during a particular day is modelled by the equation

$$\theta = 26 + \sin\left(\frac{\pi t}{3} - 8\right) + \frac{1}{2} \cos\left(\frac{\pi t}{3} - 8\right) \quad 0 \leq t \leq 24$$

where t is the number of hours after midnight.

Using the equation of the model and the answer to part (a),

- (b) deduce the minimum temperature of the water during this day,

(1)

- (c) find the time of day when the minimum temperature occurs for the 3rd time.
Give your answer to the nearest minute.

(Solutions based entirely on calculator technology are not acceptable.)

(3)

The following day the mean temperature of the water in the fish tank increased but the variation in temperature remained the same.

- (d) State how the model could be refined to take this into account.

(1)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 8 marks)



14. The curve C has equation

$$y = \frac{e^{2x}}{x^3 + 4} \quad x > -1$$

(a) Find $\frac{dy}{dx}$

(3)

The point P with x coordinate p lies on C .

The line l is the tangent to C at P .

Given that l passes through the origin,

(b) show that $x = p$ is a solution of the equation

$$x^4 - 2x^3 + 4x - 2 = 0$$

(3)

The iteration formula

$$x_{n+1} = \frac{2x_n^3 + 2}{x_n^3 + 4}$$

with $x_1 = 0.5$ is used to find an approximation for p .

(c) Use the iteration formula to find the value, to 4 decimal places, of

(i) x_2

$$(ii) \quad p$$

(3)

(d) Hence find the gradient of l , giving your answer to 2 decimal places.

(1)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 10 marks)



15. (a) Find the values of the constants A , B and C for which

$$\frac{u^2}{u^2 - 1} \equiv A + \frac{B}{u + 1} + \frac{C}{u - 1} \quad (3)$$

(b) Use the substitution $u = \sqrt{1 + e^{3x}}$ to show that

$$\int_{\frac{1}{2} \ln 3}^{\frac{1}{3} \ln 8} \sqrt{1 + e^{3x}} \, dx = k \int_{\alpha}^{\beta} \frac{u^2}{u^2 - 1} \, du$$

where k , α and β are constants to be found. (5)

(c) Using the answers from parts (a) and (b) and algebraic integration, show that

$$\int_{\frac{1}{3} \ln 3}^{\frac{1}{3} \ln 8} \sqrt{1 + e^{3x}} \, dx = p + q \ln r$$

where p, q and r are rational numbers to be found. (4)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 12 marks)

TOTAL FOR PAPER IS 100 MARKS



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Mathematics

Advanced

Paper 1: Pure Mathematics 1

Sample Assessment Material for first teaching September 2017

Time: 2 hours

Paper Reference

9MA0/01

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
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Information

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- There are 15 questions in this question paper. The total mark for this paper is 100.
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– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer cross it out and put your new answer and any working out underneath.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. The curve C has equation

$$y = 3x^4 - 8x^3 - 3$$

(a) Find (i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$ (3)

- (b) Verify that C has a stationary point when $x = 2$ (2)

- (c) Determine the nature of this stationary point, giving a reason for your answer. (2)

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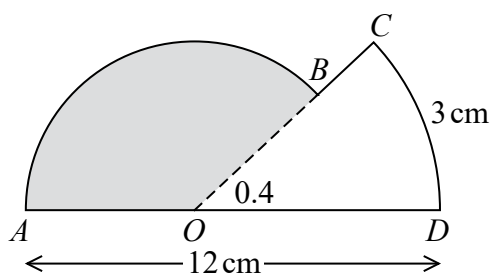
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Question 1 continued

(Total for Question 1 is 7 marks)

2.

**Figure 1**

The shape $ABCDOA$, as shown in Figure 1, consists of a sector COD of a circle centre O joined to a sector AOB of a different circle, also centre O .

Given that arc length $CD = 3$ cm, $\angle COD = 0.4$ radians and AOD is a straight line of length 12 cm,

(a) find the length of OD , (2)

(b) find the area of the shaded sector AOB . (3)

(Total for Question 2 is 5 marks)

3. A circle C has equation

$$x^2 + y^2 - 4x + 10y = k$$

where k is a constant.

- (a) Find the coordinates of the centre of C .

(2)

- (b) State the range of possible values for k .

(2)

(Total for Question 3 is 4 marks)

4. Given that a is a positive constant and

$$\int_a^{2a} \frac{t+1}{t} dt = \ln 7$$

show that $a = \ln k$, where k is a constant to be found.

(4)

(Total for Question 4 is 4 marks)

5. A curve C has parametric equations

$$x = 2t - 1, \quad y = 4t - 7 + \frac{3}{t}, \quad t \neq 0$$

Show that the Cartesian equation of the curve C can be written in the form

$$y = \frac{2x^2 + ax + b}{x + 1}, \quad x \neq -1$$

where a and b are integers to be found.

(3)

(Total for Question 5 is 3 marks)

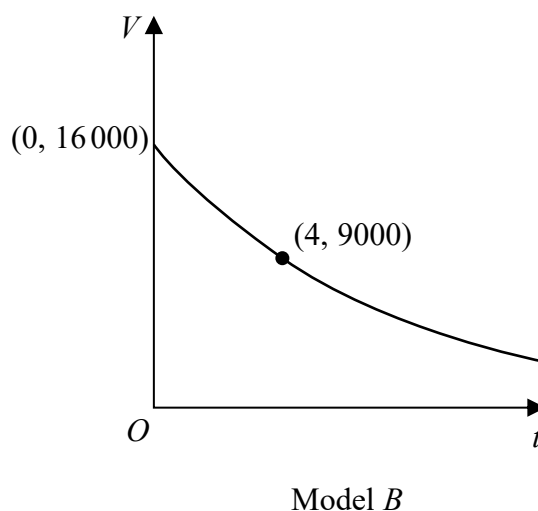
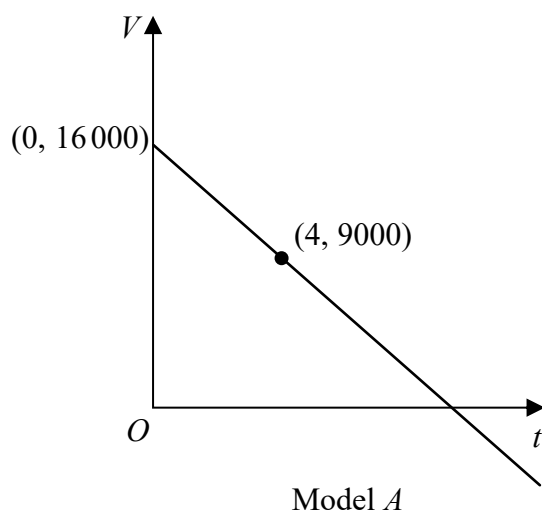
6. A company plans to extract oil from an oil field.

The daily volume of oil V , measured in barrels that the company will extract from this oil field depends upon the time, t years, after the start of drilling.

The company decides to use a model to estimate the daily volume of oil that will be extracted. The model includes the following assumptions:

- The initial daily volume of oil extracted from the oil field will be 16 000 barrels.
- The daily volume of oil that will be extracted exactly 4 years after the start of drilling will be 9000 barrels.
- The daily volume of oil extracted will decrease over time.

The diagram below shows the graphs of two possible models.



- (a) (i) Use model A to estimate the daily volume of oil that will be extracted exactly 3 years after the start of drilling.
- (ii) Write down a limitation of using model A. (2)
- (b) (i) Using an exponential model and the information given in the question, find a possible equation for model B.
- (ii) Using your answer to (b)(i) estimate the daily volume of oil that will be extracted exactly 3 years after the start of drilling. (5)

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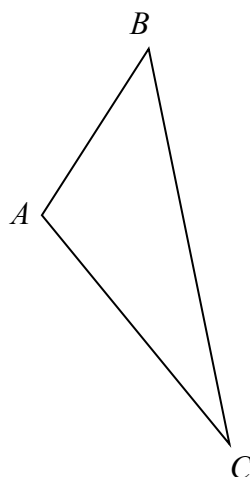
**Figure 2**

Figure 2 shows a sketch of a triangle ABC .

Given $\vec{AB} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\vec{BC} = \mathbf{i} - 9\mathbf{j} + 3\mathbf{k}$,

show that $\angle BAC = 105.9^\circ$ to one decimal place.

(5)

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Question 7 continued

(Total for Question 7 is 5 marks)

8.

(2)

(2)

(2)

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Question 8 continued

(Total for Question 8 is 6 marks)

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(Total for Question 9 is 5 marks)

10. Given that θ is measured in radians, prove, from first principles, that the derivative of $\sin \theta$ is $\cos \theta$

You may assume the formula for $\sin(A \pm B)$ and that as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$

(5)

(Total for Question 10 is 5 marks)

11. An archer shoots an arrow.

The height, H metres, of the arrow above the ground is modelled by the formula

$$H = 1.8 + 0.4d - 0.002d^2, \quad d \geq 0$$

where d is the horizontal distance of the arrow from the archer, measured in metres.

Given that the arrow travels in a vertical plane until it hits the ground,

(a) find the horizontal distance travelled by the arrow, as given by this model. (3)

(b) With reference to the model, interpret the significance of the constant 1.8 in the formula. (1)

(c) Write $1.8 + 0.4d - 0.002d^2$ in the form

$$A - B(d - C)^2$$

where A , B and C are constants to be found. (3)

It is decided that the model should be adapted for a different archer.

The adapted formula for this archer is

$$H = 2.1 + 0.4d - 0.002d^2, \quad d \geq 0$$

Hence or otherwise, find, for the adapted model

- (d) (i) the maximum height of the arrow above the ground.
(ii) the horizontal distance, from the archer, of the arrow when it is at its maximum height. (2)

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Question 11 continued

(Total for Question 11 is 9 marks)

12. In a controlled experiment, the number of microbes, N , present in a culture T days after the start of the experiment were counted.

N and T are expected to satisfy a relationship of the form

$$N = aT^b, \quad \text{where } a \text{ and } b \text{ are constants}$$

- (a) Show that this relationship can be expressed in the form

$$\log_{10} N = m \log_{10} T + c$$

giving m and c in terms of the constants a and/or b .

(2)

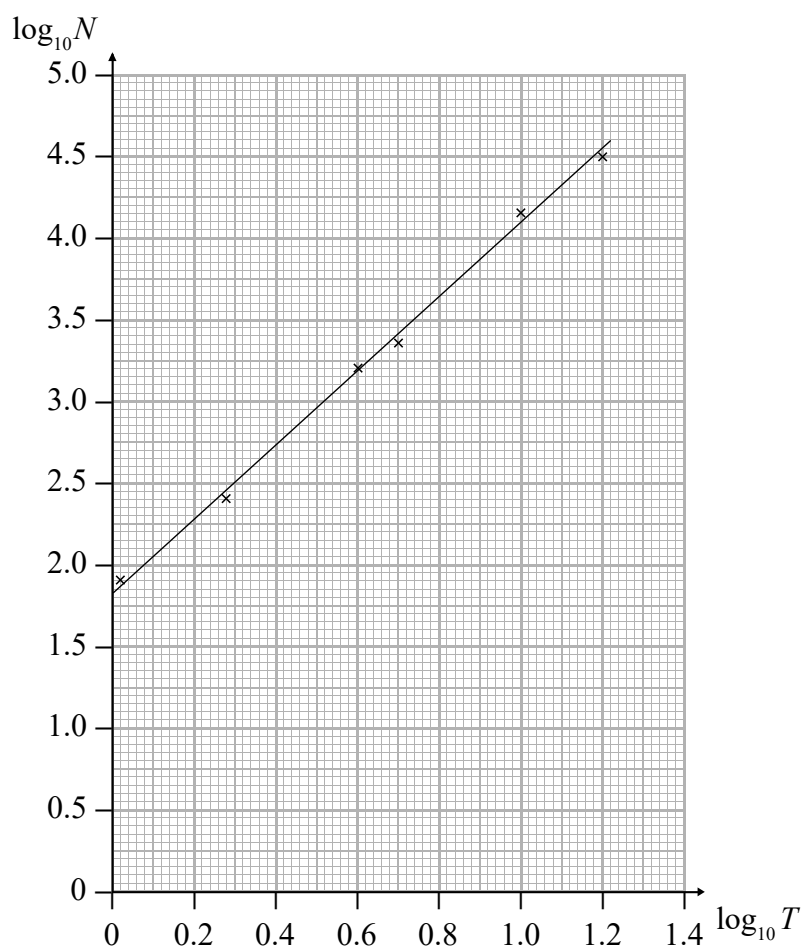


Figure 3

Figure 3 shows the line of best fit for values of $\log_{10} N$ plotted against values of $\log_{10} T$

- (b) Use the information provided to estimate the number of microbes present in the culture 3 days after the start of the experiment.
- (c) Explain why the information provided could not reliably be used to estimate the day when the number of microbes in the culture first exceeds 1 000 000.
- (d) With reference to the model, interpret the value of the constant a .

(4)

(2)

(1)

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Question 12 continued

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Question 12 continued

(Total for Question 12 is 9 marks)

13. The curve C has parametric equations

$$x = 2 \cos t, \quad y = \sqrt{3} \cos 2t, \quad 0 \leq t \leq \pi$$

- (a) Find an expression for $\frac{dy}{dx}$ in terms of t .

(2)

The point P lies on C where $t = \frac{2\pi}{3}$

The line l is the normal to C at P .

- (b) Show that an equation for l is

$$2x - 2\sqrt{3}y - 1 = 0$$

(5)

The line l intersects the curve C again at the point Q .

- (c) Find the exact coordinates of Q .

You must show clearly how you obtained your answers.

(6)

Question 13 continued

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14.

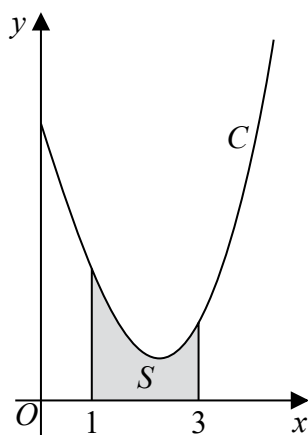


Figure 4

Figure 4 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 \ln x}{3} - 2x + 5, \quad x > 0$$

The finite region S , shown shaded in Figure 4, is bounded by the curve C , the line with equation $x = 1$, the x -axis and the line with equation $x = 3$

The table below shows corresponding values of x and y with the values of y given to 4 decimal places as appropriate.

x	1	1.5	2	2.5	3
y	3	2.3041	1.9242	1.9089	2.2958

- (a) Use the trapezium rule, with all the values of y in the table, to obtain an estimate for the area of S , giving your answer to 3 decimal places. (3)
- (b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of S . (1)
- (c) Show that the exact area of S can be written in the form $\frac{a}{b} + \ln c$, where a , b and c are integers to be found. (6)

(In part c, solutions based entirely on graphical or numerical methods are not acceptable.)

Question 14 continued

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15.

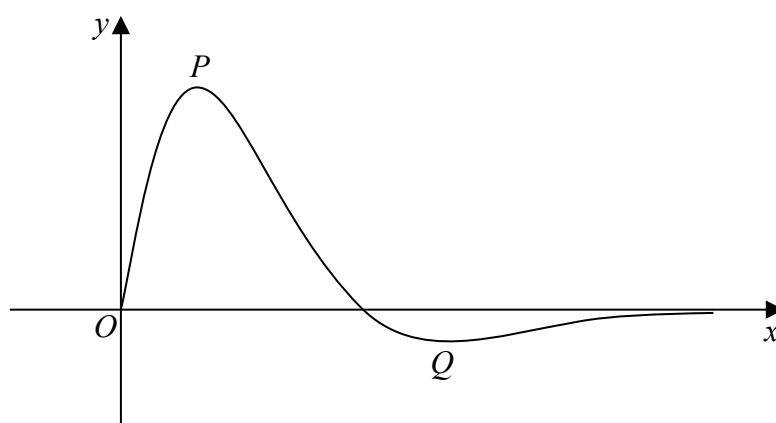
**Figure 5**

Figure 5 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \frac{4\sin 2x}{e^{\sqrt{2}x-1}}, \quad 0 \leq x \leq \pi$$

The curve has a maximum turning point at P and a minimum turning point at Q as shown in Figure 5.

- (a) Show that the x coordinates of point P and point Q are solutions of the equation

$$\tan 2x = \sqrt{2} \quad (4)$$

- (b) Using your answer to part (a), find the x -coordinate of the minimum turning point on the curve with equation

(i) $y = f(2x)$.

(ii) $y = 3 - 2f(x)$. (4)

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Question 15 continued

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(Total for Question 15 is 8 marks)

TOTAL FOR PAPER IS 100 MARKS

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Candidate Number

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Mathematics

Advanced

Paper 2: Pure Mathematics 2

Sample Assessment Material for first teaching September 2017

Time: 2 hours

Paper Reference

9MA0/02

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer cross it out and put your new answer and any working out underneath.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1.

$$f(x) = 2x^3 - 5x^2 + ax + a$$

Given that $(x + 2)$ is a factor of $f(x)$, find the value of the constant a .

(3)

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(Total for Question 1 is 3 marks)

2. Some A level students were given the following question.

Solve, for $-90^\circ < \theta < 90^\circ$, the equation

$$\cos \theta = 2 \sin \theta$$

The attempts of two of the students are shown below.

Student A

$$\cos \theta = 2 \sin \theta$$

$$\tan \theta = 2$$

$$\theta = 63.4^\circ$$

Student B

$$\cos \theta = 2 \sin \theta$$

$$\cos^2 \theta = 4 \sin^2 \theta$$

$$1 - \sin^2 \theta = 4 \sin^2 \theta$$

$$\sin^2 \theta = \frac{1}{5}$$

$$\sin \theta = \pm \frac{1}{\sqrt{5}}$$

$$\theta = \pm 26.6^\circ$$

- (a) Identify an error made by student A.

(1)

Student B gives $\theta = -26.6^\circ$ as one of the answers to $\cos \theta = 2 \sin \theta$.

- (b) (i) Explain why this answer is incorrect.

- (ii) Explain how this incorrect answer arose.

(2)

(Total for Question 2 is 3 marks)

3. Given $y = x(2x + 1)^4$, show that

$$\frac{dy}{dx} = (2x + 1)^n (Ax + B)$$

where n , A and B are constants to be found.

(4)

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(Total for Question 3 is 4 marks)

4. Given

$$f(x) = e^x, \quad x \in \mathbb{R}$$

$$g(x) = 3 \ln x, \quad x > 0, x \in \mathbb{R}$$

(a) find an expression for $gf(x)$, simplifying your answer.

(2)

(b) Show that there is only one real value of x for which $gf(x) = fg(x)$

(3)

(Total for Question 4 is 5 marks)

5. The mass, m grams, of a radioactive substance, t years after first being observed, is modelled by the equation

$$m = 25e^{-0.05t}$$

According to the model,

- (a) find the mass of the radioactive substance six months after it was first observed, (2)

- (b) show that $\frac{dm}{dt} = km$, where k is a constant to be found. (2)

(Total for Question 5 is 4 marks)

6. Complete the table below. The first one has been done for you.

For each statement you must state if it is always true, sometimes true or never true, giving a reason in each case.

Statement	Always True	Sometimes True	Never True	Reason
The quadratic equation $ax^2 + bx + c = 0$, ($a \neq 0$) has 2 real roots.		✓		It only has 2 real roots when $b^2 - 4ac > 0$. When $b^2 - 4ac = 0$ it has 1 real root and when $b^2 - 4ac < 0$ it has 0 real roots.
(i) When a real value of x is substituted into $x^2 - 6x + 10$ the result is positive. (2)				
(ii) If $ax > b$ then $x > \frac{b}{a}$ (2)				
(iii) The difference between consecutive square numbers is odd. (2)				

(Total for Question 6 is 6 marks)

7. (a) Use the binomial expansion, in ascending powers of x , to show that

$$\sqrt{4-x} = 2 - \frac{1}{4}x + kx^2 + \dots$$

where k is a rational constant to be found.

(4)

A student attempts to substitute $x = 1$ into both sides of this equation to find an approximate value for $\sqrt{3}$.

- (b) State, giving a reason, if the expansion is valid for this value of x .

(1)

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Question 7 continued

(Total for Question 7 is 5 marks)

8.

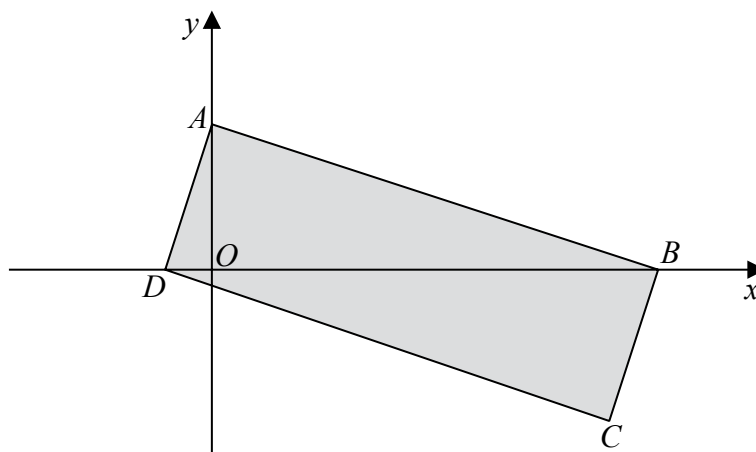
**Figure 1**

Figure 1 shows a rectangle $ABCD$.

The point A lies on the y -axis and the points B and D lie on the x -axis as shown in Figure 1.

Given that the straight line through the points A and B has equation $5y + 2x = 10$

(a) show that the straight line through the points A and D has equation $2y - 5x = 4$ (4)

(b) find the area of the rectangle $ABCD$. (3)

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Question 8 continued

(Total for Question 8 is 7 marks)

9. Given that A is constant and

$$\int_1^4 (3\sqrt{x} + A) dx = 2A^2$$

show that there are exactly two possible values for A .

(5)

(Total for Question 9 is 5 marks)

10. In a geometric series the common ratio is r and sum to n terms is S_n

Given

$$S_{\infty} = \frac{8}{7} \times S_6$$

show that $r = \pm \frac{1}{\sqrt{k}}$, where k is an integer to be found.

(4)

(Total for Question 10 is 4 marks)

Figure 2 shows a sketch of part of the graph $y = f(x)$, where

$$f(x) = 2|3 - x| + 5, \quad x \geq 0$$

- (a) State the range of f

(1)

- (b) Solve the equation

$$f(x) = \frac{1}{2}x + 30$$

(3)

Given that the equation $f(x) = k$, where k is a constant, has two distinct roots,

- (c) state the set of possible values for k .

(2)

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Question 11 continued

(Total for Question 11 is 6 marks)

12. (a) Solve, for $-180^\circ \leq x < 180^\circ$, the equation

$$3 \sin^2 x + \sin x + 8 = 9 \cos^2 x$$

giving your answers to 2 decimal places.

(6)

(b) Hence find the smallest positive solution of the equation

$$3 \sin^2(2\theta - 30^\circ) + \sin(2\theta - 30^\circ) + 8 = 9 \cos^2(2\theta - 30^\circ)$$

giving your answer to 2 decimal places.

(2)

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Question 12 continued

(Total for Question 12 is 8 marks)

13. (a) Express $10 \cos \theta - 3 \sin \theta$ in the form $R \cos (\theta + \alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$.
Give the exact value of R and give the value of α , in degrees, to 2 decimal places.

(3)

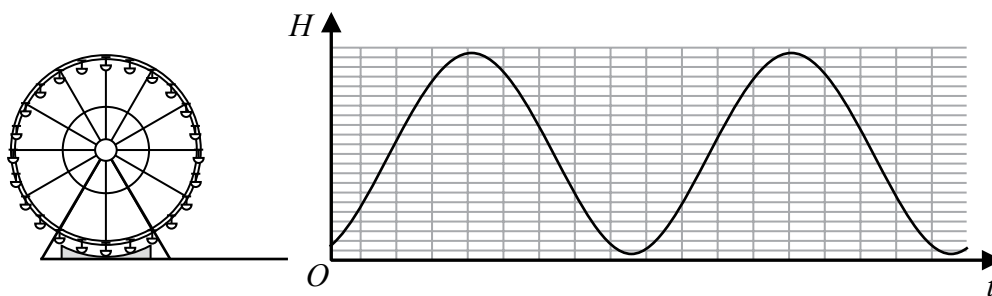


Figure 3

The height above the ground, H metres, of a passenger on a Ferris wheel t minutes after the wheel starts turning, is modelled by the equation

$$H = a - 10 \cos(80t)^\circ + 3 \sin(80t)^\circ$$

where a is a constant.

Figure 3 shows the graph of H against t for two complete cycles of the wheel.

Given that the initial height of the passenger above the ground is 1 metre,

- (b) (i) find a complete equation for the model,
(ii) hence find the maximum height of the passenger above the ground.
- (c) Find the time taken, to the nearest second, for the passenger to reach the maximum height on the second cycle.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(3)

It is decided that, to increase profits, the speed of the wheel is to be increased.

- (d) How would you adapt the equation of the model to reflect this increase in speed?

(1)

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Question 13 continued

(Total for Question 13 is 9 marks)

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Question 14 continued

(Total for Question 14 is 9 marks)

15.

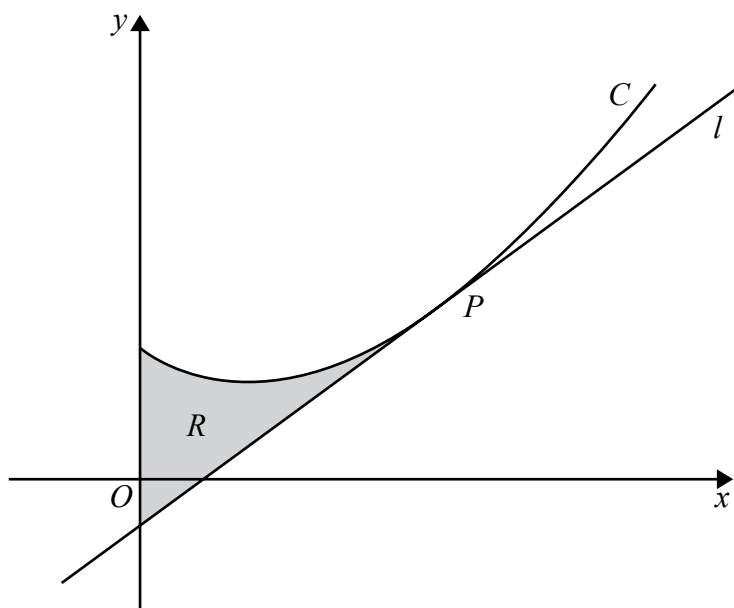
**Figure 4**

Figure 4 shows a sketch of the curve C with equation

$$y = 5x^{\frac{3}{2}} - 9x + 11, x \geq 0$$

The point P with coordinates $(4, 15)$ lies on C .

The line l is the tangent to C at the point P .

The region R , shown shaded in Figure 4, is bounded by the curve C , the line l and the y -axis.

Show that the area of R is 24, making your method clear.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(10)

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- where A , B and C are integers to be found. (3)

Question 16 continued

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(Total for Question 16 is 12 marks)

TOTAL FOR PAPER IS 100 MARKS

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Candidate Number

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Mathematics

Advanced

Paper 1: Pure Mathematics 1

Specimen Paper

Time: 2 hours

Paper Reference

9MA0/01

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

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- Check your answers if you have time at the end

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Answer ALL questions. Write your answers in the spaces provided.

1.

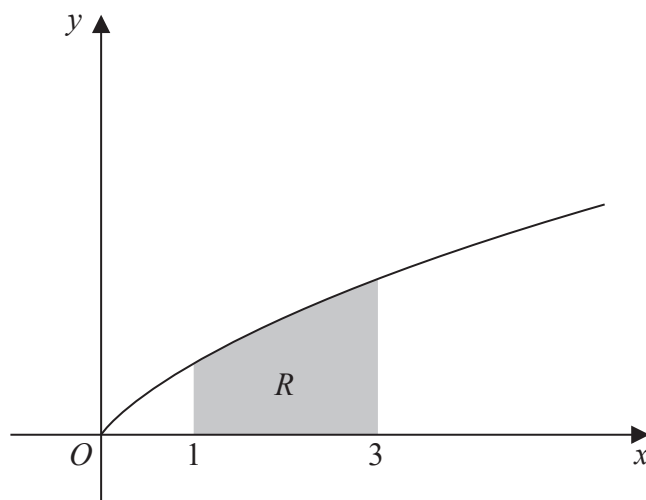


Figure 1

Figure 1 shows a sketch of the curve with equation $y = \frac{x}{1 + \sqrt{x}}$, $x \geq 0$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 1$, the x -axis and the line with equation $x = 3$

The table below shows corresponding values of x and y for $y = \frac{x}{1 + \sqrt{x}}$

x	1	1.5	2	2.5	3
y	0.5	0.6742	0.8284	0.9686	1.0981

- (a) Use the trapezium rule, with all the values of y in the table, to find an estimate for the area of R , giving your answer to 3 decimal places.

(3)

- (b) Explain how the trapezium rule can be used to give a better approximation for the area of R .

(1)

- (c) Giving your answer to 3 decimal places in each case, use your answer to part (a) to deduce an estimate for

(i) $\int_1^3 \frac{5x}{1 + \sqrt{x}} dx$

(ii) $\int_1^3 \left(6 + \frac{x}{1 + \sqrt{x}} \right) dx$

(2)

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. (a) Show that the binomial expansion of

$$(4 + 5x)^{\frac{1}{2}}$$

in ascending powers of x , up to and including the term in x^2 is

$$2 + \frac{5}{4}x + kx^2$$

giving the value of the constant k as a simplified fraction.

(4)

- (b) (i) Use the expansion from part (a), with $x = \frac{1}{10}$, to find an approximate value for $\sqrt{2}$

Give your answer in the form $\frac{p}{q}$ where p and q are integers.

- (ii) Explain why substituting $x = \frac{1}{10}$ into this binomial expansion leads to a valid approximation.

(4)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. A sequence of numbers $a_1, a_2, a_3, \dots$ is defined by

$$a_1 = 3$$

$$a_{n+1} = \frac{a_n - 3}{a_n - 2}, \quad n \in \mathbb{N}$$

(a) Find $\sum_{r=1}^{100} a_r$

(3)

(b) Hence find $\sum_{r=1}^{100} a_r + \sum_{r=1}^{99} a_r$

(1)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 4 marks)



4. Relative to a fixed origin O ,

the point A has position vector $\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}$,
the point B has position vector $4\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$,
and the point C has position vector $2\mathbf{i} + 10\mathbf{j} + 9\mathbf{k}$.

Given that $ABCD$ is a parallelogram,

(a) find the position vector of point D .

(2)

The vector $\overrightarrow{AX}$ has the same direction as $\overrightarrow{AB}$.

Given that $|\overrightarrow{AX}| = 10\sqrt{2}$,

(b) find the position vector of X .

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



5.

$$f(x) = x^3 + ax^2 - ax + 48, \text{ where } a \text{ is a constant}$$

Given that $f(-6) = 0$

(a) (i) show that $a = 4$

(ii) express $f(x)$ as a product of two algebraic factors.

(4)

Given that $2\log_2(x+2) + \log_2 x - \log_2(x-6) = 3$

(b) show that $x^3 + 4x^2 - 4x + 48 = 0$

(4)

(c) hence explain why

$$2\log_2(x+2) + \log_2 x - \log_2(x-6) = 3$$

has no real roots.

(2)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



6.

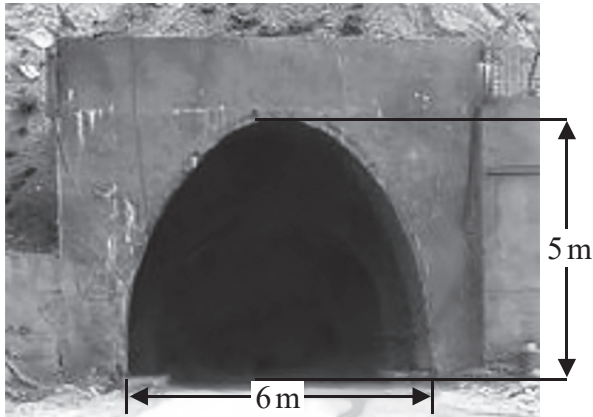


Figure 2

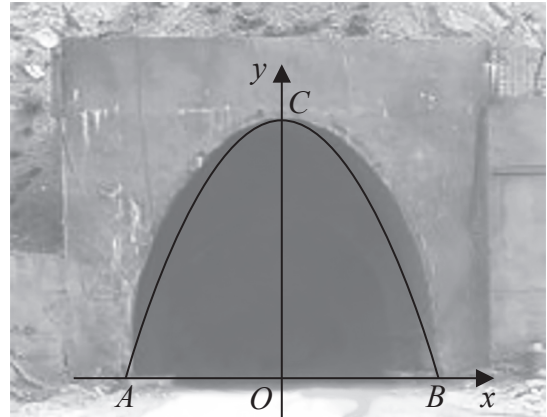


Figure 3

Figure 2 shows the entrance to a road tunnel. The maximum height of the tunnel is measured as 5 metres and the width of the base of the tunnel is measured as 6 metres.

Figure 3 shows a quadratic curve BCA used to model this entrance.

The points A , O , B and C are assumed to lie in the same vertical plane and the ground AOB is assumed to be horizontal.

- (a) Find an equation for curve BCA . (3)

A coach has height 4.1 m and width 2.4 m.

- (b) Determine whether or not it is possible for the coach to enter the tunnel. (2)

- (c) Suggest a reason why this model may not be suitable to determine whether or not the coach can pass through the tunnel. (1)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7.

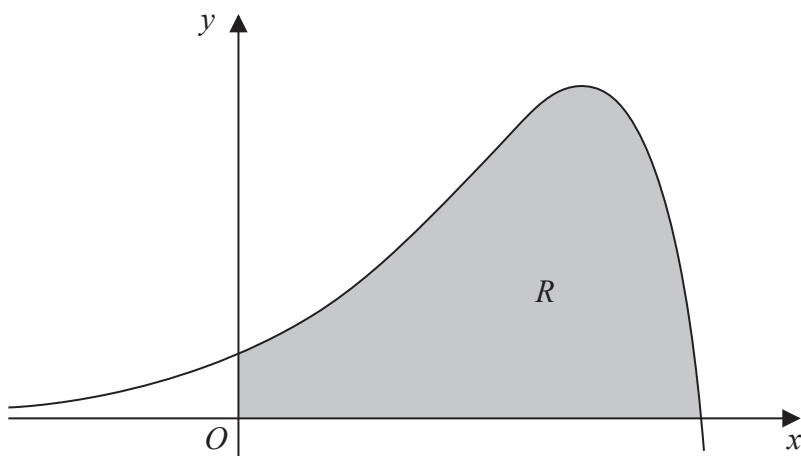


Figure 4

Figure 4 shows a sketch of part of the curve with equation

$$y = 2e^{2x} - xe^{2x}, \quad x \in \mathbb{R}$$

The finite region R , shown shaded in Figure 4, is bounded by the curve, the x -axis and the y -axis.

Use calculus to show that the exact area of R can be written in the form $pe^4 + q$, where p and q are rational constants to be found.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(5)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 5 marks)



8. There were 2100 tonnes of wheat harvested on a farm during 2017.

The mass of wheat harvested during each subsequent year is expected to increase by 1.2% per year.

- (a) Find the total mass of wheat expected to be harvested from 2017 to 2030 inclusive, giving your answer to 3 significant figures.

(2)

Each year it costs

- £5.15 per tonne to harvest the first 2000 tonnes of wheat
- £6.45 per tonne to harvest wheat in excess of 2000 tonnes

- (b) Use this information to find the expected cost of harvesting the wheat from 2017 to 2030 inclusive. Give your answer to the nearest £1000

(3)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 5 marks)



9. The curve C has equation

$$y = 2x^3 + 5$$

The curve C passes through the point $P(1, 7)$.

Use differentiation from first principles to find the value of the gradient of the tangent to C at P .

(5)



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Question 9 continued

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(Total for Question 9 is 5 marks)



10. The function f is defined by

$$f: x \mapsto \frac{3x-5}{x+1}, \quad x \in \mathbb{R}, \quad x \neq -1$$

(a) Find $f^{-1}(x)$.

(3)

(b) Show that

$$ff(x) = \frac{x+a}{x-1}, \quad x \in \mathbb{R}, \quad x \neq \pm 1$$

where a is an integer to be found.

(4)

The function g is defined by

$$g: x \mapsto x^2 - 3x, \quad x \in \mathbb{R}, \quad 0 \leq x \leq 5$$

(c) Find the value of $fg(2)$.

(2)

(d) Find the range of g .

(3)

(e) Explain why the function g does not have an inverse.

(1)



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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 13 marks)



11.

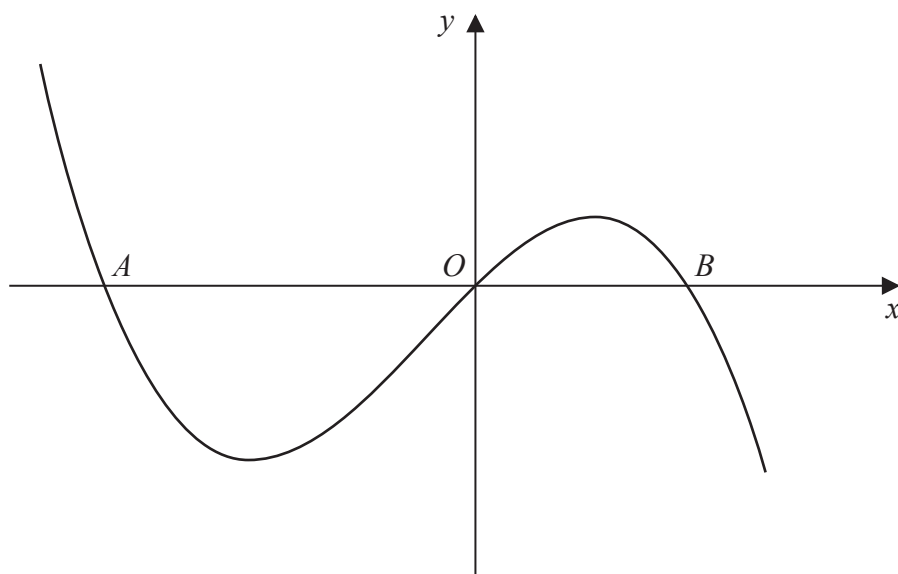


Figure 5

Figure 5 shows a sketch of the curve C with equation $y = f(x)$. The curve C crosses the x -axis at the origin, O , and at the points A and B as shown in Figure 5.

Given that

$$f'(x) = k - 4x - 3x^2$$

where k is constant,

- (a) show that C has a point of inflection at $x = -\frac{2}{3}$ (3)

Given also that the distance $AB = 4\sqrt{2}$

- (b) find, showing your working, the integer value of k . (7)



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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 10 marks)



12. Show that

$$\int_0^{\frac{\pi}{2}} \frac{\sin 2\theta}{1 + \cos \theta} d\theta = 2 - 2 \ln 2 \quad (7)$$

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 7 marks)



13. (a) Express $2 \sin \theta - 1.5 \cos \theta$ in the form $R \sin(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

State the value of R and give the value of α to 4 decimal places.

(3)

Tom models the depth of water, D metres, at Southview harbour on 18th October 2017 by the formula

$$D = 6 + 2 \sin\left(\frac{4\pi t}{25}\right) - 1.5 \cos\left(\frac{4\pi t}{25}\right), \quad 0 \leq t \leq 24$$

where t is the time, in hours, after 00:00 hours on 18th October 2017.

Use Tom's model to

- (b) find the depth of water at 00:00 hours on 18th October 2017, (1)
- (c) find the maximum depth of water, (1)
- (d) find the time, in the afternoon, when the maximum depth of water occurs. Give your answer to the nearest minute. (3)

Tom's model is supported by measurements of D taken at regular intervals on 18th October 2017. Jolene attempts to use a similar model in order to model the depth of water at Southview harbour on 19th October 2017.

Jolene models the depth of water, H metres, at Southview harbour on 19th October 2017 by the formula

$$H = 6 + 2 \sin\left(\frac{4\pi x}{25}\right) - 1.5 \cos\left(\frac{4\pi x}{25}\right), \quad 0 \leq x \leq 24$$

where x is the time, in hours, after 00:00 hours on 19th October 2017.

By considering the depth of water at 00:00 hours on 19th October 2017 for both models,

- (e) (i) explain why Jolene's model is not correct,
- (ii) hence find a suitable model for H in terms of x . (3)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 11 marks)



14.

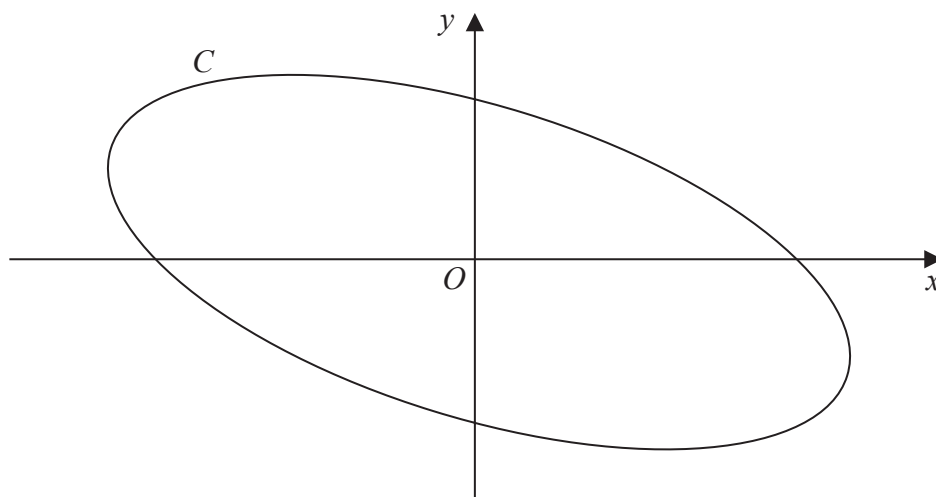


Figure 6

Figure 6 shows a sketch of the curve C with parametric equations

$$x = 4 \cos\left(t + \frac{\pi}{6}\right), \quad y = 2 \sin t, \quad 0 < t \leq 2\pi$$

Show that a Cartesian equation of C can be written in the form

$$(x + y)^2 + ay^2 = b$$

where a and b are integers to be found.

(5)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS



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Surname

Other names

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Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced

Paper 2: Pure Mathematics 2

Specimen Paper

Time: 2 hours

Paper Reference

9MA0/02

You must have:

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1.

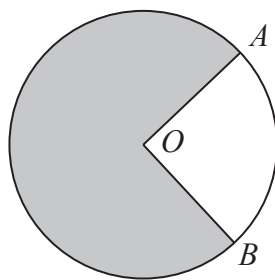


Figure 1

Figure 1 shows a circle with centre O . The points A and B lie on the circumference of the circle.

The area of the major sector, shown shaded in Figure 1, is 135 cm^2 .

The reflex angle AOB is 4.8 radians.

Find the exact length, in cm, of the minor arc AB , giving your answer in the form $a\pi + b$, where a and b are integers to be found.

(4)

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 4 marks)



$$1 + 4 \cos \theta + 3 \cos^2 \theta \approx 8 - 5\theta^2 \quad (3)$$

Adele uses $\theta = 5^\circ$ to test the approximation in part (a).

Adele's working is shown below.

Using my calculator, $1 + 4 \cos(5^\circ) + 3 \cos^2(5^\circ) = 7.962$, to 3 decimal places.

Using the approximation $8 - 5\theta^2$ gives $8 - 5(5)^2 = -117$

Therefore, $1 + 4\cos\theta + 3\cos^2\theta \approx 8 - 5\theta^2$ is not true for $\theta = 5^\circ$

(b) (i) Identify the mistake made by Adele in her working.

(ii) Show that $8 - 5\theta^2$ can be used to give a good approximation to $1 + 4\cos\theta + 3\cos^2\theta$ for an angle of size 5°

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. A cup of hot tea was placed on a table. At time t minutes after the cup was placed on the table, the temperature of the tea in the cup, θ °C, is modelled by the equation

$$\theta = 25 + Ae^{-0.03t}$$

where A is a constant.

The temperature of the tea was 75°C when the cup was placed on the table.

- (a) Find a complete equation for the model. (1)

- (b) Use the model to find the time taken for the tea to cool from 75°C to 60°C , giving your answer in minutes to one decimal place. (2)

Two hours after the cup was placed on the table, the temperature of the tea was measured as 20.3 °C.

Using this information,

- (c) evaluate the model, explaining your reasoning. (1)

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4. (a) Sketch the graph with equation

$$y = |2x - 5|$$

stating the coordinates of any points where the graph cuts or meets the coordinate axes.

(2)

- (b) Find the values of x which satisfy

$$|2x - 5| > 7$$

(2)

- (c) Find the values of x which satisfy

$$|2x - 5| > x - \frac{5}{2}$$

Write your answer in set notation.

(2)

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



5. The line l has equation

$$3x - 2y = k$$

where k is a real constant.

Given that the line l intersects the curve with equation

$$y = 2x^2 - 5$$

at two distinct points, find the range of possible values for k .

(5)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



6.

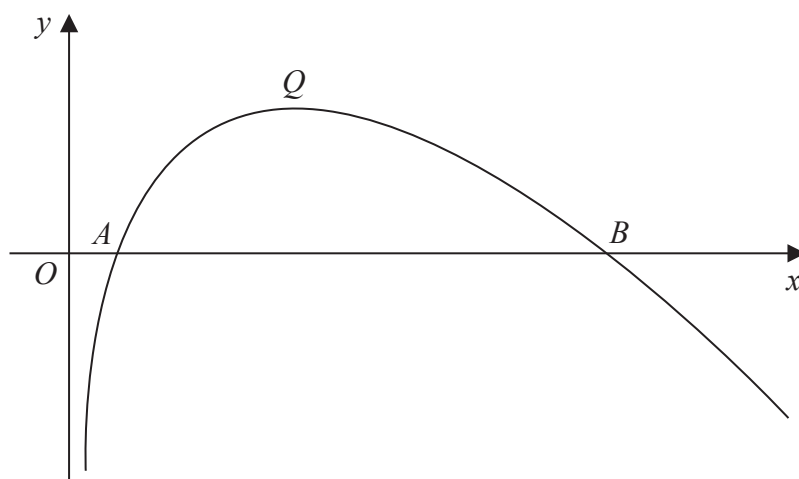


Figure 2

Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = (8 - x) \ln x, \quad x > 0$$

The curve cuts the x -axis at the points A and B and has a maximum turning point at Q , as shown in Figure 2.

- (a) Find the x coordinate of A and the x coordinate of B .

(1)

- (b) Show that the x coordinate of Q satisfies

$$x = \frac{8}{1 + \ln x}$$

(4)

- (c) Show that the x coordinate of Q lies between 3.5 and 3.6

(2)

- (d) Use the iterative formula

$$x_{n+1} = \frac{8}{1 + \ln x_n} \quad n \in \mathbb{N}$$

with $x_1 = 3.5$ to

- (i) find the value of x_5 to 4 decimal places,
 (ii) find the x coordinate of Q accurate to 2 decimal places.

(2)



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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 9 marks)



7. A bacterial culture has area $p \text{ mm}^2$ at time t hours after the culture was placed onto a circular dish.

A scientist states that at time t hours, the rate of increase of the area of the culture can be modelled as being proportional to the area of the culture.

- (a) Show that the scientist's model for p leads to the equation

$$p = ae^{kt}$$

where a and k are constants.

(4)

The scientist measures the values for p at regular intervals during the first 24 hours after the culture was placed onto the dish.

She plots a graph of $\ln p$ against t and finds that the points on the graph lie close to a straight line with gradient 0.14 and vertical intercept 3.95

- (b) Estimate, to 2 significant figures, the value of a and the value of k .

(3)

- (c) Hence show that the model for p can be rewritten as

$$p = ab^t$$

stating, to 3 significant figures, the value of the constant b .

(2)

With reference to this model,

- (d) (i) interpret the value of the constant a ,

- (ii) interpret the value of the constant b .

(2)

- (e) State a long term limitation of the model for p .

(1)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 12 marks)



8.

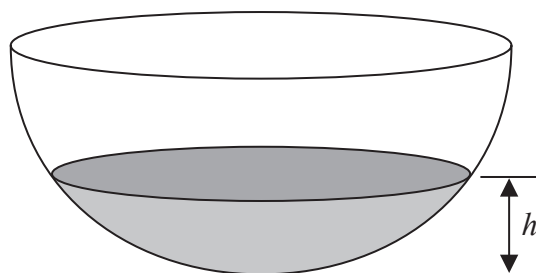


Figure 3

A bowl is modelled as a hemispherical shell as shown in Figure 3.

Initially the bowl is empty and water begins to flow into the bowl.

When the depth of the water is h cm, the volume of water, $V \text{ cm}^3$, according to the model is given by

$$V = \frac{1}{3} \pi h^2 (75 - h), \quad 0 \leq h \leq 24$$

The flow of water into the bowl is at a constant rate of $160\pi \text{ cm}^3 \text{ s}^{-1}$ for $0 \leq h \leq 12$

- (a) Find the rate of change of the depth of the water, in cm s^{-1} , when $h = 10$ (5)

Given that the flow of water into the bowl is increased to a constant rate of $300\pi \text{ cm}^3 \text{ s}^{-1}$ for $12 < h \leq 24$

- (b) find the rate of change of the depth of the water, in cm s^{-1} , when $h = 20$ (2)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 7 marks)



9. A circle with centre $A(3, -1)$ passes through the point $P(-9, 8)$ and the point $Q(15, -10)$

(a) Show that PQ is a diameter of the circle.

(2)

(b) Find an equation for the circle.

(3)

A point R also lies on the circle.

Given that the length of the chord PR is 20 units,

(c) find the length of the shortest distance from A to the chord PR .

Give your answer as a surd in its simplest form.

(2)

(d) Find the size of angle ARQ , giving your answer to the nearest 0.1 of a degree.

(2)



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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 9 marks)



10.

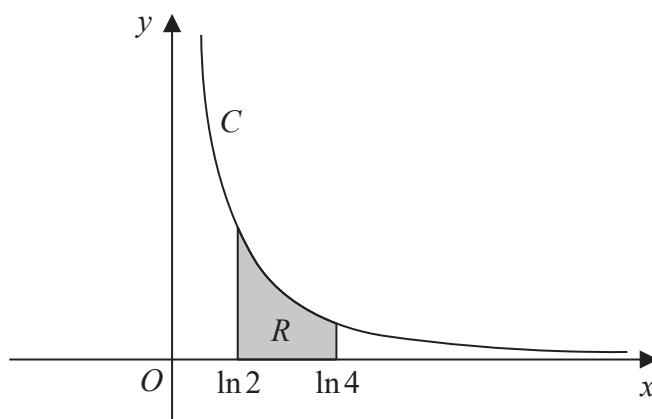


Figure 4

Figure 4 shows a sketch of the curve C with parametric equations

$$x = \ln(t + 2), \quad y = \frac{1}{t + 1}, \quad t > -\frac{2}{3}$$

(a) State the domain of values of x for the curve C .

(1)

The finite region R , shown shaded in Figure 4, is bounded by the curve C , the line with equation $x = \ln 2$, the x -axis and the line with equation $x = \ln 4$

(b) Use calculus to show that the area of R is $\ln\left(\frac{3}{2}\right)$.

(8)



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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 9 marks)



11. The second, third and fourth terms of an arithmetic sequence are $2k$, $5k - 10$ and $7k - 14$ respectively, where k is a constant.

Show that the sum of the first n terms of the sequence is a square number.

(5)



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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 5 marks)



12. A curve C is given by the equation

$$\sin x + \cos y = 0.5 \quad -\frac{\pi}{2} \leq x < \frac{3\pi}{2}, -\pi < y < \pi$$

A point P lies on C .

The tangent to C at the point P is parallel to the x -axis.

Find the exact coordinates of all possible points P , justifying your answer.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(7)



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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 7 marks)



13. (a) Show that

$$\operatorname{cosec} 2x + \cot 2x \equiv \cot x, \quad x \neq 90n^\circ, n \in \mathbb{Z} \quad (5)$$

(b) Hence, or otherwise, solve, for $0 \leq \theta < 180^\circ$,

$$\operatorname{cosec}(4\theta + 10^\circ) + \cot(4\theta + 10^\circ) = \sqrt{3}$$

You must show your working.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(5)



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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 10 marks)



14. (i) Kayden claims that

$$3^x \geq 2^x$$

Determine whether Kayden's claim is always true, sometimes true or never true, justifying your answer.

(2)

(ii) Prove that $\sqrt{3}$ is an irrational number.

(6)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 8 marks)

TOTAL FOR PAPER IS 100 MARKS

